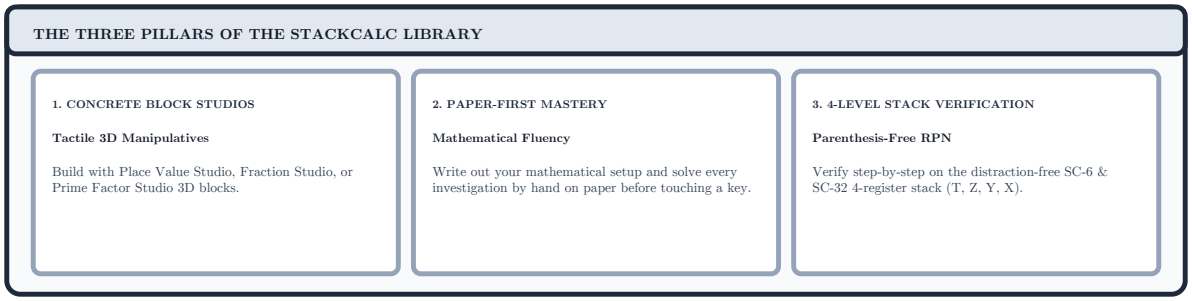
THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 5 OF 5

# STACK SPEED & FLUENCY

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**StackCalc**

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First Edition

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- Book 5 — Stack Speed & Fluency: Full-Year Speed Lessons, Playbooks & 180 Progressive Practice Tests (3,690 Problems • 128 Pages)
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- Prime Factor Studio: Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
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**STACK SPEED & FLUENCY: Full-Year Edition: 15 Speed Lessons, Shortcut Playbooks & 180 Progressive Practice Tests (3,690 Problems)**

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# Table of Contents

Directory of the Preface, 180-Test Master Tracker, 15 Seven-Page Speed Sections (Tests A1–D3), Appendices, and Online Answers.


| Section / Training Module & Test Series | Lesson & 12 Tests Included | Difficulty Progression | Pages |
|---|---|---|---|
| **Preface: The Full-Year Speed Training Cycle** | Students, Families & Teams | Lesson → Examples → 12 Tests | **p. iv** |
| **Master Score & Personal-Best Tracker (Part 1)** | Sections 1–8 (96 Timed Tests) | Score Log Across A1–A3 to D1–D3 | **p. 1** |
| **Master Score & Personal-Best Tracker (Part 2)** | Sections 9–12 & C1–C3 (84 Tests) | Score Log Across A1–A3 to D1–D3 | **p. 2** |
| **Section 1: Step Counting, Benchmarks & Sequences** | Lesson + 1A1–1D3 (246 probs) | Levels 1–4 (Foundational to Advanced) | **pp. 3–9** |
| **Section 2: Number Bonds & Missing Addends** | Lesson + 2A1–2D3 (246 probs) | Levels 1–4 (Foundational to Advanced) | **pp. 10–16** |
| **Section 3: Complements & Make-10 Bridging** | Lesson + 3A1–3D3 (246 probs) | Levels 1–4 (Foundational to Advanced) | **pp. 17–23** |
| **Section 4: Doubles, Equal Groups & Squares** | Lesson + 4A1–4D3 (246 probs) | Levels 1–4 (Foundational to Advanced) | **pp. 24–30** |
| **Cumulative Review C1: Number Sense (Sections 1–4)** | Lesson + C1A1–C1D3 (246 probs) | Levels 1–4 (Foundational to Advanced) | **pp. 31–37** |
| **Section 5: Multi-Step Operations & Grouping** | Lesson + 5A1–5D3 (246 probs) | Levels 1–4 (Grades 3–8+) | **pp. 38–44** |
| **Section 6: Fractions, Decimals & Reciprocals** | Lesson + 6A1–6D3 (246 probs) | Levels 1–4 (Grades 4–9+) | **pp. 45–51** |
| **Section 7: Primes, GCF/LCM & Nested Expressions** | Lesson + 7A1–7D3 (246 probs) | Levels 1–4 (Grades 5–10+) | **pp. 52–58** |
| **Section 8: Algebra, Quadratics & Coordinate Geometry** | Lesson + 8A1–8D3 (246 probs) | Levels 1–4 (Grades 6–10+) | **pp. 59–65** |
| **Cumulative Review C2: Intermediate Math (Sections 1–8)** | Lesson + C2A1–C2D3 (246 probs) | Levels 1–4 (Grades 5–10+) | **pp. 66–72** |
| **Section 9: Trigonometry, Vectors & Polar Coordinates** | Lesson + 9A1–9D3 (246 probs) | Levels 1–4 (Grades 8–12+) | **pp. 73–79** |
| **Section 10: Exponents, Logarithms & Growth Models** | Lesson + 10A1–10D3 (246 probs) | Levels 1–4 (Grades 8–12+) | **pp. 80–86** |
| **Section 11: Combinatorics, Probability & Statistics** | Lesson + 11A1–11D3 (246 probs) | Levels 1–4 (Grades 7–12+) | **pp. 87–93** |
| **Section 12: Base Conversions, Binary & Bitwise Logic** | Lesson + 12A1–12D3 (246 probs) | Levels 1–4 (Grades 7–12+) | **pp. 94–100** |
| **Cumulative Review C3: All-Level Championship (Sections 1–12)** | Lesson + C3A1–C3D3 (246 probs) | Levels 1–4 (Grades 7–12+) | **pp. 101–107** |
| **Appendix A: Stack Tactics & Identities** | SC-6 & SC-32 Zero-Memory RPN | Inside-Out Architectures & Rules | **pp. 108–109** |
| **Appendix B: Mental Math Sprint Drill Sheets** | 6 Timed 20-Problem Sprints | Complements, Squares, Roots, Rates | **pp. 110–115** |
| **Appendix C: Progress Graph & Streak Logs** | 180-Day Personal Best Analytics | Graph, Heatmap & Section Milestones | **pp. 116–118** |
| **Appendix D: 180-Test Mastery Award Certificate** | Framable Completion Award | Signed Recto Award & Blank Verso | **pp. 119–120** |
| **Appendix E: Speed Pipelines, Shortcuts & 3D Block Studios** | Bound Reference & Hardware Guide | SC-6/SC-32 Pipelines & Block Studios | **pp. 121–124** |
| **Online Answer Keys & Champion Pipelines** | Complete Solutions for All 180 Tests | Full 3,600 Answers + Optimal Routes | **stackcalc.io/curriculum/answers** |

# Preface: How to Train for Mathematical Speed & Fluency

Why raw test pages alone do not build speed—and how each 7-page section delivers a full year of structured practice.

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### 1. THE 7-PAGE FULL-YEAR ARCHITECTURE OF EVERY SECTION (1 LESSON + 12 TIMED TESTS = 180 TESTS)

Taking timed tests over and over without learning faster methods only makes slow habits permanent. That is why every section in this Full-Year Edition is organized as a complete 7-page training module (12 progressive tests per section = 180 tests for a 180-day school year):

- Page 1 — Speed Lesson, 'Slow Way vs. Champion Way' Playbook & 6 Guided Warm-Up Drills: Teaches the exact algebraic identities, mental shortcuts, and zero-memory stack pipelines before the clock starts.
- Pages 2–7 — Six Progressive Test Pages (Tests A1–A3, B1–B3, C1–C3, D1–D3 = 240 Timed Problems): Every test page opens with 2 Worked Examples at the top so students review the shortcut right before starting the timer!
- Physical 3D Block Studio Warm-Up (See Appendix E, p. 123): Before timing a new section, build the core pattern once with your physical Place Value Studio, Fraction Studio, or Prime Factor Studio blocks so mental math becomes automatic.

### 2. THREE WAYS TO USE THIS BOOK ACROSS A 36-WEEK SCHOOL YEAR OR CONTEST SEASON

**Mode 1 — Daily 180-Day Homeschool or Classroom Warm-Up (1 Test Per School Day):**

Complete 1 timed 20-problem test each school day (5 tests per week × 36 weeks = 180 tests). Study the 2 Worked Examples at the top of the page, run the 3-to-6-minute timer, and self-check online at https://stackcalc.io/curriculum/answers#speed-fluency.

**Champion Pipeline:** `2` `4` `ENTER` `1` `8` `+`

**Mode 2 — Curriculum Fluency Companion (3 Variants Per Level: A1–A3, B1–B3, C1–C3, D1–D3):**

Because each difficulty level provides 3 parallel test forms (e.g., A1, A2, A3), students can use Form 1 as a baseline diagnostic, Form 2 for guided practice, and Form 3 to prove personal-best mastery before advancing!

**Champion Pipeline:** `1` `4` `ENTER` `6` `+` `9` `ENTER` `4` `-` `÷`

**Mode 3 — Competitive Calculator & Number Sense Team Training (Levels 3 & 4 + Pipelines):**

Set a timer for the Goal at the top of the test strip (3 to 6 minutes). After grading in Appendix A (pp. 108–109), compare your keystrokes against the Optimal Champion Pipelines printed at the bottom of each section's key.

**Champion Pipeline:** `4` `x⁻¹` `6` `x⁻¹` `+` `x⁻¹`

### 3. SIMPLE FLUENCY SCORING OR CONTEST SCORING (20 PROBLEMS PER TEST × 180 TESTS = 3,600 PROBLEMS)

- Standard Fluency Score (Out of 20): Count 1 point for each correct answer (Score: ___ / 20), or multiply by 5 for a percentage score (18 / 20 = 90%).
- Competitive Team Scoring & Fermi Estimation (Vol. II, Ch. 14): Award +5 points per correct answer and deduct 2 points per incorrect answer (max 100 pts per test). Use Enrico Fermi's order-of-magnitude estimation (Volume II, Ch. 14) to sanity-check every result before writing!

**1. TACTILE BLOCK WARM-UP**

Build the core pattern once with your 3D Block Studios & work the 2 Page-Top Examples.

**2. TIMED SPRINT & ESTIMATE**

Run a 3-to-6-minute timer. Use inside-out RPN stack pipelines with zero parentheses.

**3. ONLINE PIPELINE AUDIT**

Check answers & optimal keystrokes at stackcalc.io/curriculum/answers#speed-fluency.

# 180-Test Speed & Fluency Progress Tracker (Part 1)

Record your scores (out of 20) across all 3 forms of Levels A, B, C, and D for Sections 1 through 8.

**HOW TO LOG YOUR PROGRESS ACROSS SECTIONS 1–8 (FOUNDATION TO PRE-ALGEBRA)**

Complete the 1-page Speed Lesson and 6 Warm-Up Drills first. Then log your score (out of 20) and time (minutes) for each of the 3 parallel forms in Level 1 (A1–A3), Level 2 (B1–B3), Level 3 (C1–C3), and Level 4 (D1–D3) across 8 core topics!

| Sec | Mathematical Topic | Lvl 1 (A1/A2/A3) | Lvl 2 (B1/B2/B3) | Lvl 3 (C1/C2/C3) | Lvl 4 (D1/D2/D3) | Pages |
|---|---|---|---|---|---|---|
| 1 | **Step Counting, Benchmarks & Sequences**<br>**Foundational to Advanced**<br>CCSS K.OA.5, 1.OA.6 • TEKS 1.3D, 2.4A • UIL Number Sense #1–10 | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 3–9 |
| 2 | **Number Bonds & Missing Addends**<br>**Foundational to Advanced**<br>CCSS 1.OA.4, 2.NBT.5 • TEKS 1.3B, 2.4B • UIL Number Sense #1–15 | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 10–16 |
| 3 | **Complements & Make-10 Bridging**<br>**Foundational to Advanced**<br>CCSS 1.NBT.4, 2.NBT.7 • TEKS 1.3D, 2.4C • UIL Number Sense #1–20 | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 17–23 |
| 4 | **Doubles, Equal Groups & Squares**<br>**Foundational to Advanced**<br>CCSS 2.OA.4, 3.OA.7 • TEKS 2.4A, 3.4F • UIL Number Sense Squares | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 24–30 |
| C1 | **Number Sense (Sections 1–4)**<br>**Foundational to Advanced**<br>CCSS 5.NBT, 6.NS • TEKS 6.3D, 7.3A • UIL Number Sense Mental Tricks | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 31–37 |
| 5 | **Multi-Step Operations & Grouping**<br>**Grades 3–8+**<br>CCSS 4.OA.3, 5.OA.1 • TEKS 4.4H, 5.4F • TMSCA / UIL Number Sense | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 38–44 |
| 6 | **Fractions, Decimals & Reciprocals**<br>**Grades 4–9+**<br>CCSS 5.NF.1, 6.NS.1 • TEKS 5.3H, 6.3E • UIL / MathCounts Sprint | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 45–51 |
| 7 | **Primes, GCF/LCM & Nested Expressions**<br>**Grades 5–10+**<br>CCSS 6.NS.4, 6.EE.2*c* • TEKS 6.7A • UIL Number Sense / MathCounts | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | 52–58 |

**36-WEEK FLUENCY SCHEDULE • 1 TEST PER SCHOOL DAY = 180 TIMED TESTS**

- Weeks 1–19: Complete Sections 1 through 8 (5 tests per week). Aim for 100% accuracy before timing.
- Self-Check Protocol: Check each day's test online at https://stackcalc.io/curriculum/answers#speed-fluency.
- If a test score falls below 16/20, review the 2 Worked Examples at the top of the test page and retry the parallel form!

MASTER SCORE & PERSONAL-BEST PROGRESS TRACKER (PART 2: SECTIONS 9–12 & C1–C3 • 84 TIMED TESTS)

# 180-Test Speed & Fluency Progress Tracker (Part 2)

Record your scores (out of 20) across all 3 forms of Levels A, B, C, and D for Sections 9 through C3.

**HOW TO LOG YOUR PROGRESS ACROSS SECTIONS 9–12 & C1–C3 (ADVANCED STEM & CONTEST SPEED)**

Sections 9 through 12 cover secondary STEM topics (vectors, logarithms, combinatorics, bitwise logic), while Sections C1 through C3 deliver competitive Number Sense, Calculator Applications, and Science Olympiad tournament speed training.

| Sec | Mathematical Topic | Lvl 1 (A1/A2/A3) | Lvl 2 (B1/B2/B3) | Lvl 3 (C1/C2/C3) | Lvl 4 (D1/D2/D3) | Pages |
|---|---|---|---|---|---|---|
| 8 | **Algebra, Quadratics & Coordinate Geometry**<br>**Grades 6–10+**<br>CCSS 8.EE.7, HSA.SSE.3 • TEKS A.5A, A.10E • UIL Calc Apps / AMC 8 | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | **59–65** |
| C2 | **Intermediate Math (Sections 1–8)**<br>**Grades 5–10+**<br>CCSS 7.EE.3, HSA.CED • TEKS 7.11A, A.5A • UIL Calculator Applications | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | **66–72** |
| 9 | **Trigonometry, Vectors & Polar Coordinates**<br>**Grades 8–12+**<br>CCSS 8.G.7, HSG.SRT.8 • TEKS 8.7C, G.9A • UIL Calc Apps Geometry | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | **73–79** |
| 10 | **Exponents, Logarithms & Growth Models**<br>**Grades 8–12+**<br>CCSS HSF.LE.4, HSF.BF.5 • TEKS 2A.5D • UIL Calc Apps Transcendentals | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | **80–86** |
| 11 | **Combinatorics, Probability & Statistics**<br>**Grades 7–12+**<br>CCSS 6.SP.5, HSS.CP.9 • TEKS 6.12C, S.4B • MathCounts / AMC 10/12 | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | **87–93** |
| 12 | **Base Conversions, Binary & Bitwise Logic**<br>**Grades 7–12+**<br>CCSS 6.NS, AP CSP • TEKS CS / Base-N • UIL Number Sense Bases | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | **94–100** |
| C3 | **All-Level Championship (Sections 1–12)**<br>**Grades 7–12+**<br>CCSS HSG.SRT, HSF.TF • TEKS G.9A, P.2P • UIL Calc Apps / Sci Olympiad | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | __ / __ / __<br>Best: ___m<br>[ ] Mastered | **101–107** |

**180-TEST FLUENCY MILESTONES & OFFICIAL ONLINE SOLUTIONS PORTAL**

- Bronze Fluency: 45 Tests Completed (Sections 1–4) • Silver Fluency: 90 Tests Completed (Sections 1–8)
- Gold Fluency: 135 Tests Completed (Sections 1–11) • Platinum Champion: All 180 Tests Completed (Certificate on page 119!)
- Online Answer Key & Keystroke Verification: Full solutions for all 3,600 test problems at https://stackcalc.io/curriculum/answers#speed-fluency

# Section 1: Step Counting, Benchmarks & Sequences

Unit steps, skip-counting, 10/100 benchmarks, and arithmetic series. • Study the Champion Way & complete W1–W6 before starting Tests 1A1–1D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 1: STEP COUNTING, BENCHMARKS & SEQUENCES

| | |
|---|---|
| RULE 1 | **Rule 1 — Pair Ends to Land on 10*s* & 100*s* First: In multi-step jumps (e.g., 68 + 12 + 25), combine the pair that completes a full 10 or 100 first (68 + 12 = 80, then + 25 = 105).** |
| RULE 2 | Rule 2 — Gauss Pairing for Arithmetic Series: To sum an evenly spaced sequence (First + ... + Last), multiply the average of the first and last terms by the number of terms: Sum = $n$ × (First + Last) ÷ 2. |
| RULE 3 | Rule 3 — Sum of First n Odd Numbers Is $n^2$: Never add 1 + 3 + 5 + ... one by one! Count how many odd numbers there are and square the count (first 5 odds = $5^2 = 25$; first 6 odds = $6^2 = 36$). |

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

### Archetype 1 — Benchmark Pairing: Evaluate 96 + 4 + 38

Stack Keys: 9 6 ENTER 4 + 3 8 +

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Add digits column by column on paper (6+4+8 = 18, carry 1).

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Spot that 96 + 4 = 100 immediately, then 100 + 38 = 138 in 1 second!**

### Archetype 2 — Gauss Rainbow Pairing: Evaluate 14 + 16 + 18 + 20 + 22

Stack Keys: 1 8 ENTER 5 ×

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Add left-to-right (14+16=30, +18=48, +20=68, +22=90) — 4 separate additions.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: For any symmetric 5-term sequence, the sum is simply 5 × Middle Term: 5 × 18 = 90!**

### Archetype 3 — Triangular Series Sum: Evaluate 1 + 2 + 3 + ... + 10

Stack Keys: 1 0 ENTER 1 1 × 2 ÷

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Type all 10 numbers into the calculator one by one (29 keystrokes).

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Use n($n$ + 1) ÷ 2: 10 × 11 ÷ 2 = 55 in 1 mental step (or 7 keys on the stack)!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** 76 + 4 + 19 = _____

Shortcut Cue: Complete 80 first, then + 19

**W2.** 150 − 25 − 25 = _____

Shortcut Cue: Combine 25 + 25 = 50, subtract from 150

**W3.** 1 + 3 + 5 + 7 = _____

Shortcut Cue: 4 consecutive odd numbers → $4^2$

**W4.** 10 + 15 + 20 + 25 + 30 = _____

Shortcut Cue: 5 terms × middle term (20)

**W5.** 1 + 2 + 3 + ... + 8 = _____

Shortcut Cue: Gauss rule: 8 × 9 ÷ 2

**W6.** 2 + 4 + 6 + 8 + 10 = _____

Shortcut Cue: n(n + 1) for 5 even terms: 5 × 6

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 1: Step Counting, Benchmarks & Sequences

Review Worked Examples 1 & 2 below, then solve Test 1A1 (Level 1: Foundation (Sprint 1)) & Test 1A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $76 + 4 + 19 = 99$ (Complete 80 first, then + 19)

**EXAMPLE 2** $150 - 25 - 25 = 100$ (Combine $25 + 25 = 50$, subtract from 150)

### TEST 1A1 • LEVEL 1: FOUNDATION (SPRINT 1) Goal: 3 min

Adding & Subtracting 1, 2, and 3 Within 20

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $3 + 1 =$ | _____ |
| 2. | $5 + 1 =$ | _____ |
| 3. | $8 + 1 =$ | _____ |
| 4. | $6 - 1 =$ | _____ |
| 5. | $10 - 1 =$ | _____ |
| 6. | $4 + 2 =$ | _____ |
| 7. | $7 + 2 =$ | _____ |
| 8. | $9 - 2 =$ | _____ |
| 9. | $2 + 1 + 1 =$ | _____ |
| 10. | $5 + 1 + 1 =$ | _____ |
| 11. | $8 - 1 - 1 =$ | _____ |
| 12. | $11 + 1 =$ | _____ |
| 13. | $14 - 1 =$ | _____ |
| 14. | $12 + 2 =$ | _____ |
| 15. | $16 - 2 =$ | _____ |
| 16. | $10 + 3 =$ | _____ |
| 17. | $15 - 3 =$ | _____ |
| 18. | $7 + 1 + 2 =$ | _____ |
| 19. | $13 + 2 + 1 =$ | _____ |
| 20. | $19 - 2 - 1 =$ | _____ |

### TEST 1A2 • LEVEL 1: FOUNDATION (SPRINT 2) Goal: 3 min

Adding & Subtracting 1, 2, and 3 Within 20 — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $7 + 2 =$ | _____ |
| 2. | $12 - 3 =$ | _____ |
| 3. | $13 + 1 + 1 =$ | _____ |
| 4. | $2 + 2 =$ | _____ |
| 5. | $7 - 3 =$ | _____ |
| 6. | $8 + 1 + 2 =$ | _____ |
| 7. | $11 + 2 =$ | _____ |
| 8. | $16 - 3 =$ | _____ |
| 9. | $3 + 1 + 1 =$ | _____ |
| 10. | $6 + 2 =$ | _____ |
| 11. | $11 - 3 =$ | _____ |
| 12. | $12 + 1 + 2 =$ | _____ |
| 13. | $15 + 2 =$ | _____ |
| 14. | $6 - 3 =$ | _____ |
| 15. | $7 + 1 + 1 =$ | _____ |
| 16. | $10 + 2 =$ | _____ |
| 17. | $15 - 3 =$ | _____ |
| 18. | $2 + 1 + 2 =$ | _____ |
| 19. | $5 + 2 =$ | _____ |
| 20. | $10 - 3 =$ | _____ |

# Section 1: Step Counting, Benchmarks & Sequences

Review Worked Examples 1 & 2 below, then solve Test 1A3 (Level 1: Foundation (Sprint 3)) & Test 1B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $1 + 3 + 5 + 7 = 16$ (4 consecutive odd numbers → $4^2$)

**EXAMPLE 2** $10 + 15 + 20 + 25 + 30 = 100$ (5 terms × middle term (20))

### TEST 1A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 3 min

Bridge from Level 1 to Level 2 (Skip-Counting Steps by 2*s*, 5*s*, a)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $12 + 3 =$ | |
| 2. | $17 - 1 =$ | |
| 3. | $4 + 2 + 1 =$ | |
| 4. | $7 + 3 =$ | |
| 5. | $12 - 1 =$ | |
| 6. | $13 + 2 + 2 =$ | |
| 7. | $2 + 3 =$ | |
| 8. | $7 - 1 =$ | |
| 9. | $8 + 2 + 1 =$ | |
| 10. | $11 + 3 =$ | |
| 11. | $16 - 1 =$ | |
| 12. | $3 + 2 + 2 =$ | |
| 13. | $6 + 3 =$ | |
| 14. | $11 - 1 =$ | |
| 15. | $12 + 2 + 1 =$ | |
| 16. | $15 + 3 =$ | |
| 17. | $6 - 1 =$ | |
| 18. | $7 + 2 + 2 =$ | |
| 19. | $10 + 3 =$ | |
| 20. | $15 - 1 =$ | |

### TEST 1B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 4 min

Skip-Counting Steps by 2*s*, 5*s*, and 10*s*

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $2 + 2 + 2 =$ | |
| 2. | $2 + 2 + 2 + 2 =$ | |
| 3. | $5 + 5 =$ | |
| 4. | $5 + 5 + 5 =$ | |
| 5. | $10 + 10 + 10 =$ | |
| 6. | $12 + 2 + 2 =$ | |
| 7. | $18 - 2 - 2 =$ | |
| 8. | $15 + 5 + 5 =$ | |
| 9. | $25 - 5 - 5 =$ | |
| 10. | $20 + 10 + 5 =$ | |
| 11. | $30 - 10 - 5 =$ | |
| 12. | $14 + 3 + 3 =$ | |
| 13. | $22 + 4 + 4 =$ | |
| 14. | $40 - 5 - 5 =$ | |
| 15. | $35 + 10 + 5 =$ | |
| 16. | $50 - 10 - 10 =$ | |
| 17. | $16 + 2 + 2 + 2 =$ | |
| 18. | $45 + 5 + 10 =$ | |
| 19. | $70 - 10 - 5 =$ | |
| 20. | $85 + 5 + 10 =$ | |

# Section 1: Step Counting, Benchmarks & Sequences

Review Worked Examples 1 & 2 below, then solve Test 1B2 (Level 2: Core Fluency (Sprint 2)) & Test 1B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Benchmark Pairing: Evaluate $96 + 4 + 38 \rightarrow$ Spot that $96 + 4 = 100$ immediately, then $100 + 38 = 138$ in 1 second!

**EXAMPLE 2** $1 + 2 + 3 + \ldots + 8 = 36$ (Gauss rule: $8 \times 9 \div 2$)

**TEST 1B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 4 min

Skip-Counting Steps by 2*s*, 5*s*, and 10*s* — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $15 + 5 + 5 =$ | |
| 2. | $70 - 10 - 10 =$ | |
| 3. | $15 + 3 + 3 =$ | |
| 4. | $36 - 4 - 4 =$ | |
| 5. | $14 + 2 + 2 =$ | |
| 6. | $55 - 5 - 5 =$ | |
| 7. | $90 + 10 + 10 =$ | |
| 8. | $18 - 3 - 3 =$ | |
| 9. | $16 + 4 + 4 =$ | |
| 10. | $16 - 2 - 2 =$ | |
| 11. | $30 + 5 + 5 =$ | |
| 12. | $100 - 10 - 10 =$ | |
| 13. | $24 + 3 + 3 =$ | |
| 14. | $48 - 4 - 4 =$ | |
| 15. | $6 + 2 + 2 =$ | |
| 16. | $35 - 5 - 5 =$ | |
| 17. | $50 + 10 + 10 =$ | |
| 18. | $27 - 3 - 3 =$ | |
| 19. | $28 + 4 + 4 =$ | |
| 20. | $22 - 2 - 2 =$ | |

**TEST 1B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 4 min

Bridge from Level 2 to Level 3 (Crossing 10*s* and 100*s* with Multi)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $40 + 10 + 10 =$ | |
| 2. | $24 - 3 - 3 =$ | |
| 3. | $24 + 4 + 4 =$ | |
| 4. | $20 - 2 - 2 =$ | |
| 5. | $40 + 5 + 5 =$ | |
| 6. | $120 - 10 - 10 =$ | |
| 7. | $30 + 3 + 3 =$ | |
| 8. | $28 - 4 - 4 =$ | |
| 9. | $10 + 2 + 2 =$ | |
| 10. | $45 - 5 - 5 =$ | |
| 11. | $70 + 10 + 10 =$ | |
| 12. | $33 - 3 - 3 =$ | |
| 13. | $36 + 4 + 4 =$ | |
| 14. | $26 - 2 - 2 =$ | |
| 15. | $20 + 5 + 5 =$ | |
| 16. | $80 - 10 - 10 =$ | |
| 17. | $18 + 3 + 3 =$ | |
| 18. | $40 - 4 - 4 =$ | |
| 19. | $16 + 2 + 2 =$ | |
| 20. | $60 - 5 - 5 =$ | |

# Section 1: Step Counting, Benchmarks & Sequences

Review Worked Examples 1 & 2 below, then solve Test 1C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 1C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Gauss Rainbow Pairing: Evaluate $14 + 16 + 18 + 20 + 22$ → For any symmetric 5-term sequence, the sum is simply 5 × Middle Term: $5 \times 18 = 90$!

**EXAMPLE 2** 1C #11 (68+12+20) → Pair $68 + 12 = 80$ first, then $+ 20 = 100$

**TEST 1C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Crossing 10*s* and 100*s* with Multi-Step Jumps

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $28 + 2 + 6 =$ | |
| 2. | $37 + 3 + 8 =$ | |
| 3. | $54 - 4 - 7 =$ | |
| 4. | $63 - 3 - 9 =$ | |
| 5. | $45 + 15 + 10 =$ | |
| 6. | $82 - 12 - 15 =$ | |
| 7. | $96 + 4 + 25 =$ | |
| 8. | $115 - 15 - 20 =$ | |
| 9. | $75 + 25 + 35 =$ | |
| 10. | $150 - 25 - 25 =$ | |
| 11. | $68 + 12 + 20 =$ | |
| 12. | $134 - 14 - 30 =$ | |
| 13. | $180 + 20 + 45 =$ | |
| 14. | $260 - 60 - 35 =$ | |
| 15. | $125 + 25 + 50 =$ | |
| 16. | $310 - 10 - 45 =$ | |
| 17. | $240 + 30 + 30 =$ | |
| 18. | $420 - 70 - 50 =$ | |
| 19. | $375 + 25 + 80 =$ | |
| 20. | $500 - 125 - 75 =$ | |

**TEST 1C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Crossing 10*s* and 100*s* with Multi-Step Jumps — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $48 + 2 + 19 =$ | |
| 2. | $89 - 5 - 24 =$ | |
| 3. | $62 + 8 + 29 =$ | |
| 4. | $116 - 2 - 34 =$ | |
| 5. | $85 + 5 + 39 =$ | |
| 6. | $152 - 8 - 44 =$ | |
| 7. | $108 + 2 + 14 =$ | |
| 8. | $144 - 5 - 19 =$ | |
| 9. | $122 + 8 + 24 =$ | |
| 10. | $171 - 2 - 29 =$ | |
| 11. | $145 + 5 + 34 =$ | |
| 12. | $207 - 8 - 39 =$ | |
| 13. | $168 + 2 + 44 =$ | |
| 14. | $199 - 5 - 14 =$ | |
| 15. | $182 + 8 + 19 =$ | |
| 16. | $226 - 2 - 24 =$ | |
| 17. | $205 + 5 + 29 =$ | |
| 18. | $262 - 8 - 34 =$ | |
| 19. | $228 + 2 + 39 =$ | |
| 20. | $289 - 5 - 44 =$ | |

# Section 1: Step Counting, Benchmarks & Sequences

Review Worked Examples 1 & 2 below, then solve Test 1C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 1D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Triangular Series Sum: Evaluate $1 + 2 + 3 + \ldots + 10$ → Use $n(n+1) \div 2$: $10 \times 11 \div 2 = 55$ in 1 mental step (or 7 keys on the stack)!

**EXAMPLE 2** 1D #3 (1+3+5+7+9) → 5 odd terms → $5^2 = 25$

**TEST 1C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Arithmetic Sequences & Fast Seri)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $67 + 3 + 26 =$ _____
2. $117 - 6 - 31 =$ _____
3. $81 + 9 + 36 =$ _____
4. $144 - 3 - 41 =$ _____
5. $104 + 6 + 46 =$ _____
6. $145 - 9 - 16 =$ _____
7. $127 + 3 + 21 =$ _____
8. $172 - 6 - 26 =$ _____
9. $141 + 9 + 31 =$ _____
10. $199 - 3 - 36 =$ _____
11. $164 + 6 + 41 =$ _____
12. $235 - 9 - 46 =$ _____
13. $187 + 3 + 16 =$ _____
14. $227 - 6 - 21 =$ _____
15. $201 + 9 + 26 =$ _____
16. $254 - 3 - 31 =$ _____
17. $224 + 6 + 36 =$ _____
18. $290 - 9 - 41 =$ _____
19. $247 + 3 + 46 =$ _____
20. $282 - 6 - 16 =$ _____

**TEST 1D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 5 min

Arithmetic Sequences & Fast Series Sums

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1 + 2 + 3 + 4 + 5 =$ _____
2. $2 + 4 + 6 + 8 + 10 =$ _____
3. $1 + 3 + 5 + 7 + 9 =$ _____
4. $3 + 6 + 9 + 12 =$ _____
5. $4 + 8 + 12 + 16 =$ _____
6. $5 + 10 + 15 + 20 + 25 =$ _____
7. $1 + 2 + 3 + 4 + 5 + 6 + 7 =$ _____
8. $10 + 12 + 14 + 16 =$ _____
9. $15 + 20 + 25 + 30 =$ _____
10. $11 + 13 + 15 + 17 =$ _____
11. $20 + 25 + 30 + 35 + 40 =$ _____
12. $1 + 3 + 5 + 7 + 9 + 11 =$ _____
13. $6 + 12 + 18 + 24 =$ _____
14. $100 - 12 - 18 - 20 =$ _____
15. $25 + 50 + 75 + 100 =$ _____
16. $14 + 16 + 18 + 20 + 22 =$ _____
17. $7 + 14 + 21 + 28 =$ _____
18. $50 + 60 + 70 + 80 + 90 =$ _____
19. $1 + 2 + 3 + \ldots + 10 =$ _____
20. $2 + 4 + 6 + \ldots + 20 =$ _____

# Section 1: Step Counting, Benchmarks & Sequences

Review Worked Examples 1 & 2 below, then solve Test 1D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 1D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 1D #16 (14+16+18+20+22) → 5 × middle term (18) = 90

**EXAMPLE 2** 1D #19 (1+2+...+10) → 10 × 11 ÷ 2 = 55 ([10] [ENTER] [11] [×] [2] [÷])

**TEST 1D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 5 min

Arithmetic Sequences & Fast Series Sums — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 1 + 3 + 5 + ... + 11 = _____
2. 1 + 3 + 5 + ... + 13 = _____
3. 1 + 3 + 5 + ... + 15 = _____
4. 1 + 3 + 5 + ... + 17 = _____
5. 1 + 3 + 5 + ... + 19 = _____
6. 1 + 3 + 5 + ... + 21 = _____
7. 15 + 19 + 23 + 27 + 31 = _____
8. 15 + 20 + 25 + 30 + 35 = _____
9. 23 + 25 + 27 + 29 + 31 = _____
10. 23 + 26 + 29 + 32 + 35 = _____
11. 23 + 27 + 31 + 35 + 39 = _____
12. 23 + 28 + 33 + 38 + 43 = _____
13. 31 + 33 + 35 + 37 + 39 = _____
14. 1 + 2 + 3 + ... + 9 = _____
15. 1 + 2 + 3 + ... + 11 = _____
16. 1 + 2 + 3 + ... + 13 = _____
17. 1 + 2 + 3 + ... + 15 = _____
18. 1 + 2 + 3 + ... + 17 = _____
19. 1 + 2 + 3 + ... + 19 = _____
20. 1 + 2 + 3 + ... + 21 = _____

**TEST 1D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 4 min

Section 1 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 1 + 3 + 5 + ... + 15 = _____
2. 1 + 3 + 5 + ... + 17 = _____
3. 1 + 3 + 5 + ... + 19 = _____
4. 1 + 3 + 5 + ... + 21 = _____
5. 1 + 3 + 5 + ... + 23 = _____
6. 1 + 3 + 5 + ... + 25 = _____
7. 18 + 22 + 26 + 30 + 34 = _____
8. 18 + 23 + 28 + 33 + 38 = _____
9. 26 + 28 + 30 + 32 + 34 = _____
10. 26 + 29 + 32 + 35 + 38 = _____
11. 26 + 30 + 34 + 38 + 42 = _____
12. 26 + 31 + 36 + 41 + 46 = _____
13. 34 + 36 + 38 + 40 + 42 = _____
14. 1 + 2 + 3 + ... + 10 = _____
15. 1 + 2 + 3 + ... + 12 = _____
16. 1 + 2 + 3 + ... + 14 = _____
17. 1 + 2 + 3 + ... + 16 = _____
18. 1 + 2 + 3 + ... + 18 = _____
19. 1 + 2 + 3 + ... + 20 = _____
20. 1 + 2 + 3 + ... + 22 = _____

# Section 2: Number Bonds & Missing Addends

Part-part-whole composition, decomposition, and missing-part equations. • Study the Champion Way & complete W1–W6 before starting Tests 2A1–2D3.

### 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 2: NUMBER BONDS & MISSING ADDENDS

| | |
|---|---|
| RULE 1 | **Rule 1 — Think Addition to Solve Missing Addends: For 28 + [ ? ] = 50, count up to the next ten (+2 → 30) and then to 50 (+20) → 22, or key [50] [ENTER] [28] [-].** |
| RULE 2 | Rule 2 — Subtract in Two Clean Chunks (Tens Then Ones): For 72 - 35, subtract 30 first (72 - 30 = 42), then subtract 5 (42 - 5 = 37). |
| RULE 3 | Rule 3 — Balanced Equation Shifting: For 85 + 45 = 100 + [ ? ], notice 100 is 15 more than 85, so the missing number is 15 less than 45 (45 - 15 = 30)! |

### 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Missing Addend to 100: Solve 63 + [ ? ] = 100** — Stack Keys: `1` `0` `0` `ENTER` `6` `3` `-`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Write vertical 100 - 63 on paper and borrow across two zeros.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Make 9 tens and 10 ones: (9 - 6 = 3 tens) and (10 - 3 = 7 ones) → 37!**

**Archetype 2 — Reordering Three-Part Sums: Evaluate 48 + 35 + 52** — Stack Keys: `4` `8` `ENTER` `5` `2` `+` `3` `5` `+`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 48 + 35 = 83 first, then add 83 + 52 = 135.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Pair the complements 48 + 52 = 100 first, then 100 + 35 = 135!**

**Archetype 3 — Balanced Equation Shortcut: Solve 76 + 54 = 90 + [ ? ]** — Stack Keys: `7` `6` `ENTER` `5` `4` `+` `9` `0` `-`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Add 76 + 54 = 130, then subtract 130 - 90 = 40.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Add 76 + 54 = 130 on the stack, then press [90] [-] → 40!**

### 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** 9 + [ ? ] = 17 _____

Shortcut Cue: 17 - 9 (or 1 + 7)

**W2.** 36 + [ ? ] = 60 _____

Shortcut Cue: Count up: +4 to 40, +20 to 60

**W3.** 82 − 35 = _____

Shortcut Cue: 82 - 30 = 52, then 52 - 5

**W4.** 64 + 29 + 36 = _____

Shortcut Cue: Pair 64 + 36 = 100 first, then + 29

**W5.** 200 − 68 − 32 = _____

Shortcut Cue: Combine 68 + 32 = 100, then 200 - 100

**W6.** 85 + 45 = 100 + [ ? ] _____

Shortcut Cue: 85 + 45 = 130, minus 100

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 2: Number Bonds & Missing Addends

Review Worked Examples 1 & 2 below, then solve Test 2A1 (Level 1: Foundation (Sprint 1)) & Test 2A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** 9 + [ ? ] = 17 8 (17 - 9 (or 1 + 7))

**EXAMPLE 2** 36 + [ ? ] = 60 24 (Count up: +4 to 40, +20 to 60)

**TEST 2A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 3 min

Combining & Decomposing Parts Within 12

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 4 + 3 = | |
| 2. | 5 + 2 = | |
| 3. | 3 + 5 = | |
| 4. | 6 + 2 = | |
| 5. | 4 + 5 = | |
| 6. | 7 + 2 = | |
| 7. | 6 + 3 = | |
| 8. | 7 − 3 = | |
| 9. | 8 − 5 = | |
| 10. | 9 − 4 = | |
| 11. | 9 − 6 = | |
| 12. | 5 + 6 = | |
| 13. | 7 + 4 = | |
| 14. | 8 + 3 = | |
| 15. | 7 + 5 = | |
| 16. | 8 + 4 = | |
| 17. | 9 + 3 = | |
| 18. | 11 − 5 = | |
| 19. | 12 − 7 = | |
| 20. | 12 − 8 = | |

**TEST 2A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 3 min

Combining & Decomposing Parts Within 12 — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 5 + 4 + 1 = | |
| 2. | 17 − 7 = | |
| 3. | 9 + 3 + 1 = | |
| 4. | 12 − 3 = | |
| 5. | 5 + 2 + 1 = | |
| 6. | 15 − 7 = | |
| 7. | 9 + 8 + 1 = | |
| 8. | 10 − 3 = | |
| 9. | 5 + 7 + 1 = | |
| 10. | 13 − 7 = | |
| 11. | 9 + 6 + 1 = | |
| 12. | 8 − 3 = | |
| 13. | 5 + 5 + 1 = | |
| 14. | 18 − 7 = | |
| 15. | 9 + 4 + 1 = | |
| 16. | 13 − 3 = | |
| 17. | 5 + 3 + 1 = | |
| 18. | 16 − 7 = | |
| 19. | 9 + 2 + 1 = | |
| 20. | 11 − 3 = | |

# Section 2: Number Bonds & Missing Addends

Review Worked Examples 1 & 2 below, then solve Test 2A3 (Level 1: Foundation (Sprint 3)) & Test 2B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** 82 - 35 = 47 (82 - 30 = 52, then 52 - 5)

**EXAMPLE 2** 64 + 29 + 36 = 129 (Pair 64 + 36 = 100 first, then + 29)

**TEST 2A3 • LEVEL 1: FOUNDATION (SPRINT 3)** Goal: 3 min

Bridge from Level 1 to Level 2 (Missing Addends & Subtraction Wi)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 8 + 6 + 2 = | |
| 2. | 8 − 2 = | |
| 3. | 4 + 5 + 2 = | |
| 4. | 18 − 6 = | |
| 5. | 8 + 4 + 2 = | |
| 6. | 13 − 2 = | |
| 7. | 4 + 3 + 2 = | |
| 8. | 16 − 6 = | |
| 9. | 8 + 2 + 2 = | |
| 10. | 11 − 2 = | |
| 11. | 4 + 8 + 2 = | |
| 12. | 14 − 6 = | |
| 13. | 8 + 7 + 2 = | |
| 14. | 9 − 2 = | |
| 15. | 4 + 6 + 2 = | |
| 16. | 12 − 6 = | |
| 17. | 8 + 5 + 2 = | |
| 18. | 14 − 2 = | |
| 19. | 4 + 4 + 2 = | |
| 20. | 17 − 6 = | |

**TEST 2B1 • LEVEL 2: CORE FLUENCY (SPRINT 1)** Goal: 4 min

Missing Addends & Subtraction Within 25

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 6 + 8 = | |
| 2. | 7 + 6 = | |
| 3. | 8 + 7 = | |
| 4. | 9 + 6 = | |
| 5. | 8 + 9 = | |
| 6. | 13 − 6 = | |
| 7. | 14 − 8 = | |
| 8. | 15 − 7 = | |
| 9. | 16 − 9 = | |
| 10. | 17 − 8 = | |
| 11. | 5 + [ ? ] = 13 | |
| 12. | 7 + [ ? ] = 16 | |
| 13. | 9 + [ ? ] = 18 | |
| 14. | [ ? ] + 6 = 15 | |
| 15. | [ ? ] + 8 = 19 | |
| 16. | 20 − 13 = | |
| 17. | 22 − 8 = | |
| 18. | 14 + [ ? ] = 21 | |
| 19. | 24 − 9 = | |
| 20. | 16 + [ ? ] = 25 | |

# Section 2: Number Bonds & Missing Addends

Review Worked Examples 1 & 2 below, then solve Test 2B2 (Level 2: Core Fluency (Sprint 2)) & Test 2B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Missing Addend to 100: Solve 63 + [ ? ] = 100 → Make 9 tens and 10 ones: (9 - 6 = 3 tens) and (10 - 3 = 7 ones) → 37!

**EXAMPLE 2** 200 - 68 - 32 = 100 (Combine 68 + 32 = 100, then 200 - 100)

**TEST 2B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 4 min

Missing Addends & Subtraction Within 25 — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 9 + [ ? ] = 18 | |
| 2. | [ ? ] + 11 = 23 | |
| 3. | 28 − 15 = | |
| 4. | 6 + [ ? ] = 10 | |
| 5. | [ ? ] + 6 = 15 | |
| 6. | 20 − 12 = | |
| 7. | 15 + [ ? ] = 25 | |
| 8. | [ ? ] + 12 = 18 | |
| 9. | 23 − 9 = | |
| 10. | 12 + [ ? ] = 17 | |
| 11. | [ ? ] + 7 = 22 | |
| 12. | 15 − 6 = | |
| 13. | 9 + [ ? ] = 20 | |
| 14. | [ ? ] + 13 = 25 | |
| 15. | 19 − 15 = | |
| 16. | 6 + [ ? ] = 12 | |
| 17. | [ ? ] + 8 = 17 | |
| 18. | 22 − 12 = | |
| 19. | 15 + [ ? ] = 27 | |
| 20. | [ ? ] + 14 = 20 | |

**TEST 2B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 4 min

Bridge from Level 2 to Level 3 (Two-Digit Number Bonds & Missing)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 13 + [ ? ] = 27 | |
| 2. | [ ? ] + 5 = 21 | |
| 3. | 14 − 7 = | |
| 4. | 10 + [ ? ] = 19 | |
| 5. | [ ? ] + 11 = 24 | |
| 6. | 29 − 16 = | |
| 7. | 7 + [ ? ] = 11 | |
| 8. | [ ? ] + 6 = 16 | |
| 9. | 21 − 13 = | |
| 10. | 16 + [ ? ] = 26 | |
| 11. | [ ? ] + 12 = 19 | |
| 12. | 24 − 10 = | |
| 13. | 13 + [ ? ] = 18 | |
| 14. | [ ? ] + 7 = 23 | |
| 15. | 16 − 7 = | |
| 16. | 10 + [ ? ] = 21 | |
| 17. | [ ? ] + 13 = 26 | |
| 18. | 20 − 16 = | |
| 19. | 7 + [ ? ] = 13 | |
| 20. | [ ? ] + 8 = 18 | |

# Section 2: Number Bonds & Missing Addends

Review Worked Examples 1 & 2 below, then solve Test 2C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 2C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Reordering Three-Part Sums: Evaluate $48 + 35 + 52 \rightarrow$ Pair the complements $48 + 52 = 100$ first, then $100 + 35 = 135$!

**EXAMPLE 2** 2C #17 ($63 + ? = 100$) → 9-from-9, 10-from-10 rule → 37

**TEST 2C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Two-Digit Number Bonds & Missing Parts

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $24 + 15 =$ | |
| 2. | $32 + 27 =$ | |
| 3. | $46 + 18 =$ | |
| 4. | $38 + 25 =$ | |
| 5. | $54 - 21 =$ | |
| 6. | $60 - 24 =$ | |
| 7. | $72 - 35 =$ | |
| 8. | $81 - 46 =$ | |
| 9. | $28 + [\ ?\ ] = 50$ | |
| 10. | $34 + [\ ?\ ] = 60$ | |
| 11. | $[\ ?\ ] + 42 = 75$ | |
| 12. | $[\ ?\ ] + 37 = 80$ | |
| 13. | $90 - 56 =$ | |
| 14. | $85 - 29 =$ | |
| 15. | $47 + 36 =$ | |
| 16. | $58 + 37 =$ | |
| 17. | $63 + [\ ?\ ] = 100$ | |
| 18. | $100 - 48 =$ | |
| 19. | $110 - 65 =$ | |
| 20. | $74 + [\ ?\ ] = 120$ | |

**TEST 2C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Two-Digit Number Bonds & Missing Parts — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $22 + [\ ?\ ] = 43$ | |
| 2. | $51 - 25 =$ | |
| 3. | $28 + 31 =$ | |
| 4. | $31 + [\ ?\ ] = 67$ | |
| 5. | $75 - 34 =$ | |
| 6. | $37 + 46 =$ | |
| 7. | $40 + [\ ?\ ] = 55$ | |
| 8. | $63 - 43 =$ | |
| 9. | $46 + 25 =$ | |
| 10. | $49 + [\ ?\ ] = 79$ | |
| 11. | $87 - 52 =$ | |
| 12. | $55 + 40 =$ | |
| 13. | $58 + [\ ?\ ] = 103$ | |
| 14. | $75 - 61 =$ | |
| 15. | $64 + 19 =$ | |
| 16. | $67 + [\ ?\ ] = 91$ | |
| 17. | $99 - 70 =$ | |
| 18. | $73 + 34 =$ | |
| 19. | $76 + [\ ?\ ] = 115$ | |
| 20. | $123 - 79 =$ | |

# Section 2: Number Bonds & Missing Addends

Review Worked Examples 1 & 2 below, then solve Test 2C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 2D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Balanced Equation Shortcut: Solve 76 + 54 = 90 + [ ? ] → Add 76 + 54 = 130 on the stack, then press [90] [-] → 40!

**EXAMPLE 2** 2D #6 (48 + 35 + 52) → Group (48 + 52) + 35 = 100 + 35 = 135

**TEST 2C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Three-Part Bonds & Balanced Equa)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 26 + [ ? ] = 54 _____
2. 62 − 29 = _____
3. 32 + 38 = _____
4. 35 + [ ? ] = 78 _____
5. 86 − 38 = _____
6. 41 + 17 = _____
7. 44 + [ ? ] = 66 _____
8. 74 − 47 = _____
9. 50 + 32 = _____
10. 53 + [ ? ] = 90 _____
11. 98 − 56 = _____
12. 59 + 47 = _____
13. 62 + [ ? ] = 78 _____
14. 86 − 65 = _____
15. 68 + 26 = _____
16. 71 + [ ? ] = 102 _____
17. 110 − 74 = _____
18. 77 + 41 = _____
19. 80 + [ ? ] = 126 _____
20. 98 − 83 = _____

**TEST 2D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 5 min

Three-Part Bonds & Balanced Equations

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 125 + 65 = _____
2. 240 − 85 = _____
3. 135 + [ ? ] = 200 _____
4. 168 + [ ? ] = 250 _____
5. 320 − 145 = _____
6. 48 + 35 + 52 = _____
7. 64 + 29 + 36 = _____
8. 150 − 45 − 35 = _____
9. 200 − 68 − 32 = _____
10. 85 + 45 = 100 + [ ? ] _____
11. 76 + 54 = 90 + [ ? ] _____
12. 140 − 55 = 50 + [ ? ] _____
13. 360 − 175 = _____
14. 245 + 155 − 80 = _____
15. 500 − 185 − 115 = _____
16. 325 + [ ? ] = 600 _____
17. 450 − [ ? ] = 185 _____
18. (165 + 85) − 70 = _____
19. (340 − 160) + 95 = _____
20. 1000 − 425 − 175 = _____

# Section 2: Number Bonds & Missing Addends

Review Worked Examples 1 & 2 below, then solve Test 2D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 2D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 2D #9 (200 - 68 - 32) → 200 - (68 + 32) = 200 - 100 = 100

**EXAMPLE 2** 2D #11 (76 + 54 = 90 + ?) → [76] [ENTER] [54] [+] [90] [-] → 40

**TEST 2D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 5 min

Three-Part Bonds & Balanced Equations — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 38 + 17 + 62 = _____
2. 42 + 20 + 58 = _____
3. 46 + 23 + 54 = _____
4. 50 + 26 + 50 = _____
5. 54 + 29 + 46 = _____
6. 58 + 32 + 42 = _____
7. 62 + 35 + 38 = _____
8. 66 + 38 + 34 = _____
9. 70 + 41 + 30 = _____
10. 74 + 44 + 26 = _____
11. 69 + 47 = 80 + [ ? ] _____
12. 74 + 50 = 85 + [ ? ] _____
13. 79 + 53 = 90 + [ ? ] _____
14. 84 + 56 = 95 + [ ? ] _____
15. 89 + 59 = 100 + [ ? ] _____
16. 94 + 62 = 105 + [ ? ] _____
17. 99 + 65 = 110 + [ ? ] _____
18. 104 + 68 = 115 + [ ? ] _____
19. 109 + 71 = 120 + [ ? ] _____
20. 114 + 74 = 125 + [ ? ] _____

**TEST 2D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 4 min

Section 2 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 44 + 19 + 56 = _____
2. 48 + 22 + 52 = _____
3. 52 + 25 + 48 = _____
4. 56 + 28 + 44 = _____
5. 60 + 31 + 40 = _____
6. 64 + 34 + 36 = _____
7. 68 + 37 + 32 = _____
8. 72 + 40 + 28 = _____
9. 76 + 43 + 24 = _____
10. 80 + 46 + 20 = _____
11. 73 + 49 = 80 + [ ? ] _____
12. 78 + 52 = 85 + [ ? ] _____
13. 83 + 55 = 90 + [ ? ] _____
14. 88 + 58 = 95 + [ ? ] _____
15. 93 + 61 = 100 + [ ? ] _____
16. 98 + 64 = 105 + [ ? ] _____
17. 103 + 67 = 110 + [ ? ] _____
18. 108 + 70 = 115 + [ ? ] _____
19. 113 + 73 = 120 + [ ? ] _____
20. 118 + 76 = 125 + [ ? ] _____

# Section 3: Complements & Make-10 Bridging

Complements to 10, 100, and 1,000, bridging across tens, and compensation. • Study the Champion Way & complete W1–W6 before starting Tests 3A1–3D3.

### 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 3: COMPLEMENTS & MAKE-10 BRIDGING

| | |
|---|---|
| RULE 1 | **Rule 1 — All-from-9 and the Last-from-10 (Instant Complements to 100 & 1,000): To find 1000 - 384 from left to right, subtract hundreds and tens from 9 (9-3=6, 9-8=1) and ones from 10 (10-4=6) → 616!** |
| RULE 2 | Rule 2 — Fast Compensation for +9, +19, +99, +199: Add the rounded ten/hundred first, then subtract the overshoot (e.g., 498 + 237 = 500 + 235 = 735). |
| RULE 3 | Rule 3 — Decimal Complements to 10 and 100: Treat 10 - 6.75 as 10.00 - 6.75: whole part is 9 - 6 = 3, decimal part is 100 - 75 = .25 → 3.25. |

### 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Left-to-Right 1,000 Complement: Evaluate 1000 - 742** — Stack Keys: `1` `0` `0` `0` `ENTER` `7` `4` `2` `-`

| SLOW / AMATEUR WAY (Avoid): | CHAMPION SPEED WAY (Use This!): |
|---|---|
| Slow Way: Write 1000 on paper and borrow across three columns from right to left. | **Champion Way: Apply 9-9-10 from left to right: (9-7=2), (9-4=5), (10-2=8) → write 258 instantly!** |

**Archetype 2 — Near-Hundred Compensation: Evaluate 498 + 237** — Stack Keys: `4` `9` `8` `ENTER` `2` `3` `7` `+`

| SLOW / AMATEUR WAY (Avoid): | CHAMPION SPEED WAY (Use This!): |
|---|---|
| Slow Way: Add column-by-column with two carries (8+7=15, 9+3+1=13, 4+2+1=7). | **Champion Way: Borrow 2 from 237 to turn 498 into 500: 500 + 235 = 735!** |

**Archetype 3 — Multi-Term Near-Hundred Sum: Evaluate 98 + 97 + 96** — Stack Keys: `3` `0` `0` `ENTER` `9` `-`

| SLOW / AMATEUR WAY (Avoid): | CHAMPION SPEED WAY (Use This!): |
|---|---|
| Slow Way: Add 98 + 97 = 195, then 195 + 96 = 291. | **Champion Way: Count down from 3 × 100 = 300 by the deficits (2 + 3 + 4 = 9): 300 - 9 = 291!** |

### 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

| | |
|---|---|
| **W1.** 100 − 64 = _____ | **W4.** 10 − 3.65 = _____ |
| Shortcut Cue: 9-from-9 (9-6=3), 10-from-10 (10-4=6) | Shortcut Cue: 9 - 3 = 6 and 100 - 65 = .35 |
| **W2.** 57 + 19 = _____ | **W5.** 299 + 156 = _____ |
| Shortcut Cue: Add 20 (77), subtract 1 | Shortcut Cue: 300 + 155 |
| **W3.** 1000 − 384 = _____ | **W6.** 99 + 98 + 97 = _____ |
| Shortcut Cue: 9-3=6, 9-8=1, 10-4=6 | Shortcut Cue: 300 - (1 + 2 + 3) |

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 3: Complements & Make-10 Bridging

Review Worked Examples 1 & 2 below, then solve Test 3A1 (Level 1: Foundation (Sprint 1)) & Test 3A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** 100 - 64 = 36 (9-from-9 (9-6=3), 10-from-10 (10-4=6))

**EXAMPLE 2** 57 + 19 = 76 (Add 20 (77), subtract 1)

**TEST 3A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 3 min

Complements to 10 & Teen Place Value

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $10 - 7 =$ | |
| 2. | $10 - 4 =$ | |
| 3. | $10 - 8 =$ | |
| 4. | $10 - 3 =$ | |
| 5. | $10 - 9 =$ | |
| 6. | $10 - 6 =$ | |
| 7. | $10 - 2 =$ | |
| 8. | $10 - 5 =$ | |
| 9. | $10 + 4 =$ | |
| 10. | $10 + 7 =$ | |
| 11. | $10 + 9 =$ | |
| 12. | $8 + 2 + 3 =$ | |
| 13. | $9 + 1 + 5 =$ | |
| 14. | $7 + 3 + 4 =$ | |
| 15. | $6 + 4 + 2 =$ | |
| 16. | $8 + 5 =$ | |
| 17. | $9 + 6 =$ | |
| 18. | $7 + 5 =$ | |
| 19. | $8 + 6 =$ | |
| 20. | $9 + 7 =$ | |

**TEST 3A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 3 min

Complements to 10 & Teen Place Value — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $10 - 4 + 1 =$ | |
| 2. | $10 - 5 + 1 =$ | |
| 3. | $10 - 6 + 1 =$ | |
| 4. | $10 - 7 + 1 =$ | |
| 5. | $10 - 8 + 1 =$ | |
| 6. | $10 - 9 + 1 =$ | |
| 7. | $10 - 1 + 1 =$ | |
| 8. | $10 - 2 + 1 =$ | |
| 9. | $3 + 7 + 5 =$ | |
| 10. | $4 + 6 + 7 =$ | |
| 11. | $5 + 5 + 2 =$ | |
| 12. | $6 + 4 + 4 =$ | |
| 13. | $7 + 3 + 6 =$ | |
| 14. | $8 + 2 + 8 =$ | |
| 15. | $9 + 7 + 1 =$ | |
| 16. | $7 + 8 + 1 =$ | |
| 17. | $8 + 9 + 1 =$ | |
| 18. | $9 + 4 + 1 =$ | |
| 19. | $7 + 5 + 1 =$ | |
| 20. | $8 + 6 + 1 =$ | |

# Section 3: Complements & Make-10 Bridging

Review Worked Examples 1 & 2 below, then solve Test 3A3 (Level 1: Foundation (Sprint 3)) & Test 3B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** 1000 - 384 = 616 (9-3=6, 9-8=1, 10-4=6)

**EXAMPLE 2** 10 - 3.65 = 6.35 (9 - 3 = 6 and 100 - 65 = .35)

**TEST 3A3 • LEVEL 1: FOUNDATION (SPRINT 3)** Goal: 3 min

Bridge from Level 1 to Level 2 (Bridging Across 10, 20, and 50)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $10 - 7 + 2 =$ | |
| 2. | $10 - 8 + 2 =$ | |
| 3. | $10 - 9 + 2 =$ | |
| 4. | $10 - 1 + 2 =$ | |
| 5. | $10 - 2 + 2 =$ | |
| 6. | $10 - 3 + 2 =$ | |
| 7. | $10 - 4 + 2 =$ | |
| 8. | $10 - 5 + 2 =$ | |
| 9. | $6 + 4 + 6 =$ | |
| 10. | $7 + 3 + 8 =$ | |
| 11. | $8 + 2 + 3 =$ | |
| 12. | $9 + 1 + 5 =$ | |
| 13. | $1 + 9 + 7 =$ | |
| 14. | $2 + 8 + 2 =$ | |
| 15. | $9 + 8 + 2 =$ | |
| 16. | $7 + 9 + 2 =$ | |
| 17. | $8 + 4 + 2 =$ | |
| 18. | $9 + 5 + 2 =$ | |
| 19. | $7 + 6 + 2 =$ | |
| 20. | $8 + 7 + 2 =$ | |

**TEST 3B1 • LEVEL 2: CORE FLUENCY (SPRINT 1)** Goal: 4 min

Bridging Across 10, 20, and 50

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $9 + 8 =$ | |
| 2. | $8 + 7 =$ | |
| 3. | $7 + 6 =$ | |
| 4. | $9 + 5 =$ | |
| 5. | $20 - 7 =$ | |
| 6. | $20 - 4 =$ | |
| 7. | $20 - 12 =$ | |
| 8. | $18 + 5 =$ | |
| 9. | $19 + 7 =$ | |
| 10. | $17 + 6 =$ | |
| 11. | $28 + 6 =$ | |
| 12. | $29 + 8 =$ | |
| 13. | $37 + 5 =$ | |
| 14. | $30 - 6 =$ | |
| 15. | $40 - 8 =$ | |
| 16. | $50 - 14 =$ | |
| 17. | $50 - 27 =$ | |
| 18. | $48 + 7 =$ | |
| 19. | $59 + 8 =$ | |
| 20. | $70 - 23 =$ | |

# Section 3: Complements & Make-10 Bridging

Review Worked Examples 1 & 2 below, then solve Test 3B2 (Level 2: Core Fluency (Sprint 2)) & Test 3B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Left-to-Right 1,000 Complement: Evaluate 1000 - 742 → Apply 9-9-10 from left to right: (9-7=2), (9-4=5), (10-2=8) → write 258 instantly!

**EXAMPLE 2** 299 + 156 = 455 (300 + 155)

**TEST 3B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 4 min

Bridging Across 10, 20, and 50 — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 30 − 5 = | |
| 2. | 38 + 6 = | |
| 3. | 50 − 11 = | |
| 4. | 59 + 8 = | |
| 5. | 70 − 17 = | |
| 6. | 77 + 4 = | |
| 7. | 30 − 7 = | |
| 8. | 38 + 6 = | |
| 9. | 50 − 13 = | |
| 10. | 59 + 8 = | |
| 11. | 70 − 3 = | |
| 12. | 77 + 4 = | |
| 13. | 30 − 9 = | |
| 14. | 38 + 6 = | |
| 15. | 50 − 15 = | |
| 16. | 59 + 8 = | |
| 17. | 70 − 5 = | |
| 18. | 77 + 4 = | |
| 19. | 30 − 11 = | |
| 20. | 38 + 6 = | |

**TEST 3B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 4 min

Bridge from Level 2 to Level 3 (Complements to 100 & Fast Compen)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | 40 − 7 = | |
| 2. | 48 + 7 = | |
| 3. | 60 − 13 = | |
| 4. | 69 + 9 = | |
| 5. | 80 − 3 = | |
| 6. | 87 + 5 = | |
| 7. | 40 − 9 = | |
| 8. | 48 + 7 = | |
| 9. | 60 − 15 = | |
| 10. | 69 + 9 = | |
| 11. | 80 − 5 = | |
| 12. | 87 + 5 = | |
| 13. | 40 − 11 = | |
| 14. | 48 + 7 = | |
| 15. | 60 − 17 = | |
| 16. | 69 + 9 = | |
| 17. | 80 − 7 = | |
| 18. | 87 + 5 = | |
| 19. | 40 − 13 = | |
| 20. | 48 + 7 = | |

# Section 3: Complements & Make-10 Bridging

Review Worked Examples 1 & 2 below, then solve Test 3C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 3C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** **Archetype 2 — Near-Hundred Compensation: Evaluate 498 + 237 → Borrow 2 from 237 to turn 498 into 500: 500 + 235 = 735!**

**EXAMPLE 2** **3C #6 (100 - 19) → 9-1=8, 10-9=1 → 81**

**TEST 3C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** **Goal: 5 min**

Complements to 100 & Fast Compensation (+9, +19, +29)

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | 100 − 30 = | |
| 2. | 100 − 45 = | |
| 3. | 100 − 62 = | |
| 4. | 100 − 28 = | |
| 5. | 100 − 74 = | |
| 6. | 100 − 19 = | |
| 7. | 100 − 83 = | |
| 8. | 100 − 57 = | |
| 9. | 36 + 9 = | |
| 10. | 58 + 9 = | |
| 11. | 47 + 19 = | |
| 12. | 65 + 19 = | |
| 13. | 38 + 29 = | |
| 14. | 54 + 39 = | |
| 15. | 73 − 9 = | |
| 16. | 82 − 19 = | |
| 17. | 95 − 29 = | |
| 18. | 67 + 33 + 25 = | |
| 19. | 48 + 52 + 39 = | |
| 20. | 200 − 64 = | |

**TEST 3C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** **Goal: 5 min**

Complements to 100 & Fast Compensation (+9, +19, +29) — Inside-Out Mastery

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | 100 − 16 = | |
| 2. | 100 − 25 = | |
| 3. | 100 − 34 = | |
| 4. | 100 − 43 = | |
| 5. | 100 − 52 = | |
| 6. | 100 − 61 = | |
| 7. | 100 − 70 = | |
| 8. | 100 − 79 = | |
| 9. | 59 + 29 = | |
| 10. | 63 + 9 = | |
| 11. | 67 + 19 = | |
| 12. | 71 + 29 = | |
| 13. | 75 + 9 = | |
| 14. | 79 + 19 = | |
| 15. | 83 + 29 = | |
| 16. | 95 − 9 = | |
| 17. | 97 − 19 = | |
| 18. | 99 − 29 = | |
| 19. | 101 − 9 = | |
| 20. | 103 − 19 = | |

# Section 3: Complements & Make-10 Bridging

Review Worked Examples 1 & 2 below, then solve Test 3C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 3D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Multi-Term Near-Hundred Sum: Evaluate $98 + 97 + 96$ → Count down from $3 \times 100 = 300$ by the deficits $(2 + 3 + 4 = 9)$: $300 - 9 = 291$!

**EXAMPLE 2** 3D #5 (1000 - 742) → Left-to-right 9-9-10 rule → 258

**TEST 3C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Complements to 1,000, Decimals &)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $100 - 20 =$ | |
| 2. | $100 - 29 =$ | |
| 3. | $100 - 38 =$ | |
| 4. | $100 - 47 =$ | |
| 5. | $100 - 56 =$ | |
| 6. | $100 - 65 =$ | |
| 7. | $100 - 74 =$ | |
| 8. | $100 - 83 =$ | |
| 9. | $62 + 29 =$ | |
| 10. | $66 + 9 =$ | |
| 11. | $70 + 19 =$ | |
| 12. | $74 + 29 =$ | |
| 13. | $78 + 9 =$ | |
| 14. | $82 + 19 =$ | |
| 15. | $86 + 29 =$ | |
| 16. | $98 - 9 =$ | |
| 17. | $100 - 19 =$ | |
| 18. | $102 - 29 =$ | |
| 19. | $104 - 9 =$ | |
| 20. | $106 - 19 =$ | |

**TEST 3D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 5 min

Complements to 1,000, Decimals & Near-Hundred Sums

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $1000 - 400 =$ | |
| 2. | $1000 - 650 =$ | |
| 3. | $1000 - 275 =$ | |
| 4. | $1000 - 384 =$ | |
| 5. | $1000 - 742 =$ | |
| 6. | $1000 - 519 =$ | |
| 7. | $1000 - 863 =$ | |
| 8. | $1000 - 97 =$ | |
| 9. | $10 - 3.4 =$ | |
| 10. | $10 - 6.75 =$ | |
| 11. | $100 - 42.5 =$ | |
| 12. | $99 + 47 =$ | |
| 13. | $98 + 65 =$ | |
| 14. | $199 + 84 =$ | |
| 15. | $299 + 156 =$ | |
| 16. | $498 + 237 =$ | |
| 17. | $99 + 99 + 99 =$ | |
| 18. | $98 + 97 + 96 =$ | |
| 19. | $198 + 297 + 105 =$ | |
| 20. | $5000 - 1645 =$ | |

# Section 3: Complements & Make-10 Bridging

Review Worked Examples 1 & 2 below, then solve Test 3D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 3D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 3D #16 (498 + 237) → 500 + 235 = 735

**EXAMPLE 2** 3D #18 (98 + 97 + 96) → 300 - (2 + 3 + 4) = 291

**TEST 3D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 5 min

Complements to 1,000, Decimals & Near-Hundred Sums — Contest Pacing

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. 1000 − 158 = _____
2. 1000 − 245 = _____
3. 1000 − 332 = _____
4. 1000 − 419 = _____
5. 1000 − 506 = _____
6. 1000 − 593 = _____
7. 1000 − 680 = _____
8. 1000 − 767 = _____
9. 599 + 273 = _____
10. 698 + 290 = _____
11. 299 + 307 = _____
12. 398 + 324 = _____
13. 499 + 341 = _____
14. 598 + 358 = _____
15. 30 − 1.5 = _____
16. 40 − 2 = _____
17. 50 − 2.5 = _____
18. 60 − 3 = _____
19. 70 − 3.5 = _____
20. 80 − 4 = _____

**TEST 3D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 4 min

Section 3 Cumulative Tournament Sprint

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. 1000 − 201 = _____
2. 1000 − 288 = _____
3. 1000 − 375 = _____
4. 1000 − 462 = _____
5. 1000 − 549 = _____
6. 1000 − 636 = _____
7. 1000 − 723 = _____
8. 1000 − 810 = _____
9. 699 + 286 = _____
10. 798 + 303 = _____
11. 399 + 320 = _____
12. 498 + 337 = _____
13. 599 + 354 = _____
14. 698 + 371 = _____
15. 40 − 1.75 = _____
16. 50 − 2.25 = _____
17. 60 − 2.75 = _____
18. 70 − 3.25 = _____
19. 80 − 3.75 = _____
20. 90 − 4.25 = _____

# Section 4: Doubles, Equal Groups & Squares

Doubles, near-doubles, equal groups, halving, and mental squaring. • Study the Champion Way & complete W1–W6 before starting Tests 4A1–4D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 4: DOUBLES, EQUAL GROUPS & SQUARES

| | |
|---|---|
| RULE 1 | **Rule 1 — Stack Doubling & Squaring Reflex: Pressing [ENTER] duplicates Line X into Line Y, so [N] [ENTER] [+] doubles any number and [N] [ENTER] [×] (or [$x^2$]) squares it in 2–3 keys!** |
| RULE 2 | Rule 2 — Squaring Numbers Ending in 5 ($n5^2$): Multiply the tens digit n by $(n + 1)$ and append 25! Example: $35^2 \rightarrow 3 \times 4 = 12$, append $25 \rightarrow 1225$. |
| RULE 3 | Rule 3 — Near-Square Product $(a - 1)(a + 1) = a^2 - 1$: Whenever two factors straddle a round number (like $19 \times 21$ or $29 \times 31$), square the middle number and subtract 1 ($20^2 - 1 = 399$; $30^2 - 1 = 899$). |

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

### Archetype 1 — Doubling & Halving Chain: Evaluate $4 \times 35$ and $130 \div 2$

Stack Keys: `3` `5` `ENTER` `+` `ENTER` `+`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Multiply $35 \times 4$ with paper carries; do long division for $130 \div 2$.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Double twice for $\times 4$ ($35 \rightarrow 70 \rightarrow 140$); halve $100 + 30 \rightarrow 50 + 15 = 65$!**

### Archetype 2 — Ends-in-5 Mental Square: Evaluate $25^2$ and $35^2$

Stack Keys: `3` `5` `ENTER` `×`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Multiply $35 \times 35$ by hand (4 partial products).

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: For $25^2$, $2 \times 3 = 6 \rightarrow 625$. For $35^2$, $3 \times 4 = 12 \rightarrow 1225$!**

### Archetype 3 — Difference of Squares Shortcut: Evaluate $25^2 - 15^2$

Stack Keys: `2` `5` `ENTER` `1` `5` `+` `1` `0` `×`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 625 and 225 separately, then subtract.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Use $(a + b)(a - b) = (25 + 15)(25 - 15) = 40 \times 10 = 400$ in 1 second!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** $2 \times 46 =$ _____

Shortcut Cue: Double 40 (80) + double 6 (12)

**W2.** $4 \times 18 =$ _____

Shortcut Cue: Double 18 → 36, double 36 → 72

**W3.** $15^2 =$ _____

Shortcut Cue: Ends-in-5 rule: $1 \times 2 = 2$, append 25

**W4.** $25^2 =$ _____

Shortcut Cue: Ends-in-5 rule: $2 \times 3 = 6$, append 25

**W5.** $19 \times 21 =$ _____

Shortcut Cue: $20^2 - 1 = 400 - 1$

**W6.** $15^2 - 5^2 =$ _____

Shortcut Cue: $(15 + 5)(15 - 5) = 20 \times 10$

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 4: Doubles, Equal Groups & Squares

Review Worked Examples 1 & 2 below, then solve Test 4A1 (Level 1: Foundation (Sprint 1)) & Test 4A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $2 \times 46 = 92$ (Double 40 (80) + double 6 (12))

**EXAMPLE 2** $4 \times 18 = 72$ (Double 18 → 36, double 36 → 72)

**TEST 4A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 3 min

Doubles and Near-Doubles Within 25

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $3 + 3 =$ | _____ |
| 2. | $4 + 4 =$ | _____ |
| 3. | $5 + 5 =$ | _____ |
| 4. | $6 + 6 =$ | _____ |
| 5. | $7 + 7 =$ | _____ |
| 6. | $8 + 8 =$ | _____ |
| 7. | $9 + 9 =$ | _____ |
| 8. | $10 + 10 =$ | _____ |
| 9. | $4 + 5 =$ | _____ |
| 10. | $5 + 6 =$ | _____ |
| 11. | $6 + 7 =$ | _____ |
| 12. | $7 + 8 =$ | _____ |
| 13. | $8 + 9 =$ | _____ |
| 14. | $9 + 10 =$ | _____ |
| 15. | $11 + 11 =$ | _____ |
| 16. | $12 + 12 =$ | _____ |
| 17. | $10 + 11 =$ | _____ |
| 18. | $11 + 12 =$ | _____ |
| 19. | $2 \times 6 =$ | _____ |
| 20. | $2 \times 9 =$ | _____ |

**TEST 4A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 3 min

Doubles and Near-Doubles Within 25 — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $5 + 5 =$ | _____ |
| 2. | $6 + 7 =$ | _____ |
| 3. | $2 \times 7 + 1 =$ | _____ |
| 4. | $8 + 8 =$ | _____ |
| 5. | $9 + 10 =$ | _____ |
| 6. | $2 \times 10 + 1 =$ | _____ |
| 7. | $11 + 11 =$ | _____ |
| 8. | $12 + 13 =$ | _____ |
| 9. | $2 \times 13 + 1 =$ | _____ |
| 10. | $14 + 14 =$ | _____ |
| 11. | $15 + 16 =$ | _____ |
| 12. | $2 \times 16 + 1 =$ | _____ |
| 13. | $17 + 17 =$ | _____ |
| 14. | $18 + 19 =$ | _____ |
| 15. | $2 \times 19 + 1 =$ | _____ |
| 16. | $20 + 20 =$ | _____ |
| 17. | $21 + 22 =$ | _____ |
| 18. | $2 \times 22 + 1 =$ | _____ |
| 19. | $23 + 23 =$ | _____ |
| 20. | $24 + 25 =$ | _____ |

# Section 4: Doubles, Equal Groups & Squares

Review Worked Examples 1 & 2 below, then solve Test 4A3 (Level 1: Foundation (Sprint 3)) & Test 4B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $15^2 = 225$ **(Ends-in-5 rule: $1 \times 2 = 2$, append 25)**

**EXAMPLE 2** $25^2 = 625$ **(Ends-in-5 rule: $2 \times 3 = 6$, append 25)**

**TEST 4A3 • LEVEL 1: FOUNDATION (SPRINT 3)** **Goal: 3 min**

Bridge from Level 1 to Level 2 (Two-Digit Doubles, Halves & Equa)

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $7 + 7 =$ | |
| 2. | $8 + 9 =$ | |
| 3. | $2 \times 9 + 2 =$ | |
| 4. | $10 + 10 =$ | |
| 5. | $11 + 12 =$ | |
| 6. | $2 \times 12 + 2 =$ | |
| 7. | $13 + 13 =$ | |
| 8. | $14 + 15 =$ | |
| 9. | $2 \times 15 + 2 =$ | |
| 10. | $16 + 16 =$ | |
| 11. | $17 + 18 =$ | |
| 12. | $2 \times 18 + 2 =$ | |
| 13. | $19 + 19 =$ | |
| 14. | $20 + 21 =$ | |
| 15. | $2 \times 21 + 2 =$ | |
| 16. | $22 + 22 =$ | |
| 17. | $23 + 24 =$ | |
| 18. | $2 \times 24 + 2 =$ | |
| 19. | $25 + 25 =$ | |
| 20. | $26 + 27 =$ | |

**TEST 4B1 • LEVEL 2: CORE FLUENCY (SPRINT 1)** **Goal: 4 min**

Two-Digit Doubles, Halves & Equal Groups

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $13 + 13 =$ | |
| 2. | $14 + 14 =$ | |
| 3. | $15 + 15 =$ | |
| 4. | $16 + 16 =$ | |
| 5. | $18 + 18 =$ | |
| 6. | $25 + 25 =$ | |
| 7. | $35 + 35 =$ | |
| 8. | $45 + 45 =$ | |
| 9. | $15 + 16 =$ | |
| 10. | $24 + 25 =$ | |
| 11. | $3 \times 4 =$ | |
| 12. | $4 \times 5 =$ | |
| 13. | $3 \times 6 =$ | |
| 14. | $5 \times 6 =$ | |
| 15. | $4 \times 8 =$ | |
| 16. | $6 \times 7 =$ | |
| 17. | $24 \div 2 =$ | |
| 18. | $36 \div 2 =$ | |
| 19. | $50 \div 2 =$ | |
| 20. | $48 \div 2 =$ | |

# Section 4: Doubles, Equal Groups & Squares

Review Worked Examples 1 & 2 below, then solve Test 4B2 (Level 2: Core Fluency (Sprint 2)) & Test 4B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Doubling & Halving Chain: Evaluate $4 \times 35$ and $130 \div 2$ → Double twice for ×4 ($35 \to 70 \to 140$); halve $100 + 30 \to 50 + 15 = 65$!

**EXAMPLE 2** $19 \times 21 = 399$ $(20^2 - 1 = 400 - 1)$

**TEST 4B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 4 min

Two-Digit Doubles, Halves & Equal Groups — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $15 + 15 =$ | _____ |
| 2. | $17 + 17 =$ | _____ |
| 3. | $19 + 19 =$ | _____ |
| 4. | $21 + 21 =$ | _____ |
| 5. | $23 + 23 =$ | _____ |
| 6. | $25 + 25 =$ | _____ |
| 7. | $27 + 27 =$ | _____ |
| 8. | $5 \times 6 + 1 =$ | _____ |
| 9. | $6 \times 7 + 1 =$ | _____ |
| 10. | $7 \times 8 + 1 =$ | _____ |
| 11. | $3 \times 9 + 1 =$ | _____ |
| 12. | $4 \times 4 + 1 =$ | _____ |
| 13. | $5 \times 5 + 1 =$ | _____ |
| 14. | $6 \times 6 + 1 =$ | _____ |
| 15. | $30 \div 2 =$ | _____ |
| 16. | $36 \div 2 =$ | _____ |
| 17. | $42 \div 2 =$ | _____ |
| 18. | $48 \div 2 =$ | _____ |
| 19. | $54 \div 2 =$ | _____ |
| 20. | $60 \div 2 =$ | _____ |

**TEST 4B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 4 min

Bridge from Level 2 to Level 3 (Doubling Chains (×2, ×4, ×8) & T)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $18 + 18 =$ | _____ |
| 2. | $20 + 20 =$ | _____ |
| 3. | $22 + 22 =$ | _____ |
| 4. | $24 + 24 =$ | _____ |
| 5. | $26 + 26 =$ | _____ |
| 6. | $28 + 28 =$ | _____ |
| 7. | $30 + 30 =$ | _____ |
| 8. | $5 \times 7 + 2 =$ | _____ |
| 9. | $6 \times 8 + 2 =$ | _____ |
| 10. | $7 \times 9 + 2 =$ | _____ |
| 11. | $3 \times 4 + 2 =$ | _____ |
| 12. | $4 \times 5 + 2 =$ | _____ |
| 13. | $5 \times 6 + 2 =$ | _____ |
| 14. | $6 \times 7 + 2 =$ | _____ |
| 15. | $34 \div 2 =$ | _____ |
| 16. | $40 \div 2 =$ | _____ |
| 17. | $46 \div 2 =$ | _____ |
| 18. | $52 \div 2 =$ | _____ |
| 19. | $58 \div 2 =$ | _____ |
| 20. | $64 \div 2 =$ | _____ |

# Section 4: Doubles, Equal Groups & Squares

Review Worked Examples 1 & 2 below, then solve Test 4C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 4C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Ends-in-5 Mental Square: Evaluate $25^2$ and $35^2$ → For $25^2$, $2 \times 3 = 6$ → 625. For $35^2$, $3 \times 4 = 12$ → 1225!

**EXAMPLE 2** 4C #10 ($4 \times 35$) → Double 35 → 70, double 70 → 140

## TEST 4C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1) Goal: 5 min

Doubling Chains (×2, ×4, ×8) & Two-Digit Multiplication

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $2 \times 27 =$ | |
| 2. | $2 \times 38 =$ | |
| 3. | $2 \times 46 =$ | |
| 4. | $2 \times 65 =$ | |
| 5. | $2 \times 78 =$ | |
| 6. | $4 \times 15 =$ | |
| 7. | $4 \times 18 =$ | |
| 8. | $4 \times 23 =$ | |
| 9. | $4 \times 25 =$ | |
| 10. | $4 \times 35 =$ | |
| 11. | $8 \times 12 =$ | |
| 12. | $8 \times 15 =$ | |
| 13. | $8 \times 25 =$ | |
| 14. | $74 \div 2 =$ | |
| 15. | $96 \div 2 =$ | |
| 16. | $130 \div 2 =$ | |
| 17. | $84 \div 4 =$ | |
| 18. | $120 \div 4 =$ | |
| 19. | $2 \times 34 + 10 =$ | |
| 20. | $4 \times 16 - 14 =$ | |

## TEST 4C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2) Goal: 5 min

Doubling Chains (×2, ×4, ×8) & Two-Digit Multiplication — Inside-Out Mastery

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $2 \times 27 =$ | |
| 2. | $2 \times 34 =$ | |
| 3. | $2 \times 41 =$ | |
| 4. | $2 \times 48 =$ | |
| 5. | $2 \times 55 =$ | |
| 6. | $2 \times 62 =$ | |
| 7. | $2 \times 69 =$ | |
| 8. | $8 \times 17 =$ | |
| 9. | $4 \times 22 =$ | |
| 10. | $8 \times 27 =$ | |
| 11. | $4 \times 32 =$ | |
| 12. | $8 \times 37 =$ | |
| 13. | $4 \times 42 =$ | |
| 14. | $8 \times 47 =$ | |
| 15. | $80 \div 4 =$ | |
| 16. | $52 \div 2 =$ | |
| 17. | $128 \div 4 =$ | |
| 18. | $76 \div 2 =$ | |
| 19. | $176 \div 4 =$ | |
| 20. | $100 \div 2 =$ | |

# Section 4: Doubles, Equal Groups & Squares

Review Worked Examples 1 & 2 below, then solve Test 4C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 4D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Difference of Squares Shortcut: Evaluate $25^2 - 15^2 \rightarrow$ Use $(a + b)(a - b) = (25 + 15)(25 - 15) = 40 \times 10 = 400$ in 1 second!

**EXAMPLE 2** 4D #10 ($35^2$) $\rightarrow 3 \times 4 = 12$, append 25 $\rightarrow$ 1225

**TEST 4C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Mental Squares, Near-Squares (a-)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $2 \times 31 =$ _____
2. $2 \times 38 =$ _____
3. $2 \times 45 =$ _____
4. $2 \times 52 =$ _____
5. $2 \times 59 =$ _____
6. $2 \times 66 =$ _____
7. $2 \times 73 =$ _____
8. $8 \times 20 =$ _____
9. $4 \times 25 =$ _____
10. $8 \times 30 =$ _____
11. $4 \times 35 =$ _____
12. $8 \times 40 =$ _____
13. $4 \times 45 =$ _____
14. $8 \times 50 =$ _____
15. $92 \div 4 =$ _____
16. $58 \div 2 =$ _____
17. $140 \div 4 =$ _____
18. $82 \div 2 =$ _____
19. $188 \div 4 =$ _____
20. $106 \div 2 =$ _____

**TEST 4D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 5 min

Mental Squares, Near-Squares (a-1)(a+1) & Powers

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $11^2 =$ _____
2. $12^2 =$ _____
3. $13^2 =$ _____
4. $14^2 =$ _____
5. $15^2 =$ _____
6. $16^2 =$ _____
7. $20^2 =$ _____
8. $25^2 =$ _____
9. $30^2 =$ _____
10. $35^2 =$ _____
11. $9 \times 11 =$ _____
12. $11 \times 13 =$ _____
13. $14 \times 16 =$ _____
14. $19 \times 21 =$ _____
15. $24 \times 26 =$ _____
16. $29 \times 31 =$ _____
17. $2 \times 15^2 =$ _____
18. $10^2 + 12^2 =$ _____
19. $15^2 - 5^2 =$ _____
20. $25^2 - 15^2 =$ _____

# Section 4: Doubles, Equal Groups & Squares

Review Worked Examples 1 & 2 below, then solve Test 4D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 4D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 4D #15 $(24 \times 26) \rightarrow 25^2 - 1 = 625 - 1 = 624$

**EXAMPLE 2** 4D #20 $(25^2 - 15^2) \rightarrow (25 + 15)(25 - 15) = 40 \times 10 = 400$

**TEST 4D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 5 min

Mental Squares, Near-Squares (a-1)(a+1) & Powers — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $12^2 + 1 =$ | |
| 2. | $14^2 + 1 =$ | |
| 3. | $16^2 + 1 =$ | |
| 4. | $18^2 + 1 =$ | |
| 5. | $20^2 + 1 =$ | |
| 6. | $22^2 + 1 =$ | |
| 7. | $24^2 + 1 =$ | |
| 8. | $25^2 =$ | |
| 9. | $35^2 =$ | |
| 10. | $45^2 =$ | |
| 11. | $55^2 =$ | |
| 12. | $65^2 =$ | |
| 13. | $75^2 =$ | |
| 14. | $12 \times 16 =$ | |
| 15. | $14 \times 20 =$ | |
| 16. | $19 \times 21 =$ | |
| 17. | $21 \times 25 =$ | |
| 18. | $23 \times 29 =$ | |
| 19. | $28 \times 30 =$ | |
| 20. | $30 \times 34 =$ | |

**TEST 4D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 4 min

Section 4 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $13^2 + 2 =$ | |
| 2. | $15^2 + 2 =$ | |
| 3. | $17^2 + 2 =$ | |
| 4. | $19^2 + 2 =$ | |
| 5. | $21^2 + 2 =$ | |
| 6. | $23^2 + 2 =$ | |
| 7. | $25^2 + 2 =$ | |
| 8. | $35^2 =$ | |
| 9. | $45^2 =$ | |
| 10. | $55^2 =$ | |
| 11. | $65^2 =$ | |
| 12. | $75^2 =$ | |
| 13. | $85^2 =$ | |
| 14. | $14 \times 18 =$ | |
| 15. | $16 \times 22 =$ | |
| 16. | $21 \times 23 =$ | |
| 17. | $23 \times 27 =$ | |
| 18. | $25 \times 31 =$ | |
| 19. | $30 \times 32 =$ | |
| 20. | $32 \times 36 =$ | |

# Cumulative Review C1: Number Sense (Sections 1–4)

Integrated review across step counting, number bonds, complements, and doubles. • Study the Champion Way & complete W1–W6 before starting Tests C1A1–C1D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — CUMULATIVE REVIEW C1: NUMBER SENSE (SECTIONS 1–4)

| | |
|---|---|
| RULE 1 | **Rule 1 — Scan Before Computing: Before pressing a key or writing a digit, check if the problem is a 10/100 complement, a Gauss series, an ends-in-5 square, or an (a - 1)(a + 1) near-square.** |
| RULE 2 | Rule 2 — Never Write Intermediate Steps on Single-Register Arithmetic: Let Register Y hold your running total while you enter the next step on Register X. |
| RULE 3 | Rule 3 — Use Factor Splitting for Mental Multiplication: Treat ×4 as double-double, ×8 as double-double-double, and ÷4 as half-half. |

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Combined Series & Complement: Evaluate 99 + 98 + 97 + 96** Stack Keys: [4] [0] [0] [ENTER] [1] [0] [-]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Add four 2-digit numbers one by one (99+98=197, +97=294, +96=390).

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: 4 × 100 minus the deficits (1 + 2 + 3 + 4 = 10): 400 - 10 = 390!**

**Archetype 2 — Difference of Squares Mental Sprint: Evaluate $18^2 - 8^2$** Stack Keys: [1] [8] [ENTER] [8] [+] [1] [0] [×]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Calculate $18^2 = 324$ and $8^2 = 64$, then subtract.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: (18 + 8) × (18 - 8) = 26 × 10 = 260 immediately!**

**Archetype 3 — Multi-Step Stack Chain: Evaluate 2 × $20^2$ - 150** Stack Keys: [2] [0] [ENTER] [×] [2] [×] [1] [5] [0] [-]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute $20^2$, write 400, multiply by 2, write 800, subtract 150.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Mental 2 × 400 - 150 = 650, or key [20] [ENTER] [×] [2] [×] [150] [-]!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** 1000 − 465 = _____

Shortcut Cue: Left-to-right 9-9-10: 5, 3, 5

**W2.** 398 + 165 = _____

Shortcut Cue: 400 + 163

**W3.** 29 × 31 = _____

Shortcut Cue: $30^2 - 1 = 900 - 1$

**W4.** 420 ÷ 4 = _____

Shortcut Cue: Halve twice: 420 → 210 → 105

**W5.** 1 + 3 + 5 + 7 + 9 + 11 + 13 = _____

Shortcut Cue: 7 odd terms → $7^2$

**W6.** 45 + 55 + 65 + 75 = _____

Shortcut Cue: Pair (45+75) + (55+65) = 120 + 120

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Cumulative Review C1: Number Sense (Sections 1–4)

Review Worked Examples 1 & 2 below, then solve Test C1A1 (Level 1: Foundation (Sprint 1)) & Test C1A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** 1000 - 465 = 535 (Left-to-right 9-9-10: 5, 3, 5)

**EXAMPLE 2** 398 + 165 = 563 (400 + 163)

**TEST C1A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 4 min

Core Addition, Subtraction, Complements to 10 & Doubles

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $6 + 1 + 1 =$ | |
| 2. | $12 - 2 =$ | |
| 3. | $5 + 4 =$ | |
| 4. | $11 - 6 =$ | |
| 5. | $10 - 8 =$ | |
| 6. | $10 - 3 =$ | |
| 7. | $10 + 6 =$ | |
| 8. | $7 + 7 =$ | |
| 9. | $8 + 8 =$ | |
| 10. | $6 + 7 =$ | |
| 11. | $8 + 5 =$ | |
| 12. | $9 + 6 =$ | |
| 13. | $14 - 7 =$ | |
| 14. | $16 - 8 =$ | |
| 15. | $5 + 5 + 5 =$ | |
| 16. | $9 + 9 =$ | |
| 17. | $7 + 8 =$ | |
| 18. | $15 - 6 =$ | |
| 19. | $10 + 10 =$ | |
| 20. | $20 - 5 =$ | |

**TEST C1A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 4 min

Core Addition, Subtraction, Complements to 10 & Doubles — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(8 + 3) \times 3 =$ | |
| 2. | $9 \times 4 + 4 =$ | |
| 3. | $40 \div 5 + 10 =$ | |
| 4. | $(11 + 6) \times 2 =$ | |
| 5. | $12 \times 7 + 3 =$ | |
| 6. | $24 \div 4 + 13 =$ | |
| 7. | $(14 + 4) \times 5 =$ | |
| 8. | $15 \times 5 + 2 =$ | |
| 9. | $27 \div 3 + 16 =$ | |
| 10. | $(17 + 7) \times 4 =$ | |
| 11. | $18 \times 3 + 5 =$ | |
| 12. | $14 \div 2 + 19 =$ | |
| 13. | $(20 + 5) \times 3 =$ | |
| 14. | $21 \times 6 + 4 =$ | |
| 15. | $50 \div 5 + 22 =$ | |
| 16. | $(23 + 3) \times 2 =$ | |
| 17. | $24 \times 4 + 3 =$ | |
| 18. | $32 \div 4 + 25 =$ | |
| 19. | $(26 + 6) \times 5 =$ | |
| 20. | $27 \times 7 + 2 =$ | |

# Cumulative Review C1: Number Sense (Sections 1–4)

Review Worked Examples 1 & 2 below, then solve Test C1A3 (Level 1: Foundation (Sprint 3)) & Test C1B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $29 \times 31 = 899$ $(30^2 - 1 = 900 - 1)$

**EXAMPLE 2** $420 \div 4 = 105$ (Halve twice: $420 \to 210 \to 105$)

**TEST C1A3 • LEVEL 1: FOUNDATION (SPRINT 3)** Goal: 4 min

Bridge from Level 1 to Level 2 (Two-Digit Fluency, Missing Adden)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(10 + 3) \times 4 =$ | |
| 2. | $11 \times 4 + 5 =$ | |
| 3. | $18 \div 2 + 12 =$ | |
| 4. | $(13 + 6) \times 3 =$ | |
| 5. | $14 \times 7 + 4 =$ | |
| 6. | $35 \div 5 + 15 =$ | |
| 7. | $(16 + 4) \times 2 =$ | |
| 8. | $17 \times 5 + 3 =$ | |
| 9. | $40 \div 4 + 18 =$ | |
| 10. | $(19 + 7) \times 5 =$ | |
| 11. | $20 \times 3 + 2 =$ | |
| 12. | $24 \div 3 + 21 =$ | |
| 13. | $(22 + 5) \times 4 =$ | |
| 14. | $23 \times 6 + 5 =$ | |
| 15. | $22 \div 2 + 24 =$ | |
| 16. | $(25 + 3) \times 3 =$ | |
| 17. | $26 \times 4 + 4 =$ | |
| 18. | $45 \div 5 + 27 =$ | |
| 19. | $(28 + 6) \times 2 =$ | |
| 20. | $29 \times 7 + 3 =$ | |

**TEST C1B1 • LEVEL 2: CORE FLUENCY (SPRINT 1)** Goal: 4 min

Two-Digit Fluency, Missing Addends & Equal Groups

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $18 + 2 + 7 =$ | |
| 2. | $35 - 5 - 10 =$ | |
| 3. | $8 + [\ ?\ ] = 17$ | |
| 4. | $14 + [\ ?\ ] = 22$ | |
| 5. | $30 - 14 =$ | |
| 6. | $50 - 18 =$ | |
| 7. | $16 + 16 =$ | |
| 8. | $24 + 24 =$ | |
| 9. | $17 + 18 =$ | |
| 10. | $3 \times 7 =$ | |
| 11. | $4 \times 6 =$ | |
| 12. | $5 \times 8 =$ | |
| 13. | $32 \div 2 =$ | |
| 14. | $46 \div 2 =$ | |
| 15. | $29 + 7 =$ | |
| 16. | $48 + 9 =$ | |
| 17. | $60 - 25 =$ | |
| 18. | $25 + 25 + 15 =$ | |
| 19. | $4 \times 9 =$ | |
| 20. | $100 - 45 =$ | |

# Cumulative Review C1: Number Sense (Sections 1–4)

Review Worked Examples 1 & 2 below, then solve Test C1B2 (Level 2: Core Fluency (Sprint 2)) & Test C1B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Combined Series & Complement: Evaluate $99 + 98 + 97 + 96 \rightarrow 4 \times 100$ minus the deficits ($1 + 2 + 3 + 4 = 10$): 400 - 10 = 390!

**EXAMPLE 2** $1 + 3 + 5 + 7 + 9 + 11 + 13 = 49$ (7 odd terms → $7^2$)

**TEST C1B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 4 min

Two-Digit Fluency, Missing Addends & Equal Groups — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(15 - 4) \times 4 =$ | |
| 2. | $42 - 5 \times 5 =$ | |
| 3. | $(19 - 6) \times 6 =$ | |
| 4. | $70 - 7 \times 7 =$ | |
| 5. | $(23 - 8) \times 3 =$ | |
| 6. | $61 - 9 \times 4 =$ | |
| 7. | $(27 - 4) \times 5 =$ | |
| 8. | $59 - 5 \times 6 =$ | |
| 9. | $(31 - 6) \times 7 =$ | |
| 10. | $54 - 7 \times 3 =$ | |
| 11. | $(35 - 8) \times 4 =$ | |
| 12. | $82 - 9 \times 5 =$ | |
| 13. | $(39 - 4) \times 6 =$ | |
| 14. | $76 - 5 \times 7 =$ | |
| 15. | $(43 - 6) \times 3 =$ | |
| 16. | $73 - 7 \times 4 =$ | |
| 17. | $(47 - 8) \times 5 =$ | |
| 18. | $103 - 9 \times 6 =$ | |
| 19. | $(51 - 4) \times 7 =$ | |
| 20. | $68 - 5 \times 3 =$ | |

**TEST C1B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 4 min

Bridge from Level 2 to Level 3 (Complements to 100, Doubling Cha)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(18 - 4) \times 5 =$ | |
| 2. | $50 - 5 \times 6 =$ | |
| 3. | $(22 - 6) \times 7 =$ | |
| 4. | $45 - 7 \times 3 =$ | |
| 5. | $(26 - 8) \times 4 =$ | |
| 6. | $73 - 9 \times 5 =$ | |
| 7. | $(30 - 4) \times 6 =$ | |
| 8. | $67 - 5 \times 7 =$ | |
| 9. | $(34 - 6) \times 3 =$ | |
| 10. | $64 - 7 \times 4 =$ | |
| 11. | $(38 - 8) \times 5 =$ | |
| 12. | $94 - 9 \times 6 =$ | |
| 13. | $(42 - 4) \times 7 =$ | |
| 14. | $59 - 5 \times 3 =$ | |
| 15. | $(46 - 6) \times 4 =$ | |
| 16. | $83 - 7 \times 5 =$ | |
| 17. | $(50 - 8) \times 6 =$ | |
| 18. | $115 - 9 \times 7 =$ | |
| 19. | $(54 - 4) \times 3 =$ | |
| 20. | $76 - 5 \times 4 =$ | |

# Cumulative Review C1: Number Sense (Sections 1–4)

Review Worked Examples 1 & 2 below, then solve Test C1C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test C1C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Difference of Squares Mental Sprint: Evaluate $18^2 - 8^2 \rightarrow (18 + 8) \times (18 - 8) = 26 \times 10 = 260$ immediately!

**EXAMPLE 2** C1C #16 $(15^2) \rightarrow 1 \times 2 = 2$, append 25 $\rightarrow$ 225

**TEST C1C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Complements to 100, Doubling Chains & Multi-Step Bonds

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $100 - 37 =$ | _____ |
| 2. | $100 - 68 =$ | _____ |
| 3. | $46 + 38 =$ | _____ |
| 4. | $83 - 47 =$ | _____ |
| 5. | $57 + 19 =$ | _____ |
| 6. | $84 - 29 =$ | _____ |
| 7. | $2 \times 48 =$ | _____ |
| 8. | $4 \times 19 =$ | _____ |
| 9. | $8 \times 15 =$ | _____ |
| 10. | $110 \div 2 =$ | _____ |
| 11. | $92 \div 4 =$ | _____ |
| 12. | $64 + [\ ?\ ] = 120$ | _____ |
| 13. | $150 - 85 =$ | _____ |
| 14. | $35 + 45 + 20 =$ | _____ |
| 15. | $12^2 =$ | _____ |
| 16. | $15^2 =$ | _____ |
| 17. | $2 \times 36 + 18 =$ | _____ |
| 18. | $100 - 24 - 36 =$ | _____ |
| 19. | $250 - 125 + 40 =$ | _____ |
| 20. | $4 \times 25 - 33 =$ | _____ |

**TEST C1C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Complements to 100, Doubling Chains & Multi-Step Bonds — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(8 + 2) \times (9 - 3) =$ | _____ |
| 2. | $(9 \times 3) + (10 \times 4) =$ | _____ |
| 3. | $(10 + 4) \times (11 - 5) =$ | _____ |
| 4. | $(11 \times 5) + (12 \times 6) =$ | _____ |
| 5. | $(12 + 6) \times (13 - 3) =$ | _____ |
| 6. | $(13 \times 2) + (14 \times 4) =$ | _____ |
| 7. | $(14 + 3) \times (15 - 5) =$ | _____ |
| 8. | $(15 \times 4) + (9 \times 6) =$ | _____ |
| 9. | $(16 + 5) \times (10 - 3) =$ | _____ |
| 10. | $(17 \times 6) + (11 \times 4) =$ | _____ |
| 11. | $(18 + 2) \times (12 - 5) =$ | _____ |
| 12. | $(19 \times 3) + (13 \times 6) =$ | _____ |
| 13. | $(20 + 4) \times (14 - 3) =$ | _____ |
| 14. | $(21 \times 5) + (15 \times 4) =$ | _____ |
| 15. | $(22 + 6) \times (9 - 5) =$ | _____ |
| 16. | $(23 \times 2) + (10 \times 6) =$ | _____ |
| 17. | $(24 + 3) \times (11 - 3) =$ | _____ |
| 18. | $(25 \times 4) + (12 \times 4) =$ | _____ |
| 19. | $(26 + 5) \times (13 - 5) =$ | _____ |
| 20. | $(27 \times 6) + (14 \times 6) =$ | _____ |

# Cumulative Review C1: Number Sense (Sections 1–4)

Review Worked Examples 1 & 2 below, then solve Test C1C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test C1D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Multi-Step Stack Chain: Evaluate $2 \times 20^2 - 150 \rightarrow$ Mental $2 \times 400 - 150 = 650$, or key [20] [ENTER] [×] [2] [×] [150] [-]!

**EXAMPLE 2** C1D #10 $(29 \times 31) \rightarrow 30^2 - 1 = 899$

**TEST C1C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Series Sums, Near-Squares & Ment)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(10 + 2) \times (10 - 3) =$ | _____ |
| 2. | $(11 \times 3) + (11 \times 4) =$ | _____ |
| 3. | $(12 + 4) \times (12 - 5) =$ | _____ |
| 4. | $(13 \times 5) + (13 \times 6) =$ | _____ |
| 5. | $(14 + 6) \times (14 - 3) =$ | _____ |
| 6. | $(15 \times 2) + (15 \times 4) =$ | _____ |
| 7. | $(16 + 3) \times (16 - 5) =$ | _____ |
| 8. | $(17 \times 4) + (10 \times 6) =$ | _____ |
| 9. | $(18 + 5) \times (11 - 3) =$ | _____ |
| 10. | $(19 \times 6) + (12 \times 4) =$ | _____ |
| 11. | $(20 + 2) \times (13 - 5) =$ | _____ |
| 12. | $(21 \times 3) + (14 \times 6) =$ | _____ |
| 13. | $(22 + 4) \times (15 - 3) =$ | _____ |
| 14. | $(23 \times 5) + (16 \times 4) =$ | _____ |
| 15. | $(24 + 6) \times (10 - 5) =$ | _____ |
| 16. | $(25 \times 2) + (11 \times 6) =$ | _____ |
| 17. | $(26 + 3) \times (12 - 3) =$ | _____ |
| 18. | $(27 \times 4) + (13 \times 4) =$ | _____ |
| 19. | $(28 + 5) \times (14 - 5) =$ | _____ |
| 20. | $(29 \times 6) + (15 \times 6) =$ | _____ |

**TEST C1D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 5 min

Series Sums, Near-Squares & Mental Compensation

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $1 + 2 + 3 + 4 + 5 + 6 =$ | _____ |
| 2. | $2 + 4 + 6 + 8 + 10 + 12 =$ | _____ |
| 3. | $1000 - 465 =$ | _____ |
| 4. | $1000 - 782 =$ | _____ |
| 5. | $199 + 247 =$ | _____ |
| 6. | $398 + 165 =$ | _____ |
| 7. | $14^2 =$ | _____ |
| 8. | $25^2 =$ | _____ |
| 9. | $19 \times 21 =$ | _____ |
| 10. | $29 \times 31 =$ | _____ |
| 11. | $10 - 4.35 =$ | _____ |
| 12. | $100 - 63.8 =$ | _____ |
| 13. | $99 + 98 + 97 + 96 =$ | _____ |
| 14. | $18^2 - 8^2 =$ | _____ |
| 15. | $45 + 55 + 65 + 75 =$ | _____ |
| 16. | $8 \times 35 =$ | _____ |
| 17. | $420 \div 4 =$ | _____ |
| 18. | $350 + [\ ?\ ] = 825$ | _____ |
| 19. | $2 \times 20^2 - 150 =$ | _____ |
| 20. | $1 + 3 + 5 + 7 + 9 + 11 + 13 =$ | _____ |

# Cumulative Review C1: Number Sense (Sections 1–4)

Review Worked Examples 1 & 2 below, then solve Test C1D2 (Level 4: Advanced Challenge (Sprint 2)) & Test C1D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** C1D #14 $(18^2 - 8^2) \rightarrow (18 + 8)(18 - 8) = 26 \times 10 = 260$

**EXAMPLE 2** C1D #20 $(1+3+...+13) \rightarrow$ 7 odd numbers $\rightarrow 7^2 = 49$

**TEST C1D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 5 min

Series Sums, Near-Squares & Mental Compensation — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(21 \times 4) - (10 \times 4) =$ _____
2. $(50 - 25) \div 5 =$ _____
3. $(25 \times 6) - (12 \times 6) =$ _____
4. $(98 - 49) \div 7 =$ _____
5. $(29 \times 8) - (14 \times 8) =$ _____
6. $(72 - 36) \div 4 =$ _____
7. $(33 \times 10) - (16 \times 10) =$ _____
8. $(60 - 30) \div 6 =$ _____
9. $(37 \times 5) - (18 \times 5) =$ _____
10. $(77 - 56) \div 3 =$ _____
11. $(41 \times 7) - (20 \times 7) =$ _____
12. $(90 - 45) \div 5 =$ _____
13. $(45 \times 9) - (22 \times 9) =$ _____
14. $(70 - 35) \div 7 =$ _____
15. $(49 \times 4) - (24 \times 4) =$ _____
16. $(56 - 28) \div 4 =$ _____
17. $(53 \times 6) - (26 \times 6) =$ _____
18. $(108 - 54) \div 6 =$ _____
19. $(57 \times 8) - (28 \times 8) =$ _____
20. $(55 - 40) \div 3 =$ _____

**TEST C1D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 4 min

Section C1 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(24 \times 4) - (12 \times 4) =$ _____
2. $(66 - 30) \div 6 =$ _____
3. $(28 \times 6) - (14 \times 6) =$ _____
4. $(80 - 56) \div 3 =$ _____
5. $(32 \times 8) - (16 \times 8) =$ _____
6. $(90 - 40) \div 5 =$ _____
7. $(36 \times 10) - (18 \times 10) =$ _____
8. $(78 - 36) \div 7 =$ _____
9. $(40 \times 5) - (20 \times 5) =$ _____
10. $(96 - 64) \div 4 =$ _____
11. $(44 \times 7) - (22 \times 7) =$ _____
12. $(110 - 50) \div 6 =$ _____
13. $(48 \times 9) - (24 \times 9) =$ _____
14. $(60 - 42) \div 3 =$ _____
15. $(52 \times 4) - (26 \times 4) =$ _____
16. $(72 - 32) \div 5 =$ _____
17. $(56 \times 6) - (28 \times 6) =$ _____
18. $(130 - 60) \div 7 =$ _____
19. $(60 \times 8) - (30 \times 8) =$ _____
20. $(72 - 48) \div 4 =$ _____

# Section 5: Multi-Step Operations & Grouping

Multiplication, division, inside-out parentheses, and two-branch expressions. • Study the Champion Way & complete W1–W6 before starting Tests 5A1–5D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 5: MULTI-STEP OPERATIONS & GROUPING

| | |
|---|---|
| RULE 1 | **Rule 1 — Start at the Deepest Inner Parentheses: Never type left-to-right when an inner group is on the right! Evaluate the deepest inner group first so it sits on the stack.** |
| RULE 2 | Rule 2 — Automatic Stack Lift Saves [ENTER] Keys: After any operation (+, -, ×, ÷), typing the next number automatically lifts the result to Shelf Y—never press [ENTER] after an operator! |
| RULE 3 | Rule 3 — Two-Branch Expression Pipeline $(a \pm b) \times (c \pm d)$: Compute $(a \pm b)$ first, then immediately type c [ENTER] d [±] (which lifts the first branch to Y), and press [×] or [÷]. |

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

### Archetype 1 — Right-Hand Inner Group: Evaluate 100 - (16 × 5 - 12)

Stack Keys: [1] [6] [ENTER] [5] [×] [1] [2] [-] [1] [0] [0] [⇅] [-]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Type 100 [ENTER] 16 [ENTER] 5 [×] [12] [-] [-] (extra stack depth & mental tracking).

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Compute inner 16 × 5 - 12 = 68 first, then subtract from 100: 100 - 68 = 32!**

### Archetype 2 — Two Parallel Branches: Evaluate (50 - 14) ÷ (3 + 6)

Stack Keys: [5] [0] [ENTER] [1] [4] [-] [3] [ENTER] [6] [+] [÷]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 50 - 14 = 36, write 36 on paper, compute 3 + 6 = 9, re-type 36 ÷ 9.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Key [50] [ENTER] [14] [-] (36 in X), then [3] [ENTER] [6] [+] (36 lifts to Y, 9 in X), then [÷] → 4!**

### Archetype 3 — Common Factor Distribution: Evaluate (28 × 9) - (18 × 9)

Stack Keys: [2] [8] [ENTER] [1] [8] [-] [9] [×]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Multiply 28 × 9 = 252 and 18 × 9 = 162, then subtract.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Factor out ×9 mentally: (28 - 18) × 9 = 10 × 9 = 90 in 1 second!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** $(18 - 6) \times 3 =$ _____

Shortcut Cue: [18] [ENTER] [6] [-] [3] [×] (no extra ENTER!)

**W2.** $45 \div (12 - 3) =$ _____

Shortcut Cue: Inner 12 - 3 = 9 first, then 45 ÷ 9

**W3.** $(15 \times 4) + (18 \div 2) =$ _____

Shortcut Cue: 60 + 9 (or two-branch stack)

**W4.** $((20 - 8) \times 4) \div 6 =$ _____

Shortcut Cue: 12 × 4 = 48, then ÷ 6

**W5.** $(17 \times 6) - (13 \times 6) =$ _____

Shortcut Cue: Factor out 6: (17 - 13) × 6

**W6.** $(45 \times 12) \div 18 =$ _____

Shortcut Cue: Cancel 9: (5 × 12) ÷ 2

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 5: Multi-Step Operations & Grouping

Review Worked Examples 1 & 2 below, then solve Test 5A1 (Level 1: Foundation (Sprint 1)) & Test 5A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $(18 - 6) \times 3 = 36$ ([18] [ENTER] [6] [-] [3] [×] (no extra ENTER!))

**EXAMPLE 2** $45 \div (12 - 3) = 5$ (Inner 12 - 3 = 9 first, then 45 ÷ 9)

### TEST 5A1 • LEVEL 1: FOUNDATION (SPRINT 1) Goal: 4 min

Core Multiplication, Division & Two-Step Expressions

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $6 \times 4 =$ | |
| 2. | $7 \times 5 =$ | |
| 3. | $8 \times 6 =$ | |
| 4. | $9 \times 7 =$ | |
| 5. | $24 \div 3 =$ | |
| 6. | $36 \div 4 =$ | |
| 7. | $45 \div 5 =$ | |
| 8. | $56 \div 8 =$ | |
| 9. | $(10 - 4) \times 3 =$ | |
| 10. | $(12 + 8) \div 4 =$ | |
| 11. | $(15 - 6) \times 2 =$ | |
| 12. | $(18 + 6) \div 3 =$ | |
| 13. | $5 \times 4 + 7 =$ | |
| 14. | $6 \times 5 - 8 =$ | |
| 15. | $20 - 3 \times 4 =$ | |
| 16. | $30 - 4 \times 5 =$ | |
| 17. | $14 + 18 \div 3 =$ | |
| 18. | $25 - 20 \div 4 =$ | |
| 19. | $(9 + 6) \times 2 =$ | |
| 20. | $40 \div (3 + 5) =$ | |

### TEST 5A2 • LEVEL 1: FOUNDATION (SPRINT 2) Goal: 4 min

Core Multiplication, Division & Two-Step Expressions — Fluency Builder

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $(8 + 3) \times 3 =$ | |
| 2. | $9 \times 4 + 4 =$ | |
| 3. | $40 \div 5 + 10 =$ | |
| 4. | $(11 + 6) \times 2 =$ | |
| 5. | $12 \times 7 + 3 =$ | |
| 6. | $24 \div 4 + 13 =$ | |
| 7. | $(14 + 4) \times 5 =$ | |
| 8. | $15 \times 5 + 2 =$ | |
| 9. | $27 \div 3 + 16 =$ | |
| 10. | $(17 + 7) \times 4 =$ | |
| 11. | $18 \times 3 + 5 =$ | |
| 12. | $14 \div 2 + 19 =$ | |
| 13. | $(20 + 5) \times 3 =$ | |
| 14. | $21 \times 6 + 4 =$ | |
| 15. | $50 \div 5 + 22 =$ | |
| 16. | $(23 + 3) \times 2 =$ | |
| 17. | $24 \times 4 + 3 =$ | |
| 18. | $32 \div 4 + 25 =$ | |
| 19. | $(26 + 6) \times 5 =$ | |
| 20. | $27 \times 7 + 2 =$ | |

# Section 5: Multi-Step Operations & Grouping

Review Worked Examples 1 & 2 below, then solve Test 5A3 (Level 1: Foundation (Sprint 3)) & Test 5B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $(15 \times 4) + (18 \div 2) = 69$ $(60 + 9$ (or two-branch stack))

**EXAMPLE 2** $((20 - 8) \times 4) \div 6 = 8$ $(12 \times 4 = 48$, then $\div 6)$

### TEST 5A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 4 min

Bridge from Level 1 to Level 2 (Order of Operations & Inner Pare)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(10 + 3) \times 4 =$ | _____ |
| 2. | $11 \times 4 + 5 =$ | _____ |
| 3. | $18 \div 2 + 12 =$ | _____ |
| 4. | $(13 + 6) \times 3 =$ | _____ |
| 5. | $14 \times 7 + 4 =$ | _____ |
| 6. | $35 \div 5 + 15 =$ | _____ |
| 7. | $(16 + 4) \times 2 =$ | _____ |
| 8. | $17 \times 5 + 3 =$ | _____ |
| 9. | $40 \div 4 + 18 =$ | _____ |
| 10. | $(19 + 7) \times 5 =$ | _____ |
| 11. | $20 \times 3 + 2 =$ | _____ |
| 12. | $24 \div 3 + 21 =$ | _____ |
| 13. | $(22 + 5) \times 4 =$ | _____ |
| 14. | $23 \times 6 + 5 =$ | _____ |
| 15. | $22 \div 2 + 24 =$ | _____ |
| 16. | $(25 + 3) \times 3 =$ | _____ |
| 17. | $26 \times 4 + 4 =$ | _____ |
| 18. | $45 \div 5 + 27 =$ | _____ |
| 19. | $(28 + 6) \times 2 =$ | _____ |
| 20. | $29 \times 7 + 3 =$ | _____ |

### TEST 5B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 4 min

Order of Operations & Inner Parentheses

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(18 - 6) \times 3 =$ | _____ |
| 2. | $(24 + 16) \div 8 =$ | _____ |
| 3. | $45 \div (12 - 3) =$ | _____ |
| 4. | $60 \div (7 + 5) =$ | _____ |
| 5. | $12 + 6 \times 4 =$ | _____ |
| 6. | $50 - 7 \times 6 =$ | _____ |
| 7. | $8 \times 7 - 16 =$ | _____ |
| 8. | $9 \times 8 + 18 =$ | _____ |
| 9. | $(32 - 12) \times 4 =$ | _____ |
| 10. | $(48 + 12) \div 6 =$ | _____ |
| 11. | $72 \div (4 + 5) =$ | _____ |
| 12. | $100 - 15 \times 4 =$ | _____ |
| 13. | $24 + 36 \div 6 =$ | _____ |
| 14. | $64 \div 8 + 17 =$ | _____ |
| 15. | $(14 + 7) \div 3 \times 5 =$ | _____ |
| 16. | $(25 - 10) \times 4 \div 6 =$ | _____ |
| 17. | $90 \div (15 - 6) \times 4 =$ | _____ |
| 18. | $120 - (8 \times 9) =$ | _____ |
| 19. | $(16 + 24) \div (12 - 4) =$ | _____ |
| 20. | $(15 - 3) \times (2 + 4) =$ | _____ |

# Section 5: Multi-Step Operations & Grouping

Review Worked Examples 1 & 2 below, then solve Test 5B2 (Level 2: Core Fluency (Sprint 2)) & Test 5B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Right-Hand Inner Group: Evaluate $100 - (16 \times 5 - 12) \rightarrow$ Compute inner $16 \times 5 - 12 = 68$ first, then subtract from 100: $100 - 68 = 32$!

**EXAMPLE 2** $(17 \times 6) - (13 \times 6) = 24$ (Factor out 6: $(17 - 13) \times 6$)

**TEST 5B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 4 min

Order of Operations & Inner Parentheses — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(15 - 4) \times 4 =$ | |
| 2. | $42 - 5 \times 5 =$ | |
| 3. | $(19 - 6) \times 6 =$ | |
| 4. | $70 - 7 \times 7 =$ | |
| 5. | $(23 - 8) \times 3 =$ | |
| 6. | $61 - 9 \times 4 =$ | |
| 7. | $(27 - 4) \times 5 =$ | |
| 8. | $59 - 5 \times 6 =$ | |
| 9. | $(31 - 6) \times 7 =$ | |
| 10. | $54 - 7 \times 3 =$ | |
| 11. | $(35 - 8) \times 4 =$ | |
| 12. | $82 - 9 \times 5 =$ | |
| 13. | $(39 - 4) \times 6 =$ | |
| 14. | $76 - 5 \times 7 =$ | |
| 15. | $(43 - 6) \times 3 =$ | |
| 16. | $73 - 7 \times 4 =$ | |
| 17. | $(47 - 8) \times 5 =$ | |
| 18. | $103 - 9 \times 6 =$ | |
| 19. | $(51 - 4) \times 7 =$ | |
| 20. | $68 - 5 \times 3 =$ | |

**TEST 5B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 4 min

Bridge from Level 2 to Level 3 (Two-Branch Expressions $(a \pm b) \times$)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(18 - 4) \times 5 =$ | |
| 2. | $50 - 5 \times 6 =$ | |
| 3. | $(22 - 6) \times 7 =$ | |
| 4. | $45 - 7 \times 3 =$ | |
| 5. | $(26 - 8) \times 4 =$ | |
| 6. | $73 - 9 \times 5 =$ | |
| 7. | $(30 - 4) \times 6 =$ | |
| 8. | $67 - 5 \times 7 =$ | |
| 9. | $(34 - 6) \times 3 =$ | |
| 10. | $64 - 7 \times 4 =$ | |
| 11. | $(38 - 8) \times 5 =$ | |
| 12. | $94 - 9 \times 6 =$ | |
| 13. | $(42 - 4) \times 7 =$ | |
| 14. | $59 - 5 \times 3 =$ | |
| 15. | $(46 - 6) \times 4 =$ | |
| 16. | $83 - 7 \times 5 =$ | |
| 17. | $(50 - 8) \times 6 =$ | |
| 18. | $115 - 9 \times 7 =$ | |
| 19. | $(54 - 4) \times 3 =$ | |
| 20. | $76 - 5 \times 4 =$ | |

# Section 5: Multi-Step Operations & Grouping

Review Worked Examples 1 & 2 below, then solve Test 5C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 5C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Two Parallel Branches: Evaluate (50 - 14) ÷ (3 + 6) → Key [50] [ENTER] [14] [-] (36 in X), then [3] [ENTER] [6] [+] (36 lifts to Y, 9 in X), then [÷] → 4!

**EXAMPLE 2** 5C #5 ((8+4)×(9-3)) → [8] [ENTER] [4] [+] [9] [ENTER] [3] [-] [×] → 72

**TEST 5C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Two-Branch Expressions $(a \pm b) \times (c \pm d)$ & Quotients

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(15 \times 4) + (18 \div 2) =$ | |
| 2. | $(12 \times 5) - (8 \times 4) =$ | |
| 3. | $(50 - 14) \div (3 + 6) =$ | |
| 4. | $(42 + 18) \div (11 - 5) =$ | |
| 5. | $(8 + 4) \times (9 - 3) =$ | |
| 6. | $(14 - 5) \times (6 + 2) =$ | |
| 7. | $(20 \times 3) + (15 \times 2) =$ | |
| 8. | $(16 \times 4) - (12 \times 3) =$ | |
| 9. | $((20 - 8) \times 4) \div 6 =$ | |
| 10. | $((18 + 6) \div 3) \times 7 =$ | |
| 11. | $(72 \div (4 + 5)) \times 3 =$ | |
| 12. | $100 - (16 \times 5 - 12) =$ | |
| 13. | $(84 \div 7) + (45 \div 9) =$ | |
| 14. | $(65 - 25) \times (18 \div 6) =$ | |
| 15. | $((8 + 4) \times (9 - 3)) \div 12 =$ | |
| 16. | $(96 \div 8) \times (20 - 15) =$ | |
| 17. | $150 - (12 \times 6 + 18) =$ | |
| 18. | $(35 + 25) \div (4 \times 3) =$ | |
| 19. | $(18 \times 5) \div (14 - 5) =$ | |
| 20. | $(24 - 9) \times (16 - 8) =$ | |

**TEST 5C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Two-Branch Expressions $(a \pm b) \times (c \pm d)$ & Quotients — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(8 + 2) \times (9 - 3) =$ | |
| 2. | $(9 \times 3) + (10 \times 4) =$ | |
| 3. | $(10 + 4) \times (11 - 5) =$ | |
| 4. | $(11 \times 5) + (12 \times 6) =$ | |
| 5. | $(12 + 6) \times (13 - 3) =$ | |
| 6. | $(13 \times 2) + (14 \times 4) =$ | |
| 7. | $(14 + 3) \times (15 - 5) =$ | |
| 8. | $(15 \times 4) + (9 \times 6) =$ | |
| 9. | $(16 + 5) \times (10 - 3) =$ | |
| 10. | $(17 \times 6) + (11 \times 4) =$ | |
| 11. | $(18 + 2) \times (12 - 5) =$ | |
| 12. | $(19 \times 3) + (13 \times 6) =$ | |
| 13. | $(20 + 4) \times (14 - 3) =$ | |
| 14. | $(21 \times 5) + (15 \times 4) =$ | |
| 15. | $(22 + 6) \times (9 - 5) =$ | |
| 16. | $(23 \times 2) + (10 \times 6) =$ | |
| 17. | $(24 + 3) \times (11 - 3) =$ | |
| 18. | $(25 \times 4) + (12 \times 4) =$ | |
| 19. | $(26 + 5) \times (13 - 5) =$ | |
| 20. | $(27 \times 6) + (14 \times 6) =$ | |

# Section 5: Multi-Step Operations & Grouping

Review Worked Examples 1 & 2 below, then solve Test 5C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 5D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Common Factor Distribution: Evaluate (28 × 9) - (18 × 9) → Factor out ×9 mentally: (28 - 18) × 9 = 10 × 9 = 90 in 1 second!

**EXAMPLE 2** 5C #12 (100-(16×5-12)) → Inner 80 - 12 = 68 first, 100 - 68 = 32

**TEST 5C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Multi-Digit Chained Operations &)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(10 + 2) \times (10 - 3) =$ | _____ |
| 2. | $(11 \times 3) + (11 \times 4) =$ | _____ |
| 3. | $(12 + 4) \times (12 - 5) =$ | _____ |
| 4. | $(13 \times 5) + (13 \times 6) =$ | _____ |
| 5. | $(14 + 6) \times (14 - 3) =$ | _____ |
| 6. | $(15 \times 2) + (15 \times 4) =$ | _____ |
| 7. | $(16 + 3) \times (16 - 5) =$ | _____ |
| 8. | $(17 \times 4) + (10 \times 6) =$ | _____ |
| 9. | $(18 + 5) \times (11 - 3) =$ | _____ |
| 10. | $(19 \times 6) + (12 \times 4) =$ | _____ |
| 11. | $(20 + 2) \times (13 - 5) =$ | _____ |
| 12. | $(21 \times 3) + (14 \times 6) =$ | _____ |
| 13. | $(22 + 4) \times (15 - 3) =$ | _____ |
| 14. | $(23 \times 5) + (16 \times 4) =$ | _____ |
| 15. | $(24 + 6) \times (10 - 5) =$ | _____ |
| 16. | $(25 \times 2) + (11 \times 6) =$ | _____ |
| 17. | $(26 + 3) \times (12 - 3) =$ | _____ |
| 18. | $(27 \times 4) + (13 \times 4) =$ | _____ |
| 19. | $(28 + 5) \times (14 - 5) =$ | _____ |
| 20. | $(29 \times 6) + (15 \times 6) =$ | _____ |

**TEST 5D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Multi-Digit Chained Operations & Deep Nesting

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $(25 \times 8) - (15 \times 6) =$ | _____ |
| 2. | $(144 \div 12) \times (36 \div 4) =$ | _____ |
| 3. | $(125 + 75) \div (32 - 7) =$ | _____ |
| 4. | $((45 - 15) \times 6) \div (12 + 6) =$ | _____ |
| 5. | $250 - (18 \times 7 + 24) =$ | _____ |
| 6. | $(36 \times 5) \div (27 - 9) =$ | _____ |
| 7. | $((64 \div 4) + 9) \times 4 =$ | _____ |
| 8. | $(180 \div (12 + 3)) \times 8 =$ | _____ |
| 9. | $(22 + 18) \times (35 - 20) =$ | _____ |
| 10. | $(160 - 40) \div (6 \times 4) =$ | _____ |
| 11. | $((32 - 14) \times 5) \div 6 =$ | _____ |
| 12. | $80 - (15 \times 4) + 12 =$ | _____ |
| 13. | $(240 \div 15) + (120 \div 8) =$ | _____ |
| 14. | $(17 \times 6) - (13 \times 6) =$ | _____ |
| 15. | $(28 \times 9) - (18 \times 9) =$ | _____ |
| 16. | $(45 \times 12) \div 18 =$ | _____ |
| 17. | $((100 - 28) \div 9) \times 15 =$ | _____ |
| 18. | $(360 \div (18 - 6)) - 14 =$ | _____ |
| 19. | $((15 + 5) \times (8 - 3)) \div 10 =$ | _____ |
| 20. | $(500 - 140) \div (9 \times 4) =$ | _____ |

# Section 5: Multi-Step Operations & Grouping

Review Worked Examples 1 & 2 below, then solve Test 5D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 5D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 5D #15 ((28×9)-(18×9)) → Distributive shortcut: (28 - 18) × 9 = 90

**EXAMPLE 2** 5D #16 ((45×12)÷18) → Cancel common factors: 45 × (12/18) = 45 × 2/3 = 30

**TEST 5D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Multi-Digit Chained Operations & Deep Nesting — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. (21 × 4) − (10 × 4) = _____
2. (50 − 25) ÷ 5 = _____
3. (25 × 6) − (12 × 6) = _____
4. (98 − 49) ÷ 7 = _____
5. (29 × 8) − (14 × 8) = _____
6. (72 − 36) ÷ 4 = _____
7. (33 × 10) − (16 × 10) = _____
8. (60 − 30) ÷ 6 = _____
9. (37 × 5) − (18 × 5) = _____
10. (77 − 56) ÷ 3 = _____
11. (41 × 7) − (20 × 7) = _____
12. (90 − 45) ÷ 5 = _____
13. (45 × 9) − (22 × 9) = _____
14. (70 − 35) ÷ 7 = _____
15. (49 × 4) − (24 × 4) = _____
16. (56 − 28) ÷ 4 = _____
17. (53 × 6) − (26 × 6) = _____
18. (108 − 54) ÷ 6 = _____
19. (57 × 8) − (28 × 8) = _____
20. (55 − 40) ÷ 3 = _____

**TEST 5D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 5 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. (24 × 4) − (12 × 4) = _____
2. (66 − 30) ÷ 6 = _____
3. (28 × 6) − (14 × 6) = _____
4. (80 − 56) ÷ 3 = _____
5. (32 × 8) − (16 × 8) = _____
6. (90 − 40) ÷ 5 = _____
7. (36 × 10) − (18 × 10) = _____
8. (78 − 36) ÷ 7 = _____
9. (40 × 5) − (20 × 5) = _____
10. (96 − 64) ÷ 4 = _____
11. (44 × 7) − (22 × 7) = _____
12. (110 − 50) ÷ 6 = _____
13. (48 × 9) − (24 × 9) = _____
14. (60 − 42) ÷ 3 = _____
15. (52 × 4) − (26 × 4) = _____
16. (72 − 32) ÷ 5 = _____
17. (56 × 6) − (28 × 6) = _____
18. (130 − 60) ÷ 7 = _____
19. (60 × 8) − (30 × 8) = _____
20. (72 − 48) ÷ 4 = _____

# Section 6: Fractions, Decimals & Reciprocals

Fraction equivalence, unlike denominators, mixed numbers, decimals, and reciprocals. • Study the Champion Way & complete W1–W6 before starting Tests 6A1–6D3.

### 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 6: FRACTIONS, DECIMALS & RECIPROCALS

| | |
|---|---|
| RULE 1 | **Rule 1 — Harmonic & Reciprocal Sums via [1/x]: To evaluate 1 ÷ (1/a + 1/b) on the calculator, never find a common denominator by hand—key [a] [1/x] [b] [1/x] [+] [1/x] in 6 keys!** |
| RULE 2 | Rule 2 — Distribute Whole Multipliers to Clear Denominators: For 24 × (5/6 - 3/8), multiply 24 into each fraction directly: (24 × 5/6) - (24 × 3/8) = 20 - 9 = 11! |
| RULE 3 | Rule 3 — Memorize Eighths & Thirds as Instant Decimals: 1/8 = 0.125, 3/8 = 0.375, 5/8 = 0.625, 7/8 = 0.875; and use Two-Dot entry ([2][.][1][.][2] = 2 1/2) on the SC-6. |

### 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Harmonic Parallel Sum: Evaluate 1 ÷ (1/4 + 1/6)** Stack Keys: 


SLOW / AMATEUR WAY (Avoid):

Slow Way: Convert to twelfths (3/12 + 2/12 = 5/12), invert to 12/5, divide 12 ÷ 5 = 2.4.

CHAMPION SPEED WAY (Use This!):

**Champion Way: Use the 6-key Reciprocal Pipeline: [4] [1/x] [6] [1/x] [+] [1/x] → 2.4!**

**Archetype 2 — Clearing Denominators by Distribution: Evaluate 24 × (5/6 - 3/8)** Stack Keys: 


SLOW / AMATEUR WAY (Avoid):

Slow Way: Subtract 20/24 - 9/24 = 11/24, then multiply by 24.

CHAMPION SPEED WAY (Use This!):

**Champion Way: Distribute 24 immediately: 24(5/6) = 20 and 24(3/8) = 9, so 20 - 9 = 11!**

**Archetype 3 — Telescoping Product Cancellation: Evaluate 1/2 × 2/3 × 3/4 × 4/5** Stack Keys: 


SLOW / AMATEUR WAY (Avoid):

Slow Way: Multiply numerators (1×2×3×4 = 24) and denominators (2×3×4×5 = 120) and reduce.

CHAMPION SPEED WAY (Use This!):

**Champion Way: Cancel interior 2, 3, 4 diagonally—only first numerator (1) and last denominator (5) survive → 1/5!**

### 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** $3/4 \div 1/2$ (decimal) = _____

Shortcut Cue: 3/4 is 1/2 of tower → double to 6/4 = 1.5

**W2.** $3/4 + (1/2 \times 1/4)$ (decimal) = _____

Shortcut Cue: 0.75 + 0.125

**W3.** $1 \div (1/3 + 1/6) =$ _____

Shortcut Cue: 1/3 + 1/6 = 1/2, reciprocal is 2

**W4.** $(1/3 + 1/4) \times 24 =$ _____

Shortcut Cue: Distribute 24: 8 + 6

**W5.** $2/3 \times 3/4 \times 4/5 =$ _____

Shortcut Cue: Telescoping cancel 3 and 4 → 2/5

**W6.** $36 \times (7/9 - 1/4) =$ _____

Shortcut Cue: Distribute 36: 28 - 9

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 6: Fractions, Decimals & Reciprocals

Review Worked Examples 1 & 2 below, then solve Test 6A1 (Level 1: Foundation (Sprint 1)) & Test 6A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $3/4 \div 1/2$ (decimal) $= 1.5$ ($3/4$ is $1/2$ of tower → double to $6/4 = 1.5$)

**EXAMPLE 2** $3/4 + (1/2 \times 1/4)$ (decimal) $= 0.875$ ($0.75 + 0.125$)

### TEST 6A1 • LEVEL 1: FOUNDATION (SPRINT 1) Goal: 4 min

Like Fractions, Halves/Fourths & Exact Decimals

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1/4 + 2/4 =$ _____
2. $3/5 - 1/5 =$ _____
3. $3/8 + 2/8 =$ _____
4. $7/10 - 4/10 =$ _____
5. $1/2 + 1/4 =$ _____
6. $3/4 - 1/2 =$ _____
7. $1/2 + 1/2 =$ _____
8. $1 - 1/4 =$ _____
9. $1 - 3/8 =$ _____
10. $1/2$ as a decimal = _____
11. $1/4$ as a decimal = _____
12. $3/4$ as a decimal = _____
13. $1/5$ as a decimal = _____
14. $3/5$ as a decimal = _____
15. $1/10 + 3/10$ (simplified) = _____
16. $2/6 + 1/6$ (simplified) = _____
17. $1/2$ of $16 =$ _____
18. $1/4$ of $20 =$ _____
19. $3/4$ of $12 =$ _____
20. $2/3$ of $15 =$ _____

### TEST 6A2 • LEVEL 1: FOUNDATION (SPRINT 2) Goal: 4 min

Like Fractions, Halves/Fourths & Exact Decimals — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1/5 + 2/5 =$ _____
2. $2/6 + 3/6 =$ _____
3. $3/8 + 4/8 =$ _____
4. $4/10 + 5/10 =$ _____
5. $5/12 + 6/12 =$ _____
6. $2/4 + 1/4 =$ _____
7. $1/5 + 4/5 =$ _____
8. $4/6 + 1/6 =$ _____
9. $3/8 + 5/8 =$ _____
10. $2/10 + 3/10 =$ _____
11. $1/12 + 1/12 =$ _____
12. $2/4 + 1/4 =$ _____
13. $1/5$ of $15 =$ _____
14. $2/6$ of $24 =$ _____
15. $3/8$ of $40 =$ _____
16. $8/10$ of $60 =$ _____
17. $7/12$ of $84 =$ _____
18. $2/4$ of $32 =$ _____
19. $1/5$ of $45 =$ _____
20. $4/6$ of $60 =$ _____

# Section 6: Fractions, Decimals & Reciprocals

Review Worked Examples 1 & 2 below, then solve Test 6A3 (Level 1: Foundation (Sprint 3)) & Test 6B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $1 \div (1/3 + 1/6) = 2$ **($1/3 + 1/6 = 1/2$, reciprocal is 2)**

**EXAMPLE 2** $(1/3 + 1/4) \times 24 = 14$ **(Distribute 24: 8 + 6)**

### TEST 6A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 4 min

Bridge from Level 1 to Level 2 (Unlike Denominators, Multiplicat)

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $1/6 + 3/6 =$ | |
| 2. | $2/8 + 4/8 =$ | |
| 3. | $3/10 + 5/10 =$ | |
| 4. | $4/12 + 6/12 =$ | |
| 5. | $1/4 + 1/4 =$ | |
| 6. | $3/5 + 2/5 =$ | |
| 7. | $3/6 + 3/6 =$ | |
| 8. | $2/8 + 4/8 =$ | |
| 9. | $1/10 + 2/10 =$ | |
| 10. | $10/12 + 2/12 =$ | |
| 11. | $1/4 + 1/4 =$ | |
| 12. | $3/5 + 2/5 =$ | |
| 13. | $1/6$ of $24 =$ | |
| 14. | $2/8$ of $40 =$ | |
| 15. | $7/10$ of $60 =$ | |
| 16. | $6/12$ of $84 =$ | |
| 17. | $1/4$ of $32 =$ | |
| 18. | $3/5$ of $45 =$ | |
| 19. | $3/6$ of $60 =$ | |
| 20. | $2/8$ of $88 =$ | |

### TEST 6B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 5 min

Unlike Denominators, Multiplication & Reciprocals

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | $1/3 + 1/6 =$ | |
| 2. | $2/3 - 1/6 =$ | |
| 3. | $3/8 + 1/4 =$ | |
| 4. | $5/6 - 1/3 =$ | |
| 5. | $1/2 + 1/3 =$ | |
| 6. | $1/2 - 1/3 =$ | |
| 7. | $1/3 + 1/4 =$ | |
| 8. | $3/4 - 1/3 =$ | |
| 9. | $1/2 \times 3/4 =$ | |
| 10. | $2/3 \times 3/4 =$ | |
| 11. | $3/5 \times 5/6 =$ | |
| 12. | $4/5 \times 15 =$ | |
| 13. | Reciprocal of $3/4 =$ | |
| 14. | Reciprocal of $4/5$ (decimal) $=$ | |
| 15. | Reciprocal of $2/5$ (decimal) $=$ | |
| 16. | $1/2 \div 1/4 =$ | |
| 17. | $3/4 \div 1/4 =$ | |
| 18. | $3/4 \div 1/2$ (decimal) $=$ | |
| 19. | $2/3 \div 2/3 =$ | |
| 20. | $6 \div 1/2 =$ | |

# Section 6: Fractions, Decimals & Reciprocals

Review Worked Examples 1 & 2 below, then solve Test 6B2 (Level 2: Core Fluency (Sprint 2)) & Test 6B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Harmonic Parallel Sum: Evaluate $1 \div (1/4 + 1/6)$ → Use the 6-key Reciprocal Pipeline: [4] [1/x] [6] [1/x] [+] [1/x] → 2.4!

**EXAMPLE 2** $2/3 \times 3/4 \times 4/5 = 2/5$ (Telescoping cancel 3 and 4 → 2/5)

**TEST 6B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 5 min

Unlike Denominators, Multiplication & Reciprocals — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $1/3 + 1/2 =$ | _____ |
| 2. | $1/2 \times 3/5 =$ | _____ |
| 3. | $1/3 \div 4/5 =$ | _____ |
| 4. | $1/4 + 1/5 =$ | _____ |
| 5. | $1/3 \times 3/4 =$ | _____ |
| 6. | $2/5 \div 1/3 =$ | _____ |
| 7. | $3/5 + 1/8 =$ | _____ |
| 8. | $1/2 \times 1/3 =$ | _____ |
| 9. | $1/3 \div 1/4 =$ | _____ |
| 10. | $1/2 + 3/5 =$ | _____ |
| 11. | $1/3 \times 4/5 =$ | _____ |
| 12. | $3/4 \div 1/5 =$ | _____ |
| 13. | $1/3 + 7/8 =$ | _____ |
| 14. | $2/5 \times 5/6 =$ | _____ |
| 15. | $3/5 \div 1/4 =$ | _____ |
| 16. | $1/2 + 1/3 =$ | _____ |
| 17. | $1/3 \times 3/4 =$ | _____ |
| 18. | $1/2 \div 3/5 =$ | _____ |
| 19. | $1/3 + 4/5 =$ | _____ |
| 20. | $1/2 \times 1/5 =$ | _____ |

**TEST 6B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 5 min

Bridge from Level 2 to Level 3 (Mixed Numbers, Decimal Equivalen)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $1/2 + 3/5 =$ | _____ |
| 2. | $2/3 \times 4/5 =$ | _____ |
| 3. | $3/4 \div 1/5 =$ | _____ |
| 4. | $2/3 + 3/4 =$ | _____ |
| 5. | $1/5 \times 1/3 =$ | _____ |
| 6. | $2/5 \div 1/8 =$ | _____ |
| 7. | $1/2 + 1/3 =$ | _____ |
| 8. | $2/3 \times 1/4 =$ | _____ |
| 9. | $1/2 \div 3/5 =$ | _____ |
| 10. | $2/3 + 4/5 =$ | _____ |
| 11. | $1/2 \times 1/5 =$ | _____ |
| 12. | $2/3 \div 7/8 =$ | _____ |
| 13. | $1/5 + 5/6 =$ | _____ |
| 14. | $2/5 \times 1/4 =$ | _____ |
| 15. | $1/2 \div 1/3 =$ | _____ |
| 16. | $2/3 + 3/4 =$ | _____ |
| 17. | $1/2 \times 3/5 =$ | _____ |
| 18. | $2/3 \div 4/5 =$ | _____ |
| 19. | $1/4 + 1/5 =$ | _____ |
| 20. | $2/3 \times 1/8 =$ | _____ |

# Section 6: Fractions, Decimals & Reciprocals

Review Worked Examples 1 & 2 below, then solve Test 6C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 6C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Clearing Denominators by Distribution: Evaluate $24 \times (5/6 - 3/8) \rightarrow$ Distribute 24 immediately: $24(5/6) = 20$ and $24(3/8) = 9$, so $20 - 9 = 11$!

**EXAMPLE 2** 6C #12 $(3/4 + 1/2 \times 1/4) \rightarrow 0.75 + 0.125 = 0.875$ ([.1.2] [ENTER] [.1.4] [×] [.3.4] [+])

**TEST 6C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Mixed Numbers, Decimal Equivalents & Fraction Chains

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $2\ 1/2 + 1\ 1/4$ (decimal) = _____
2. $3\ 3/4 - 1\ 1/2$ (decimal) = _____
3. $5 - 2\ 3/4$ (decimal) = _____
4. $3\ 1/2 - 1\ 3/4$ (decimal) = _____
5. $1\ 1/2 \times 4$ = _____
6. $2\ 1/2 \times 6$ = _____
7. $1\ 1/4 \times 8$ = _____
8. $3\ 1/3 \times 3$ = _____
9. $(1/3 + 1/6) \times 12$ = _____
10. $(1/2 + 1/4) \times 8$ = _____
11. $(3/4 - 1/8) \times 16$ = _____
12. $3/4 + (1/2 \times 1/4)$ (decimal) = _____
13. $1/8$ as a decimal = _____
14. $3/8$ as a decimal = _____
15. $5/8$ as a decimal = _____
16. $7/8$ as a decimal = _____
17. $4\ 1/2 \div 1\ 1/2$ = _____
18. $2\ 1/4 \div 3/4$ = _____
19. $0.25 + 0.5 + 0.75$ = _____
20. $1.25 \times 8$ = _____

**TEST 6C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Mixed Numbers, Decimal Equivalents & Fraction Chains — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $2\ 1/2 \times 4$ = _____
2. $3.5 + 1.5$ = _____
3. $4\ 3/5 \times 5$ = _____
4. $5.5 + 2.5$ = _____
5. $2\ 1/2 \times 6$ = _____
6. $3.75 + 1$ = _____
7. $4\ 3/5 \times 10$ = _____
8. $5.125 + 2$ = _____
9. $2\ 1/2 \times 2$ = _____
10. $3.25 + 3$ = _____
11. $4\ 3/5 \times 15$ = _____
12. $5.625 + 1.5$ = _____
13. $2\ 1/2 \times 4$ = _____
14. $3.5 + 2.5$ = _____
15. $4\ 3/5 \times 5$ = _____
16. $5.25 + 1$ = _____
17. $2\ 1/2 \times 6$ = _____
18. $3.75 + 2$ = _____
19. $4\ 3/5 \times 10$ = _____
20. $5.75 + 3$ = _____

# Section 6: Fractions, Decimals & Reciprocals

Review Worked Examples 1 & 2 below, then solve Test 6C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 6D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Telescoping Product Cancellation: Evaluate $1/2 \times 2/3 \times 3/4 \times 4/5 \rightarrow$ Cancel interior 2, 3, 4 diagonally—only first numerator (1) and last denominator (5) survive $\rightarrow 1/5$!

**EXAMPLE 2** 6D #4 $(1 \div (1/4 + 1/6)) \rightarrow$ Reciprocal pipeline: [4] [1/x] [6] [1/x] [+] [1/x] $\rightarrow 2.4$

**TEST 6C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Harmonic Means, Complex Fraction)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $3\ 1/2 \times 6 =$ _____
2. $4.5 + 2 =$ _____
3. $5\ 3/5 \times 10 =$ _____
4. $6.5 + 3 =$ _____
5. $3\ 1/2 \times 2 =$ _____
6. $4.75 + 1.5 =$ _____
7. $5\ 3/5 \times 15 =$ _____
8. $6.125 + 2.5 =$ _____
9. $3\ 1/2 \times 4 =$ _____
10. $4.25 + 3.5 =$ _____
11. $5\ 3/5 \times 5 =$ _____
12. $6.625 + 2 =$ _____
13. $3\ 1/2 \times 6 =$ _____
14. $4.5 + 3 =$ _____
15. $5\ 3/5 \times 10 =$ _____
16. $6.25 + 1.5 =$ _____
17. $3\ 1/2 \times 2 =$ _____
18. $4.75 + 2.5 =$ _____
19. $5\ 3/5 \times 15 =$ _____
20. $6.75 + 3.5 =$ _____

**TEST 6D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Harmonic Means, Complex Fractions & Multi-Step Rationals

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1 \div (1/2 + 1/2) =$ _____
2. $1 \div (1/3 + 1/6) =$ _____
3. $1 \div (1/4 + 1/4) =$ _____
4. $1 \div (1/4 + 1/6)$ (decimal) $=$ _____
5. $1 \div (1/3 + 1/3 + 1/3) =$ _____
6. $(1/2 + 1/3) \times 12 =$ _____
7. $(1/3 + 1/4) \times 24 =$ _____
8. $(3/4 - 2/3) \times 12 =$ _____
9. $(2/3 + 3/4) \times 12 =$ _____
10. $(1/2 + 1/3 + 1/6) \times 18 =$ _____
11. $2/3 \times 3/4 \times 4/5 =$ _____
12. $1/2 \times 2/3 \times 3/4 \times 4/5 =$ _____
13. $(3/4) \div (3/8) =$ _____
14. $(5/6) \div (5/12) =$ _____
15. $(2\ 1/2 + 1\ 1/2) \div (1/4) =$ _____
16. $1 + 1/(1 + 1/2) =$ _____
17. $1 + 1/(1 + 1/3) =$ _____
18. $(3\ 1/3 \div 1\ 1/3)$ (decimal) $=$ _____
19. $24 \times (5/6 - 3/8) =$ _____
20. $36 \times (7/9 - 1/4) =$ _____

# Section 6: Fractions, Decimals & Reciprocals

Review Worked Examples 1 & 2 below, then solve Test 6D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 6D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 6D #12 (1/2×2/3×3/4×4/5) → Telescoping cancellation leaves 1/5

**EXAMPLE 2** 6D #19 (24 × (5/6 - 3/8)) → Distribute 24: 20 - 9 = 11

**TEST 6D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Harmonic Means, Complex Fractions & Multi-Step Rationals — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1 \div (1/3 + 1/3) =$ _____
2. $1 \div (1/4 + 1/8) =$ _____
3. $1 \div (1/5 + 1/5) =$ _____
4. $1 \div (1/6 + 1/12) =$ _____
5. $1 \div (1/7 + 1/7) =$ _____
6. $1 \div (1/3 + 1/6) =$ _____
7. $1 \div (1/4 + 1/4) =$ _____
8. $1 \div (1/5 + 1/10) =$ _____
9. $48 \times (3/4 + 1/6) =$ _____
10. $24 \times (1/4 + 1/3) =$ _____
11. $36 \times (1/2 + 1/6) =$ _____
12. $48 \times (3/4 + 1/3) =$ _____
13. $24 \times (1/4 + 1/6) =$ _____
14. $36 \times (1/2 + 1/3) =$ _____
15. $48 \times (3/4 + 1/6) =$ _____
16. $3/4 \times 4/5 \times 5/6 =$ _____
17. $4/5 \times 5/6 \times 6/7 =$ _____
18. $5/6 \times 6/7 \times 7/8 =$ _____
19. $6/7 \times 7/8 \times 8/9 =$ _____
20. $7/8 \times 8/9 \times 9/10 =$ _____

**TEST 6D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 6 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1 \div (1/4 + 1/4) =$ _____
2. $1 \div (1/5 + 1/10) =$ _____
3. $1 \div (1/6 + 1/6) =$ _____
4. $1 \div (1/7 + 1/14) =$ _____
5. $1 \div (1/8 + 1/8) =$ _____
6. $1 \div (1/4 + 1/8) =$ _____
7. $1 \div (1/5 + 1/5) =$ _____
8. $1 \div (1/6 + 1/12) =$ _____
9. $60 \times (3/4 + 1/6) =$ _____
10. $36 \times (1/4 + 1/3) =$ _____
11. $48 \times (1/2 + 1/6) =$ _____
12. $60 \times (3/4 + 1/3) =$ _____
13. $36 \times (1/4 + 1/6) =$ _____
14. $48 \times (1/2 + 1/3) =$ _____
15. $60 \times (3/4 + 1/6) =$ _____
16. $4/5 \times 5/6 \times 6/7 =$ _____
17. $5/6 \times 6/7 \times 7/8 =$ _____
18. $6/7 \times 7/8 \times 8/9 =$ _____
19. $7/8 \times 8/9 \times 9/10 =$ _____
20. $8/9 \times 9/10 \times 10/11 =$ _____

# Section 7: Primes, GCF/LCM & Nested Expressions

Prime factorization, GCF, LCM, powers, and multi-branch expression trees. • Study the Champion Way & complete W1–W6 before starting Tests 7A1–7D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 7: PRIMES, GCF/LCM & NESTED EXPRESSIONS

**RULE 1** **Rule 1 — Product Identity GCF(a, b) × LCM(a, b) = $a \times b$: Never compute GCF and LCM separately when asked for their product—just multiply $a \times b$ directly!**

**RULE 2** Rule 2 — Divisor Count Formula from Prime Exponents: If $N = p^a \times q^b$, the number of positive factors is $(a + 1)(b + 1)$. Example: $24 = 2^3 \times 3^1 \rightarrow (3 + 1)(1 + 1) = 8$ factors.

**RULE 3** Rule 3 — Difference of Squares Quotients $(a^2 - b^2) \div (a \pm b)$: Simplify algebraically before touching the calculator: $(a^2 - b^2) \div (a - b) = a + b$, and $(a^2 - b^2) \div (a + b) = a - b$!

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — GCF × LCM Identity: Evaluate GCF(15, 20) × LCM(15, 20)** Stack Keys: `1` `5` `ENTER` `2` `0` `×`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Find GCF = 5 and LCM = 60, then multiply 5 × 60 = 300.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Use GCF(a, b) × LCM(a, b) = $a \times b \rightarrow 15 \times 20 = 300$ in 2 seconds!**

**Archetype 2 — Counting Positive Divisors: How many positive factors does 36 have?** Stack Keys: `3` `ENTER` `3` `×`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: List 1, 2, 3, 4, 6, 9, 12, 18, 36 by trial division (easy to miss one).

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Prime factor $36 = 2^2 \times 3^2$, then add 1 to each exponent: $(2 + 1)(2 + 1) = 9$!**

**Archetype 3 — Algebraic Quotient Shortcut: Evaluate $(25^2 - 15^2) \div (25 - 15)$** Stack Keys: `2` `5` `ENTER` `1` `5` `+`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 625 - 225 = 400, then divide by 10 = 40.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Cancel (25 - 15) immediately to leave 25 + 15 = 40!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** **GCF(12, 18) × LCM(12, 18) =** _____

Shortcut Cue: Use $a \times b = 12 \times 18$

**W2.** **Number of positive factors of 24 =** _____

Shortcut Cue: $24 = 2^3 \times 3^1 \rightarrow (3+1)(1+1)$

**W3.** **$(15^2 - 5^2) \div (15 + 5) =$** _____

Shortcut Cue: Simplifies to 15 - 5

**W4.** **$(30 - (4 \times 5)) \times (12 \div (2 + 4)) =$** _____

Shortcut Cue: (30 - 20) × (12 ÷ 6) = 10 × 2

**W5.** **$(2^5 - 2^3) \div (3^2 - 3) =$** _____

Shortcut Cue: (32 - 8) ÷ (9 - 3) = 24 ÷ 6

**W6.** **$(12^2 + 5^2) \div 13 =$** _____

Shortcut Cue: 169 ÷ 13 (since 5-12-13 → $13^2 \div 13$)

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 7: Primes, GCF/LCM & Nested Expressions

Review Worked Examples 1 & 2 below, then solve Test 7A1 (Level 1: Foundation (Sprint 1)) & Test 7A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** GCF(12, 18) × LCM(12, 18) = 216 (Use $a \times b = 12 \times 18$)

**EXAMPLE 2** Number of positive factors of 24 = 8 ($24 = 2^3 \times 3^1 \rightarrow (3+1)(1+1)$)

### TEST 7A1 • LEVEL 1: FOUNDATION (SPRINT 1) Goal: 4 min

Prime Powers, Squares, Cubes & Factor Pairs

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $2 \times 2 \times 3 =$ _____
2. $2 \times 3 \times 3 =$ _____
3. $2 \times 2 \times 5 =$ _____
4. $2 \times 2 \times 2 \times 3 =$ _____
5. $2 \times 3 \times 5 =$ _____
6. $2 \times 2 \times 3 \times 3 =$ _____
7. $2^3 =$ _____
8. $2^4 =$ _____
9. $3^3 =$ _____
10. $4^3 =$ _____
11. $5^3 =$ _____
12. Smallest prime number = _____
13. 7th prime number = _____
14. GCF of 8 and 12 = _____
15. GCF of 15 and 25 = _____
16. GCF of 18 and 24 = _____
17. LCM of 3 and 4 = _____
18. LCM of 4 and 6 = _____
19. LCM of 6 and 8 = _____
20. LCM of 5 and 10 = _____

### TEST 7A2 • LEVEL 1: FOUNDATION (SPRINT 2) Goal: 4 min

Prime Powers, Squares, Cubes & Factor Pairs — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF of 6 and 12 = _____
2. LCM of 12 and 20 = _____
3. $4^2 + 1 =$ _____
4. GCF of 12 and 18 = _____
5. LCM of 21 and 28 = _____
6. $3^2 + 1 =$ _____
7. GCF of 18 and 54 = _____
8. LCM of 9 and 9 = _____
9. $2^2 + 1 =$ _____
10. GCF of 10 and 25 = _____
11. LCM of 18 and 36 = _____
12. $5^2 + 1 =$ _____
13. GCF of 16 and 32 = _____
14. LCM of 27 and 45 = _____
15. $4^2 + 1 =$ _____
16. GCF of 8 and 12 = _____
17. LCM of 15 and 20 = _____
18. $3^2 + 1 =$ _____
19. GCF of 14 and 42 = _____
20. LCM of 24 and 24 = _____

# Section 7: Primes, GCF/LCM & Nested Expressions

Review Worked Examples 1 & 2 below, then solve Test 7A3 (Level 1: Foundation (Sprint 3)) & Test 7B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $(15^2 - 5^2) \div (15 + 5) = 10$ (Simplifies to 15 - 5)

**EXAMPLE 2** $(30 - (4 \times 5)) \times (12 \div (2 + 4)) = 20$ ((30 - 20) × (12 ÷ 6) = 10 × 2)

**TEST 7A3 • LEVEL 1: FOUNDATION (SPRINT 3)** Goal: 4 min

Bridge from Level 1 to Level 2 (GCF, LCM & Order of Operations W)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF of 8 and 16 = _____
2. LCM of 15 and 25 = _____
3. $4^3 + 2 =$ _____
4. GCF of 14 and 21 = _____
5. LCM of 24 and 32 = _____
6. $3^3 + 2 =$ _____
7. GCF of 20 and 60 = _____
8. LCM of 12 and 12 = _____
9. $2^3 + 2 =$ _____
10. GCF of 12 and 30 = _____
11. LCM of 21 and 42 = _____
12. $5^3 + 2 =$ _____
13. GCF of 18 and 36 = _____
14. LCM of 30 and 50 = _____
15. $4^3 + 2 =$ _____
16. GCF of 10 and 15 = _____
17. LCM of 18 and 24 = _____
18. $3^3 + 2 =$ _____
19. GCF of 16 and 48 = _____
20. LCM of 27 and 27 = _____

**TEST 7B1 • LEVEL 2: CORE FLUENCY (SPRINT 1)** Goal: 5 min

GCF, LCM & Order of Operations Without Parentheses

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF of 24 and 36 = _____
2. GCF of 30 and 45 = _____
3. GCF of 32 and 48 = _____
4. GCF of 42 and 56 = _____
5. LCM of 8 and 12 = _____
6. LCM of 9 and 12 = _____
7. LCM of 12 and 15 = _____
8. LCM of 10 and 14 = _____
9. $10 + 5 \times 2 =$ _____
10. $25 - 5 \times 3 =$ _____
11. $4 + 6 \div 2 =$ _____
12. $12 - 2 \times 4 =$ _____
13. $15 - 3 \times 3 =$ _____
14. $30 - 4 \times 5 =$ _____
15. $2^3 \times 3^2 =$ _____
16. $2^2 \times 3 \times 5 =$ _____
17. $2^3 \times 5^2 =$ _____
18. $3^2 + 4^2 =$ _____
19. $5^2 - 3^2 =$ _____
20. $10^2 - 8^2 =$ _____

# Section 7: Primes, GCF/LCM & Nested Expressions

Review Worked Examples 1 & 2 below, then solve Test 7B2 (Level 2: Core Fluency (Sprint 2)) & Test 7B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — GCF × LCM Identity: Evaluate GCF(15, 20) × LCM(15, 20) → Use GCF(a, b) × LCM(a, b) = $a \times b$ → $15 \times 20 = 300$ in 2 seconds!

**EXAMPLE 2** $(2^5 - 2^3) \div (3^2 - 3) = 4$ $((32 - 8) \div (9 - 3) = 24 \div 6)$

**TEST 7B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 5 min

GCF, LCM & Order of Operations Without Parentheses — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $11 + 3 \times 3 =$ | |
| 2. | $7^2 - 3^2 =$ | |
| 3. | $15 + 5 \times 5 =$ | |
| 4. | $9^2 - 5^2 =$ | |
| 5. | $19 + 7 \times 7 =$ | |
| 6. | $11^2 - 3^2 =$ | |
| 7. | $23 + 4 \times 3 =$ | |
| 8. | $13^2 - 5^2 =$ | |
| 9. | $27 + 6 \times 5 =$ | |
| 10. | $7^2 - 3^2 =$ | |
| 11. | $31 + 3 \times 7 =$ | |
| 12. | $9^2 - 5^2 =$ | |
| 13. | $35 + 5 \times 3 =$ | |
| 14. | $11^2 - 3^2 =$ | |
| 15. | $39 + 7 \times 5 =$ | |
| 16. | $13^2 - 5^2 =$ | |
| 17. | $43 + 4 \times 7 =$ | |
| 18. | $7^2 - 3^2 =$ | |
| 19. | $47 + 6 \times 3 =$ | |
| 20. | $9^2 - 5^2 =$ | |

**TEST 7B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 5 min

Bridge from Level 2 to Level 3 (Nested Two-Branch Expression Tre)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $14 + 3 \times 4 =$ | |
| 2. | $8^2 - 3^2 =$ | |
| 3. | $18 + 5 \times 6 =$ | |
| 4. | $10^2 - 5^2 =$ | |
| 5. | $22 + 7 \times 2 =$ | |
| 6. | $12^2 - 3^2 =$ | |
| 7. | $26 + 4 \times 4 =$ | |
| 8. | $14^2 - 5^2 =$ | |
| 9. | $30 + 6 \times 6 =$ | |
| 10. | $8^2 - 3^2 =$ | |
| 11. | $34 + 3 \times 2 =$ | |
| 12. | $10^2 - 5^2 =$ | |
| 13. | $38 + 5 \times 4 =$ | |
| 14. | $12^2 - 3^2 =$ | |
| 15. | $42 + 7 \times 6 =$ | |
| 16. | $14^2 - 5^2 =$ | |
| 17. | $46 + 4 \times 2 =$ | |
| 18. | $8^2 - 3^2 =$ | |
| 19. | $50 + 6 \times 4 =$ | |
| 20. | $10^2 - 5^2 =$ | |

# Section 7: Primes, GCF/LCM & Nested Expressions

Review Worked Examples 1 & 2 below, then solve Test 7C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 7C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Counting Positive Divisors: How many positive factors does 36 have? → Prime factor $36 = 2^2 \times 3^2$, then add 1 to each exponent: $(2+1)(2+1) = 9$!

**EXAMPLE 2** 7D #2 (GCF×LCM of 15,20) → Identity $a \times b = 15 \times 20 = 300$

**TEST 7C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Nested Two-Branch Expression Trees

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(8 + 2) \times (9 - 4) =$ _____
2. $((14 - 6) \times 3) + (24 \div 4) =$ _____
3. $(30 - (4 \times 5)) \times (12 \div (2 + 4)) =$ _____
4. $((5 + 3) \times (8 - 2)) \div (6 - 2) =$ _____
5. $((18 \div 3) + 4) \times ((15 - 7) \div 2) =$ _____
6. $(10 + 5 \times 2) \div (12 - 2 \times 4) =$ _____
7. $(25 - 5 \times 3) \times (4 + 6 \div 2) =$ _____
8. $((7 + 3) \times 4 - 8) \div 4 =$ _____
9. $(48 \div (2 + 6)) \times (15 - 3 \times 3) =$ _____
10. $(16 - (4 + 2)) \times (18 \div (3 \times 2)) =$ _____
11. $(3^2 + 1) \times (4^2 - 11) =$ _____
12. $(5^2 - 5) \div (2^3 - 4) =$ _____
13. $((20 - 5) \div 3) \times (12 - 4) =$ _____
14. $(36 \div (3 \times 3)) + (5 \times 4) =$ _____
15. $(50 - 2 \times 15) \div (3 + 2) =$ _____
16. $((9 + 3) \times 5) \div (10 - 4) =$ _____
17. $(2^4 + 4) \times (3^2 - 6) =$ _____
18. $(100 - 6^2) \div (2^3) =$ _____
19. $(7^2 - 9) \div (2^2 + 4) =$ _____
20. $((12 + 8) \div 4)^3 =$ _____

**TEST 7C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Nested Two-Branch Expression Trees — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(7 + 3) \times (10 - 2) =$ _____
2. $(16 + 8) \div 4 =$ _____
3. $(9 + 5) \times (12 - 4) =$ _____
4. $(36 + 12) \div 6 =$ _____
5. $(11 + 7) \times (14 - 6) =$ _____
6. $(12 + 8) \div 4 =$ _____
7. $(13 + 3) \times (16 - 3) =$ _____
8. $(30 + 12) \div 6 =$ _____
9. $(15 + 5) \times (11 - 5) =$ _____
10. $(28 + 8) \div 4 =$ _____
11. $(17 + 7) \times (13 - 2) =$ _____
12. $(24 + 12) \div 6 =$ _____
13. $(19 + 3) \times (15 - 4) =$ _____
14. $(24 + 8) \div 4 =$ _____
15. $(21 + 5) \times (10 - 6) =$ _____
16. $(18 + 12) \div 6 =$ _____
17. $(23 + 7) \times (12 - 3) =$ _____
18. $(20 + 8) \div 4 =$ _____
19. $(25 + 3) \times (14 - 5) =$ _____
20. $(42 + 12) \div 6 =$ _____

# Section 7: Primes, GCF/LCM & Nested Expressions

Review Worked Examples 1 & 2 below, then solve Test 7C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 7D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Algebraic Quotient Shortcut: Evaluate $(25^2 - 15^2) \div (25 - 15)$ → Cancel $(25 - 15)$ immediately to leave $25 + 15 = 40$!

**EXAMPLE 2** 7D #9 (Factors of 36) → $36 = 2^2 \times 3^2$ → $(2+1)(2+1) = 9$ factors

**TEST 7C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Number Theory: GCF × LCM, Diviso)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(9 + 3) \times (11 - 2) =$ _____
2. $(20 + 8) \div 4 =$ _____
3. $(11 + 5) \times (13 - 4) =$ _____
4. $(42 + 12) \div 6 =$ _____
5. $(13 + 7) \times (15 - 6) =$ _____
6. $(16 + 8) \div 4 =$ _____
7. $(15 + 3) \times (17 - 3) =$ _____
8. $(36 + 12) \div 6 =$ _____
9. $(17 + 5) \times (12 - 5) =$ _____
10. $(32 + 8) \div 4 =$ _____
11. $(19 + 7) \times (14 - 2) =$ _____
12. $(30 + 12) \div 6 =$ _____
13. $(21 + 3) \times (16 - 4) =$ _____
14. $(28 + 8) \div 4 =$ _____
15. $(23 + 5) \times (11 - 6) =$ _____
16. $(24 + 12) \div 6 =$ _____
17. $(25 + 7) \times (13 - 3) =$ _____
18. $(24 + 8) \div 4 =$ _____
19. $(27 + 3) \times (15 - 5) =$ _____
20. $(48 + 12) \div 6 =$ _____

**TEST 7D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Number Theory: GCF × LCM, Divisor Counts & Deep Trees

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF(12, 18) × LCM(12, 18) = _____
2. GCF(15, 20) × LCM(15, 20) = _____
3. GCF of 72, 108, and 144 = _____
4. LCM of 6, 8, and 12 = _____
5. LCM of 4, 9, and 12 = _____
6. LCM of 5, 6, and 8 = _____
7. Number of positive factors of 12 = _____
8. Number of positive factors of 24 = _____
9. Number of positive factors of 36 = _____
10. Sum of distinct prime factors of 30 = _____
11. Sum of all positive factors of 12 = _____
12. $(6^2 - 4^2) \div (6 - 4) =$ _____
13. $(15^2 - 5^2) \div (15 + 5) =$ _____
14. $(25^2 - 15^2) \div (25 - 15) =$ _____
15. $((35 - 7) \times 4) \div (14 + 2) =$ _____
16. $(2^5 - 2^3) \div (3^2 - 3) =$ _____
17. $(3^4 - 1) \div (4^2 + 0) =$ _____
18. $(10^3 - 200) \div (4^2 \times 5) =$ _____
19. $((8^2 - 4) \div 15) \times (5^2 - 10) =$ _____
20. $(12^2 + 5^2) \div 13 =$ _____

# Section 7: Primes, GCF/LCM & Nested Expressions

Review Worked Examples 1 & 2 below, then solve Test 7D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 7D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 7D #14 $((25^2\text{-}15^2)\div(25\text{-}15)) \rightarrow (a^2 - b^2)/(a - b) = a + b = 25 + 15 = 40$

**EXAMPLE 2** 7D #20 $((12^2+5^2)\div13) \rightarrow$ Pythagorean $5^2 + 12^2 = 13^2$, so $13^2 \div 13 = 13$

**TEST 7D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Number Theory: GCF × LCM, Divisor Counts & Deep Trees — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF(13, 14) × LCM(13, 14) = _____
2. GCF(15, 17) × LCM(15, 17) = _____
3. GCF(17, 20) × LCM(17, 20) = _____
4. GCF(19, 23) × LCM(19, 23) = _____
5. GCF(21, 26) × LCM(21, 26) = _____
6. GCF(23, 29) × LCM(23, 29) = _____
7. GCF(25, 32) × LCM(25, 32) = _____
8. $(31^2 - 5^2) \div (31 - 5) =$ _____
9. $(33^2 - 6^2) \div (33 - 6) =$ _____
10. $(35^2 - 7^2) \div (35 - 7) =$ _____
11. $(37^2 - 8^2) \div (37 - 8) =$ _____
12. $(39^2 - 9^2) \div (39 - 9) =$ _____
13. $(41^2 - 4^2) \div (41 - 4) =$ _____
14. $(43^2 - 5^2) \div (43 - 5) =$ _____
15. Number of positive factors of 24 = _____
16. Number of positive factors of 144 = _____
17. Number of positive factors of 864 = _____
18. Number of positive factors of 192 = _____
19. Number of positive factors of 1152 = _____
20. Number of positive factors of 6912 = _____

**TEST 7D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 7 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF(16, 16) × LCM(16, 16) = _____
2. GCF(18, 19) × LCM(18, 19) = _____
3. GCF(20, 22) × LCM(20, 22) = _____
4. GCF(22, 25) × LCM(22, 25) = _____
5. GCF(24, 28) × LCM(24, 28) = _____
6. GCF(26, 31) × LCM(26, 31) = _____
7. GCF(28, 34) × LCM(28, 34) = _____
8. $(34^2 - 5^2) \div (34 - 5) =$ _____
9. $(36^2 - 6^2) \div (36 - 6) =$ _____
10. $(38^2 - 7^2) \div (38 - 7) =$ _____
11. $(40^2 - 8^2) \div (40 - 8) =$ _____
12. $(42^2 - 9^2) \div (42 - 9) =$ _____
13. $(44^2 - 4^2) \div (44 - 4) =$ _____
14. $(46^2 - 5^2) \div (46 - 5) =$ _____
15. Number of positive factors of 144 = _____
16. Number of positive factors of 864 = _____
17. Number of positive factors of 192 = _____
18. Number of positive factors of 1152 = _____
19. Number of positive factors of 6912 = _____
20. Number of positive factors of 1536 = _____

# Section 8: Algebra, Quadratics & Coordinate Geometry

Linear equations, slope, Pythagorean distance, and quadratic roots. • Study the Champion Way & complete W1–W6 before starting Tests 8A1–8D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 8: ALGEBRA, QUADRATICS & COORDINATE GEOMETRY

**RULE 1** — **Rule 1 — Vieta's Root Formulas (Zero Factoring Required!): For $ax^2 + bx + c = 0$, the Sum of Roots is $-b/a$ and the Product of Roots is $c/a$. Never solve for individual roots first!**

**RULE 2** — Rule 2 — Horner's Nested Form for Polynomials: Evaluate $ax^2 + bx + c$ as $(ax + b)x + c$ so you only type x without squaring separately (or use [POLY] on SC-32 for instant roots).

**RULE 3** — Rule 3 — Pythagorean Triples & 3D Space Diagonals: Memorize (3,4,5), (5,12,13), (8,15,17), (7,24,25), (9,40,41) and 3D triples (3,4,12 → 13), (2,3,6 → 7), (1,4,8 → 9).

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Vieta's Sum & Product of Roots: For $2x^2 - 10x + 7 = 0$, find Sum of Roots** Stack Keys: 1 0 ENTER 2 ÷

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Apply the quadratic formula $(-b \pm \sqrt{b^2 - 4ac}) / 2a$ and add the two radicals.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Sum of roots $= -b / a = -(-10) / 2 = 5$ in 1 second!**

**Archetype 2 — 2×2 Linear System by Elimination: Solve $x + y = 10$ and $x - y = 4$** Stack Keys: 1 0 ENTER 4 + 2 ÷

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Substitute $y = 10 - x$ into $x - (10 - x) = 4$ and solve.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: For $x + y = S$ and $x - y = D$, x is the average $(S + D) \div 2 = 14 \div 2 = 7$, and y is $(S - D) \div 2 = 3$!**

**Archetype 3 — 3D Box Diagonal Pipeline: Evaluate $\sqrt{3^2 + 4^2 + 12^2}$** Stack Keys: 3 ↶ x² 4 ↶ x² + 1 2 ↶ x² + √x

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute $9 + 16 + 144 = 169$ on paper and take $\sqrt{169}$.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Spot chained triples (3,4 → 5, then 5,12 → 13), or key [3][$x^2$][4][$x^2$][+][12][$x^2$][+][$\sqrt{x}$]!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** **Sum of roots of $x^2 - 9x + 20 = 0$ :** _____

Shortcut Cue: Vieta's rule: $-b / a = 9 / 1$

**W2.** **Product of roots of $x^2 + 11x + 24 = 0$ :** _____

Shortcut Cue: Vieta's rule: $c / a = 24 / 1$

**W3.** **Vertex $x$-coord of $y = x^2 - 8x + 19$ :** _____

Shortcut Cue: Axis of symmetry $x = -b / (2a) = 8 / 2$

**W4.** **Solve for x: $x + y = 12$, $x - y = 6$ :** _____

Shortcut Cue: Average $(12 + 6) \div 2$

**W5.** **Hypotenuse for legs 7 and 24 =** _____

Shortcut Cue: Pythagorean triple (7, 24, 25)

**W6.** **3D diagonal $\sqrt{2^2 + 3^2 + 6^2}$ =** _____

Shortcut Cue: $\sqrt{4 + 9 + 36} = \sqrt{49}$

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 8: Algebra, Quadratics & Coordinate Geometry

Review Worked Examples 1 & 2 below, then solve Test 8A1 (Level 1: Foundation (Sprint 1)) & Test 8A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** Sum of roots of $x^2 - 9x + 20 = 0$ : 9 (Vieta's rule: $-b / a = 9 / 1$)

**EXAMPLE 2** Product of roots of $x^2 + 11x + 24 = 0$ : 24 (Vieta's rule: $c / a = 24 / 1$)

## TEST 8A1 • LEVEL 1: FOUNDATION (SPRINT 1) Goal: 4 min

Evaluating Expressions & Solving Linear Equations

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $3x + 5$ when $x = 4$ : _____
2. $4x - 7$ when $x = 5$ : _____
3. $2x^2$ when $x = 3$ : _____
4. $x^2 + 4x$ when $x = 2$ : _____
5. $5a - 2b$ when $a = 4$, $b = 3$ : _____
6. Solve: $x + 9 = 21$ _____
7. Solve: $x - 8 = 15$ _____
8. Solve: $4x = 36$ _____
9. Solve: $x \div 5 = 7$ _____
10. Solve: $2x + 3 = 15$ _____
11. Solve: $3x - 4 = 14$ _____
12. Solve: $5x + 10 = 45$ _____
13. Solve: $4x - 9 = 11$ _____
14. Solve: $6x + 2 = 32$ _____
15. Solve: $(x \div 3) + 2 = 6$ _____
16. Evaluate $-8 + 15 =$ _____
17. Evaluate $6 - 14 =$ _____
18. Evaluate $(-4) \times (-6) =$ _____
19. Evaluate $(-36) \div 9 =$ _____
20. Evaluate $(-5)^2 =$ _____

## TEST 8A2 • LEVEL 1: FOUNDATION (SPRINT 2) Goal: 4 min

Evaluating Expressions & Solving Linear Equations — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Solve: $3x + 4 = 16$ _____
2. Evaluate $4x - 5$ when $x = 5$ : _____
3. Solve: $5x + 6 = 36$ _____
4. Evaluate $6x - 7$ when $x = 7$ : _____
5. Solve: $7x + 8 = 64$ _____
6. Evaluate $8x - 9$ when $x = 9$ : _____
7. Solve: $3x + 10 = 40$ _____
8. Evaluate $4x - 11$ when $x = 11$ : _____
9. Solve: $5x + 12 = 32$ _____
10. Evaluate $6x - 4$ when $x = 5$ : _____
11. Solve: $7x + 5 = 47$ _____
12. Evaluate $8x - 6$ when $x = 7$ : _____
13. Solve: $3x + 7 = 31$ _____
14. Evaluate $4x - 8$ when $x = 9$ : _____
15. Solve: $5x + 9 = 59$ _____
16. Evaluate $6x - 10$ when $x = 11$ : _____
17. Solve: $7x + 11 = 39$ _____
18. Evaluate $8x - 12$ when $x = 5$ : _____
19. Solve: $3x + 4 = 22$ _____
20. Evaluate $4x - 5$ when $x = 7$ : _____

# Section 8: Algebra, Quadratics & Coordinate Geometry

Review Worked Examples 1 & 2 below, then solve Test 8A3 (Level 1: Foundation (Sprint 3)) & Test 8B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** Vertex $x$-coord of $y = x^2 - 8x + 19$ : 4 (Axis of symmetry $x = -b / (2a) = 8 / 2$)

**EXAMPLE 2** Solve for x: $x + y = 12$, $x - y = 6$ : 9 (Average $(12 + 6) \div 2$)

### TEST 8A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 4 min

Bridge from Level 1 to Level 2 (Slope, Midpoints & Pythagorean R)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Solve: $4x + 4 = 24$ _____
2. Evaluate $5x - 5$ when $x = 6$ : _____
3. Solve: $6x + 6 = 48$ _____
4. Evaluate $7x - 7$ when $x = 8$ : _____
5. Solve: $8x + 8 = 80$ _____
6. Evaluate $9x - 9$ when $x = 10$ : _____
7. Solve: $4x + 10 = 54$ _____
8. Evaluate $5x - 11$ when $x = 12$ : _____
9. Solve: $6x + 12 = 42$ _____
10. Evaluate $7x - 4$ when $x = 6$ : _____
11. Solve: $8x + 5 = 61$ _____
12. Evaluate $9x - 6$ when $x = 8$ : _____
13. Solve: $4x + 7 = 43$ _____
14. Evaluate $5x - 8$ when $x = 10$ : _____
15. Solve: $6x + 9 = 75$ _____
16. Evaluate $7x - 10$ when $x = 12$ : _____
17. Solve: $8x + 11 = 51$ _____
18. Evaluate $9x - 12$ when $x = 6$ : _____
19. Solve: $4x + 4 = 32$ _____
20. Evaluate $5x - 5$ when $x = 8$ : _____

### TEST 8B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 5 min

Slope, Midpoints & Pythagorean Right Triangles

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Slope from (0, 0) to (4, 12) = _____
2. Slope from (1, 2) to (5, 10) = _____
3. Slope from (2, 3) to (6, 15) = _____
4. Slope from (-1, -2) to (3, 10) = _____
5. Slope from (0, 10) to (5, 0) = _____
6. Slope from (2, 8) to (6, 6) = _____
7. Midpoint $x$-value of (2, 5) & (8, 9) = _____
8. Midpoint $y$-value of (2, 5) & (8, 9) = _____
9. Hypotenuse for legs 3 and 4 = _____
10. Hypotenuse for legs 6 and 8 = _____
11. Hypotenuse for legs 5 and 12 = _____
12. Hypotenuse for legs 8 and 15 = _____
13. Hypotenuse for legs 9 and 12 = _____
14. Hypotenuse for legs 12 and 16 = _____
15. Missing leg: hyp = 10, leg = 6 → _____
16. Missing leg: hyp = 13, leg = 5 → _____
17. Distance from (0, 0) to (3, 4) = _____
18. Distance from (1, 1) to (4, 5) = _____
19. Distance from (3, 4) to (9, 12) = _____
20. Distance from (-3, 5) to (5, -10) = _____

# Section 8: Algebra, Quadratics & Coordinate Geometry

Review Worked Examples 1 & 2 below, then solve Test 8B2 (Level 2: Core Fluency (Sprint 2)) & Test 8B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Vieta's Sum & Product of Roots: For $2x^2 - 10x + 7 = 0$, find Sum of Roots → Sum of roots = $-b / a = -(-10) / 2 = 5$ in 1 second!

**EXAMPLE 2** Hypotenuse for legs 7 and 24 = 25 (Pythagorean triple (7, 24, 25))

**TEST 8B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 5 min

Slope, Midpoints & Pythagorean Right Triangles — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Slope from (2, 2) to (4, 8) = _____
2. Slope from (3, 4) to (6, 16) = _____
3. Slope from (4, 6) to (8, 26) = _____
4. Slope from (5, 8) to (10, 38) = _____
5. Slope from (6, 10) to (8, 14) = _____
6. Slope from (7, 12) to (10, 21) = _____
7. Slope from (8, 14) to (12, 30) = _____
8. Slope from (9, 16) to (14, 41) = _____
9. Slope from (10, 18) to (12, 30) = _____
10. Slope from (11, 20) to (14, 26) = _____
11. Hypotenuse for legs 6 and 8 = _____
12. Hypotenuse for legs 15 and 36 = _____
13. Hypotenuse for legs 8 and 15 = _____
14. Hypotenuse for legs 14 and 48 = _____
15. Hypotenuse for legs 27 and 120 = _____
16. Hypotenuse for legs 3 and 4 = _____
17. Hypotenuse for legs 10 and 24 = _____
18. Hypotenuse for legs 24 and 45 = _____
19. Hypotenuse for legs 7 and 24 = _____
20. Hypotenuse for legs 18 and 80 = _____

**TEST 8B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 5 min

Bridge from Level 2 to Level 3 (Polynomial Evaluation & Quadrati)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Slope from (3, 2) to (5, 10) = _____
2. Slope from (4, 4) to (7, 19) = _____
3. Slope from (5, 6) to (9, 30) = _____
4. Slope from (6, 8) to (11, 18) = _____
5. Slope from (7, 10) to (9, 16) = _____
6. Slope from (8, 12) to (11, 24) = _____
7. Slope from (9, 14) to (13, 34) = _____
8. Slope from (10, 16) to (15, 46) = _____
9. Slope from (11, 18) to (13, 22) = _____
10. Slope from (12, 20) to (15, 29) = _____
11. Hypotenuse for legs 9 and 12 = _____
12. Hypotenuse for legs 5 and 12 = _____
13. Hypotenuse for legs 16 and 30 = _____
14. Hypotenuse for legs 21 and 72 = _____
15. Hypotenuse for legs 9 and 40 = _____
16. Hypotenuse for legs 6 and 8 = _____
17. Hypotenuse for legs 15 and 36 = _____
18. Hypotenuse for legs 8 and 15 = _____
19. Hypotenuse for legs 14 and 48 = _____
20. Hypotenuse for legs 27 and 120 = _____

# Section 8: Algebra, Quadratics & Coordinate Geometry

Review Worked Examples 1 & 2 below, then solve Test 8C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 8C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — 2×2 Linear System by Elimination: Solve $x + y = 10$ and $x - y = 4$ → For $x + y = S$ and $x - y = D$, x is the average $(S + D) \div 2 = 14 \div 2 = 7$, and y is $(S - D) \div 2 = 3$!

**EXAMPLE 2** 8C #5 (Roots of $x^2+5x+6=0$) → [1] [ENTER] [5] [ENTER] [6] [POLY] → -2, -3

**TEST 8C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Polynomial Evaluation & Quadratic Factoring

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Evaluate $x^2 + 5x + 6$ at $x = 2$ : _____
2. Evaluate $2x^2 + 4x - 5$ at $x = 3$ : _____
3. Evaluate $x^2 - 6x + 9$ at $x = 5$ : _____
4. Evaluate $3x^2 - 2x + 1$ at $x = -2$ : _____
5. Greater root of $x^2 + 5x + 6 = 0$ : _____
6. Lesser root of $x^2 + 5x + 6 = 0$ : _____
7. Greater root of $x^2 + 7x + 12 = 0$ : _____
8. Greater root of $x^2 + 8x + 15 = 0$ : _____
9. Positive root of $x^2 - x - 6 = 0$ : _____
10. Lesser root of $x^2 - 5x + 6 = 0$ : _____
11. Greater root of $x^2 - 5x + 6 = 0$ : _____
12. Positive root of $x^2 - 7x + 10 = 0$ (max) : _____
13. Positive root of $x^2 - 9 = 0$ : _____
14. Positive root of $x^2 - 16 = 0$ : _____
15. Positive root of $x^2 - 25 = 0$ : _____
16. Repeated root of $x^2 + 10x + 25 = 0$ : _____
17. Repeated root of $x^2 - 8x + 16 = 0$ : _____
18. Greater root of $2x^2 + 5x + 2 = 0$ : _____
19. Positive root of $x^2 + 2x - 15 = 0$ : _____
20. Positive root of $x^2 - 3x - 10 = 0$ : _____

**TEST 8C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Polynomial Evaluation & Quadratic Factoring — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Greater root of $x^2 - 5x + 6 = 0$ : _____
2. Lesser root of $x^2 - 8x + 15 = 0$ : _____
3. Greater root of $x^2 - 11x + 28 = 0$ : _____
4. Lesser root of $x^2 - 14x + 45 = 0$ : _____
5. Greater root of $x^2 - 13x + 42 = 0$ : _____
6. Lesser root of $x^2 - 16x + 63 = 0$ : _____
7. Greater root of $x^2 - 7x + 10 = 0$ : _____
8. Lesser root of $x^2 - 10x + 21 = 0$ : _____
9. Greater root of $x^2 - 9x + 20 = 0$ : _____
10. Lesser root of $x^2 - 12x + 35 = 0$ : _____
11. Greater root of $x^2 - 15x + 54 = 0$ : _____
12. Lesser root of $x^2 - 18x + 77 = 0$ : _____
13. Greater root of $x^2 - 5x + 6 = 0$ : _____
14. Lesser root of $x^2 - 8x + 15 = 0$ : _____
15. Greater root of $x^2 - 11x + 28 = 0$ : _____
16. Lesser root of $x^2 - 14x + 45 = 0$ : _____
17. Greater root of $x^2 - 13x + 42 = 0$ : _____
18. Lesser root of $x^2 - 16x + 63 = 0$ : _____
19. Greater root of $x^2 - 7x + 10 = 0$ : _____
20. Lesser root of $x^2 - 10x + 21 = 0$ : _____

# Section 8: Algebra, Quadratics & Coordinate Geometry

Review Worked Examples 1 & 2 below, then solve Test 8C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 8D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — 3D Box Diagonal Pipeline: Evaluate $\sqrt{3^2 + 4^2 + 12^2} \rightarrow$ Spot chained triples (3,4 → 5, then 5,12 → 13), or key [3][$x^2$][4][$x^2$][+][12][$x^2$][+][$\sqrt{x}$]!

**EXAMPLE 2** 8D #5 (Sum of roots $2x^2$-$10x$+7) → Vieta's $-b/a = 10 / 2 = 5$

**TEST 8C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Vieta's Root Sums/Products, Line)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Greater root of $x^2 - 7x + 12 = 0$ : _____
2. Lesser root of $x^2 - 10x + 24 = 0$ : _____
3. Greater root of $x^2 - 13x + 40 = 0$ : _____
4. Lesser root of $x^2 - 16x + 60 = 0$ : _____
5. Greater root of $x^2 - 15x + 56 = 0$ : _____
6. Lesser root of $x^2 - 18x + 80 = 0$ : _____
7. Greater root of $x^2 - 9x + 18 = 0$ : _____
8. Lesser root of $x^2 - 12x + 32 = 0$ : _____
9. Greater root of $x^2 - 11x + 30 = 0$ : _____
10. Lesser root of $x^2 - 14x + 48 = 0$ : _____
11. Greater root of $x^2 - 17x + 70 = 0$ : _____
12. Lesser root of $x^2 - 20x + 96 = 0$ : _____
13. Greater root of $x^2 - 7x + 12 = 0$ : _____
14. Lesser root of $x^2 - 10x + 24 = 0$ : _____
15. Greater root of $x^2 - 13x + 40 = 0$ : _____
16. Lesser root of $x^2 - 16x + 60 = 0$ : _____
17. Greater root of $x^2 - 15x + 56 = 0$ : _____
18. Lesser root of $x^2 - 18x + 80 = 0$ : _____
19. Greater root of $x^2 - 9x + 18 = 0$ : _____
20. Lesser root of $x^2 - 12x + 32 = 0$ : _____

**TEST 8D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Vieta's Root Sums/Products, Linear Systems & 3D Distance

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Sum of roots of $x^2 - 9x + 20 = 0$ : _____
2. Product of roots of $x^2 - 9x + 20 = 0$ : _____
3. Sum of roots of $x^2 + 11x + 24 = 0$ : _____
4. Product of roots of $x^2 + 11x + 24 = 0$ : _____
5. Sum of roots of $2x^2 - 10x + 7 = 0$ : _____
6. Product of roots of $2x^2 - 10x + 7 = 0$ : _____
7. Discriminant $b^2 - 4ac$ of $x^2 + 6x + 5$ : _____
8. Discriminant $b^2 - 4ac$ of $x^2 - 4x + 4$ : _____
9. Vertex $x$-coord of $y = x^2 - 8x + 19$ : _____
10. Minimum value of $y = x^2 - 8x + 19$ : _____
11. Solve for x: $x + y = 10$, $x - y = 4$ : _____
12. Solve for y: $x + y = 10$, $x - y = 4$ : _____
13. Solve for x: $2x + y = 13$, $x + y = 8$ : _____
14. Solve for x: $3x - 4 = x + 10$ : _____
15. Solve for x: $5x + 2 = 2x + 17$ : _____
16. Hypotenuse for legs 7 and 24 = _____
17. Hypotenuse for legs 9 and 40 = _____
18. 3D diagonal $\sqrt{3^2 + 4^2 + 12^2}$ = _____
19. 3D diagonal $\sqrt{2^2 + 3^2 + 6^2}$ = _____
20. 3D diagonal $\sqrt{1^2 + 4^2 + 8^2}$ = _____

# Section 8: Algebra, Quadratics & Coordinate Geometry

Review Worked Examples 1 & 2 below, then solve Test 8D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 8D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 8D #10 (Min of $x^2$-8$x$+19) → Complete square: $(x - 4)^2 + 3$ → min is 3

**EXAMPLE 2** 8D #18 ($\sqrt{3^2+4^2+12^2}$) → Chained (3,4,5) + (5,12,13) → 13

**TEST 8D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Vieta's Root Sums/Products, Linear Systems & 3D Distance — Contest Pacing

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. Sum of roots of $x^2 - 10x + 12 = 0$ : _____
2. Product of roots of $x^2 - 12x + 15 = 0$ : _____
3. Sum of roots of $x^2 - 14x + 18 = 0$ : _____
4. Product of roots of $x^2 - 16x + 21 = 0$ : _____
5. Sum of roots of $x^2 - 18x + 24 = 0$ : _____
6. Product of roots of $x^2 - 20x + 27 = 0$ : _____
7. Sum of roots of $x^2 - 22x + 30 = 0$ : _____
8. Product of roots of $x^2 - 24x + 33 = 0$ : _____
9. Solve for x: $x + y = 10$, $x - y = 4$ : _____
10. Solve for x: $x + y = 13$, $x - y = 5$ : _____
11. Solve for x: $x + y = 16$, $x - y = 6$ : _____
12. Solve for x: $x + y = 19$, $x - y = 7$ : _____
13. Solve for x: $x + y = 22$, $x - y = 8$ : _____
14. Solve for x: $x + y = 25$, $x - y = 9$ : _____
15. Minimum value of $y = x^2 - 8x + 19$ : _____
16. Minimum value of $y = x^2 - 10x + 30$ : _____
17. Minimum value of $y = x^2 - 12x + 43$ : _____
18. Minimum value of $y = x^2 - 14x + 58$ : _____
19. Minimum value of $y = x^2 - 16x + 75$ : _____
20. Minimum value of $y = x^2 - 18x + 94$ : _____

**TEST 8D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 8 Cumulative Tournament Sprint

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. Sum of roots of $x^2 - 13x + 14 = 0$ : _____
2. Product of roots of $x^2 - 15x + 17 = 0$ : _____
3. Sum of roots of $x^2 - 17x + 20 = 0$ : _____
4. Product of roots of $x^2 - 19x + 23 = 0$ : _____
5. Sum of roots of $x^2 - 21x + 26 = 0$ : _____
6. Product of roots of $x^2 - 23x + 29 = 0$ : _____
7. Sum of roots of $x^2 - 25x + 32 = 0$ : _____
8. Product of roots of $x^2 - 27x + 35 = 0$ : _____
9. Solve for x: $x + y = 12$, $x - y = 4$ : _____
10. Solve for x: $x + y = 15$, $x - y = 5$ : _____
11. Solve for x: $x + y = 18$, $x - y = 6$ : _____
12. Solve for x: $x + y = 21$, $x - y = 7$ : _____
13. Solve for x: $x + y = 24$, $x - y = 8$ : _____
14. Solve for x: $x + y = 27$, $x - y = 9$ : _____
15. Minimum value of $y = x^2 - 10x + 29$ : _____
16. Minimum value of $y = x^2 - 12x + 42$ : _____
17. Minimum value of $y = x^2 - 14x + 57$ : _____
18. Minimum value of $y = x^2 - 16x + 74$ : _____
19. Minimum value of $y = x^2 - 18x + 93$ : _____
20. Minimum value of $y = x^2 - 20x + 114$ : _____

# Cumulative Review C2: Intermediate Math (Sections 1–8)

Comprehensive review across fractions, expressions, number theory, and algebra. • Study the Champion Way & complete W1–W6 before starting Tests C2A1–C2D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — CUMULATIVE REVIEW C2: INTERMEDIATE MATH (SECTIONS 1–8)

**RULE 1** **Rule 1 — Match the Fastest Tool to the Problem: Use mental identities for ($a^2$ - $b^2$), Vieta's formulas, and GCF×LCM, and use inside-out RPN chains for multi-branch arithmetic.**

**RULE 2** Rule 2 — Clear Fractions Before Solving Equations: Multiply or distribute the LCD first to turn fraction expressions into 1-step integer subtraction.

**RULE 3** Rule 3 — Never Re-Type Intermediate Results: Keep intermediate values on Line Y (Waiting Shelf) while computing the second branch on Line X.

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Euclidean Distance Between Points: Find distance from (-3, 5) to (5, -10)** Stack Keys: [1] [5] [ENTER] [8] [↶] [▸θ,r]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Write $d = \sqrt{(5 - (-3))^2 + (-10 - 5)^2}$ on paper and compute step-by-step.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Find $dy$ = -15 and $dx$ = 8 mentally, then spot the (8, 15, 17) triple (or press [15] [ENTER] [8] [→$\vartheta$,$r$] → 17)!**

**Archetype 2 — Parallel Fraction Reciprocal: Evaluate 1 ÷ (1/4 + 1/6)** Stack Keys: [4] [$x^{-1}$] [6] [$x^{-1}$] [+] [$x^{-1}$]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Common denominator 12 → 5/12 → 12/5 = 2.4.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Key [4] [1/$x$] [6] [1/$x$] [+] [1/$x$] → 2.4 in 6 keys!**

**Archetype 3 — Continued Fraction Evaluation: Evaluate 1 + 1/(1 + 1/2)** Stack Keys: [1] [ENTER] [•] [5] [+] [$x^{-1}$] [1] [+]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Convert layers on paper.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Start at bottom 1 + 1/2 = 3/2, invert to 2/3, add 1 → 5/3!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** **$3/4 + (1/2 \times 1/4)$ (decimal) =** _____

Shortcut Cue: 0.75 + 0.125

**W2.** **$(2/3 + 3/4) \times 12 =$** _____

Shortcut Cue: Distribute 12: 8 + 9

**W3.** **Distance from (3, 4) to (9, 12) =** _____

Shortcut Cue: $dx = 6$, $dy = 8$ → (6, 8, 10) triple

**W4.** **Sum of roots of $2x^2 - 14x + 9 = 0$ :** _____

Shortcut Cue: Vieta's $-b$ / $a$ = 14 / 2

**W5.** **Minimum value of $y = x^2 - 6x + 14$ :** _____

Shortcut Cue: $(x - 3)^2 + 5$ → min is 5

**W6.** **$2/3 \times 3/4 \times 4/5 \times 5/6 =$** _____

Shortcut Cue: Telescoping cancel → 2/6 = 1/3

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Cumulative Review C2: Intermediate Math (Sections 1–8)

Review Worked Examples 1 & 2 below, then solve Test C2A1 (Level 1: Foundation (Sprint 1)) & Test C2A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $3/4 + (1/2 \times 1/4)$ (decimal) $= 0.875$ $(0.75 + 0.125)$

**EXAMPLE 2** $(2/3 + 3/4) \times 12 = 17$ (Distribute 12: $8 + 9$)

**TEST C2A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 4 min

Multi-Step Arithmetic, Fractions & Basic Algebra

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $18 + 18 - 9 =$ _____
2. $100 - 36 =$ _____
3. $(18 - 6) \times 3 =$ _____
4. $(24 + 16) \div 8 =$ _____
5. $1/2 + 1/4$ (decimal) $=$ _____
6. $3/4 - 1/2$ (decimal) $=$ _____
7. $2/3$ of $18 =$ _____
8. $3/4$ of $24 =$ _____
9. $2^3 \times 3 =$ _____
10. GCF of 12 and 18 = _____
11. LCM of 4 and 6 = _____
12. $10 + 5 \times 2 =$ _____
13. $30 - 4 \times 5 =$ _____
14. Solve: $x + 12 = 30$ _____
15. Solve: $3x = 42$ _____
16. Solve: $2x + 5 = 19$ _____
17. Evaluate $3x - 4$ at $x = 6$ : _____
18. Hypotenuse for legs 3 and 4 = _____
19. Hypotenuse for legs 6 and 8 = _____
20. Slope from (0, 0) to (3, 12) = _____

**TEST C2A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 4 min

Multi-Step Arithmetic, Fractions & Basic Algebra — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF of 6 and 12 = _____
2. LCM of 12 and 20 = _____
3. $4^2 + 1 =$ _____
4. GCF of 12 and 18 = _____
5. LCM of 21 and 28 = _____
6. $3^2 + 1 =$ _____
7. GCF of 18 and 54 = _____
8. LCM of 9 and 9 = _____
9. $2^2 + 1 =$ _____
10. GCF of 10 and 25 = _____
11. LCM of 18 and 36 = _____
12. $5^2 + 1 =$ _____
13. GCF of 16 and 32 = _____
14. LCM of 27 and 45 = _____
15. $4^2 + 1 =$ _____
16. GCF of 8 and 12 = _____
17. LCM of 15 and 20 = _____
18. $3^2 + 1 =$ _____
19. GCF of 14 and 42 = _____
20. LCM of 24 and 24 = _____

# Cumulative Review C2: Intermediate Math (Sections 1–8)

Review Worked Examples 1 & 2 below, then solve Test C2A3 (Level 1: Foundation (Sprint 3)) & Test C2B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** Distance from (3, 4) to (9, 12) = 10 ($dx = 6$, $dy = 8 \rightarrow$ (6, 8, 10) triple)

**EXAMPLE 2** Sum of roots of $2x^2 - 14x + 9 = 0$ : 7 (Vieta's $-b$ / $a$ = 14 / 2)

## TEST C2A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 4 min

Bridge from Level 1 to Level 2 (Rational Operations, Expression )

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF of 8 and 16 = _____
2. LCM of 15 and 25 = _____
3. $4^3 + 2$ = _____
4. GCF of 14 and 21 = _____
5. LCM of 24 and 32 = _____
6. $3^3 + 2$ = _____
7. GCF of 20 and 60 = _____
8. LCM of 12 and 12 = _____
9. $2^3 + 2$ = _____
10. GCF of 12 and 30 = _____
11. LCM of 21 and 42 = _____
12. $5^3 + 2$ = _____
13. GCF of 18 and 36 = _____
14. LCM of 30 and 50 = _____
15. $4^3 + 2$ = _____
16. GCF of 10 and 15 = _____
17. LCM of 18 and 24 = _____
18. $3^3 + 2$ = _____
19. GCF of 16 and 48 = _____
20. LCM of 27 and 27 = _____

## TEST C2B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 5 min

Rational Operations, Expression Trees & Coordinate Slope

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 3 1/2 − 1 3/4 (decimal) = _____
2. 2 1/2 + 1 1/4 (decimal) = _____
3. Reciprocal of 4/5 (decimal) = _____
4. 3/4 ÷ 1/2 (decimal) = _____
5. ((32 − 14) × 5) ÷ 6 = _____
6. ((15 + 5) × (8 − 3)) ÷ 10 = _____
7. 80 − (15 × 4) + 12 = _____
8. (8 + 2) × (9 − 4) = _____
9. GCF of 24 and 36 = _____
10. LCM of 8 and 12 = _____
11. Slope from (-1, -2) to (3, 10) = _____
12. Distance from (0, 0) to (12, 16) = _____
13. Distance from (3, 4) to (9, 12) = _____
14. Positive root of $x^2 - 9 = 0$ : _____
15. Positive root of $x^2 - 16 = 0$ : _____
16. Greater root of $x^2 + 5x + 6 = 0$ : _____
17. Lesser root of $x^2 - 5x + 6 = 0$ : _____
18. Evaluate $x^2 + 3x$ at $x = 4$ : _____
19. $15^2$ = _____
20. $25^2$ = _____

# Cumulative Review C2: Intermediate Math (Sections 1–8)

Review Worked Examples 1 & 2 below, then solve Test C2B2 (Level 2: Core Fluency (Sprint 2)) & Test C2B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Euclidean Distance Between Points: Find distance from (-3, 5) to (5, -10) → Find $dy = -15$ and $dx = 8$ mentally, then spot the (8, 15, 17) triple (or press [15] [ENTER] [8] [→ $\vartheta$,$r$] → 17)!

**EXAMPLE 2** Minimum value of $y = x^2 - 6x + 14$ : 5 $((x - 3)^2 + 5 \rightarrow$ min is 5)

**TEST C2B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 5 min

Rational Operations, Expression Trees & Coordinate Slope — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $11 + 3 \times 3 =$ | |
| 2. | $7^2 - 3^2 =$ | |
| 3. | $15 + 5 \times 5 =$ | |
| 4. | $9^2 - 5^2 =$ | |
| 5. | $19 + 7 \times 7 =$ | |
| 6. | $11^2 - 3^2 =$ | |
| 7. | $23 + 4 \times 3 =$ | |
| 8. | $13^2 - 5^2 =$ | |
| 9. | $27 + 6 \times 5 =$ | |
| 10. | $7^2 - 3^2 =$ | |
| 11. | $31 + 3 \times 7 =$ | |
| 12. | $9^2 - 5^2 =$ | |
| 13. | $35 + 5 \times 3 =$ | |
| 14. | $11^2 - 3^2 =$ | |
| 15. | $39 + 7 \times 5 =$ | |
| 16. | $13^2 - 5^2 =$ | |
| 17. | $43 + 4 \times 7 =$ | |
| 18. | $7^2 - 3^2 =$ | |
| 19. | $47 + 6 \times 3 =$ | |
| 20. | $9^2 - 5^2 =$ | |

**TEST C2B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 5 min

Bridge from Level 2 to Level 3 (Complex Fractions, Quadratics & )

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $14 + 3 \times 4 =$ | |
| 2. | $8^2 - 3^2 =$ | |
| 3. | $18 + 5 \times 6 =$ | |
| 4. | $10^2 - 5^2 =$ | |
| 5. | $22 + 7 \times 2 =$ | |
| 6. | $12^2 - 3^2 =$ | |
| 7. | $26 + 4 \times 4 =$ | |
| 8. | $14^2 - 5^2 =$ | |
| 9. | $30 + 6 \times 6 =$ | |
| 10. | $8^2 - 3^2 =$ | |
| 11. | $34 + 3 \times 2 =$ | |
| 12. | $10^2 - 5^2 =$ | |
| 13. | $38 + 5 \times 4 =$ | |
| 14. | $12^2 - 3^2 =$ | |
| 15. | $42 + 7 \times 6 =$ | |
| 16. | $14^2 - 5^2 =$ | |
| 17. | $46 + 4 \times 2 =$ | |
| 18. | $8^2 - 3^2 =$ | |
| 19. | $50 + 6 \times 4 =$ | |
| 20. | $10^2 - 5^2 =$ | |

# Cumulative Review C2: Intermediate Math (Sections 1–8)

Review Worked Examples 1 & 2 below, then solve Test C2C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test C2C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Parallel Fraction Reciprocal: Evaluate $1 \div (1/4 + 1/6)$ → Key [4] [1/x] [6] [1/x] [+] [1/x] → 2.4 in 6 keys!

**EXAMPLE 2** C2C #2 $(1 \div (1/4 + 1/6))$ → [4] [1/x] [6] [1/x] [+] [1/x] → 2.4

**TEST C2C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Complex Fractions, Quadratics & Multi-Branch Evaluation

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $3/4 + (1/2 \times 1/4)$ (decimal) = _____
2. $1 \div (1/4 + 1/6)$ (decimal) = _____
3. $(1/3 + 1/4) \times 24 =$ _____
4. $(2/3 + 3/4) \times 12 =$ _____
5. $((5 + 3) \times (8 - 2)) \div (6 - 2) =$ _____
6. $((18 \div 3) + 4) \times ((15 - 7) \div 2) =$ _____
7. $(25 - 5 \times 3) \times (4 + 6 \div 2) =$ _____
8. Evaluate $2x^2 + 4x - 5$ at $x = 3$ : _____
9. Positive root of $x^2 - x - 6 = 0$ : _____
10. Greater root of $2x^2 + 5x + 2 = 0$ : _____
11. Repeated root of $x^2 + 10x + 25 = 0$ : _____
12. Sum of roots of $x^2 - 7x + 12 = 0$ : _____
13. Product of roots of $x^2 - 7x + 12 = 0$ : _____
14. Solve: $4x - 5 = 2x + 9$ _____
15. Missing leg: hyp = 13, leg = 5 → _____
16. Missing leg: hyp = 17, leg = 8 → _____
17. Number of positive factors of 24 = _____
18. LCM of 6, 8, and 12 = _____
19. $19 \times 21 =$ _____
20. $24 \times 26 =$ _____

**TEST C2C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Complex Fractions, Quadratics & Multi-Branch Evaluation — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(7 + 3) \times (10 - 2) =$ _____
2. $(16 + 8) \div 4 =$ _____
3. $(9 + 5) \times (12 - 4) =$ _____
4. $(36 + 12) \div 6 =$ _____
5. $(11 + 7) \times (14 - 6) =$ _____
6. $(12 + 8) \div 4 =$ _____
7. $(13 + 3) \times (16 - 3) =$ _____
8. $(30 + 12) \div 6 =$ _____
9. $(15 + 5) \times (11 - 5) =$ _____
10. $(28 + 8) \div 4 =$ _____
11. $(17 + 7) \times (13 - 2) =$ _____
12. $(24 + 12) \div 6 =$ _____
13. $(19 + 3) \times (15 - 4) =$ _____
14. $(24 + 8) \div 4 =$ _____
15. $(21 + 5) \times (10 - 6) =$ _____
16. $(18 + 12) \div 6 =$ _____
17. $(23 + 7) \times (12 - 3) =$ _____
18. $(20 + 8) \div 4 =$ _____
19. $(25 + 3) \times (14 - 5) =$ _____
20. $(42 + 12) \div 6 =$ _____

# Cumulative Review C2: Intermediate Math (Sections 1–8)

Review Worked Examples 1 & 2 below, then solve Test C2C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test C2D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Continued Fraction Evaluation: Evaluate $1 + 1/(1 + 1/2) \rightarrow$ Start at bottom $1 + 1/2 = 3/2$, invert to $2/3$, add $1 \rightarrow 5/3$!

**EXAMPLE 2** C2D #5 $(24 \times (5/6 - 3/8)) \rightarrow$ Distribute 24: $20 - 9 = 11$

**TEST C2C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Competition Number Sense, Algebr)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(9 + 3) \times (11 - 2) =$ _____
2. $(20 + 8) \div 4 =$ _____
3. $(11 + 5) \times (13 - 4) =$ _____
4. $(42 + 12) \div 6 =$ _____
5. $(13 + 7) \times (15 - 6) =$ _____
6. $(16 + 8) \div 4 =$ _____
7. $(15 + 3) \times (17 - 3) =$ _____
8. $(36 + 12) \div 6 =$ _____
9. $(17 + 5) \times (12 - 5) =$ _____
10. $(32 + 8) \div 4 =$ _____
11. $(19 + 7) \times (14 - 2) =$ _____
12. $(30 + 12) \div 6 =$ _____
13. $(21 + 3) \times (16 - 4) =$ _____
14. $(28 + 8) \div 4 =$ _____
15. $(23 + 5) \times (11 - 6) =$ _____
16. $(24 + 12) \div 6 =$ _____
17. $(25 + 7) \times (13 - 3) =$ _____
18. $(24 + 8) \div 4 =$ _____
19. $(27 + 3) \times (15 - 5) =$ _____
20. $(48 + 12) \div 6 =$ _____

**TEST C2D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Competition Number Sense, Algebra & Geometry

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1 + 2 + 3 + \ldots + 10 =$ _____
2. $1 + 3 + 5 + 7 + 9 + 11 =$ _____
3. $35^2 =$ _____
4. $25^2 - 15^2 =$ _____
5. $24 \times (5/6 - 3/8) =$ _____
6. $1 + 1/(1 + 1/2) =$ _____
7. $\text{GCF}(15, 20) \times \text{LCM}(15, 20) =$ _____
8. Sum of positive factors of 12 = _____
9. $((35 - 7) \times 4) \div (14 + 2) =$ _____
10. $(12^2 + 5^2) \div 13 =$ _____
11. Distance from (-3, 5) to (5, -10) = _____
12. 3D diagonal $\sqrt{3^2 + 4^2 + 12^2} =$ _____
13. Solve for x: $x + y = 12$, $x - y = 6$ : _____
14. Minimum value of $y = x^2 - 6x + 14$ : _____
15. Sum of roots of $2x^2 - 14x + 9 = 0$ : _____
16. Hypotenuse for legs 7 and 24 = _____
17. $(45 \times 12) \div 18 =$ _____
18. $1000 - 384 =$ _____
19. $99 + 98 + 97 =$ _____
20. $2/3 \times 3/4 \times 4/5 \times 5/6 =$ _____

# Cumulative Review C2: Intermediate Math (Sections 1–8)

Review Worked Examples 1 & 2 below, then solve Test C2D2 (Level 4: Advanced Challenge (Sprint 2)) & Test C2D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** C2D #11 (Dist (-3,5) to (5,-10)) → $dx = 8$, $dy = 15$ → (8,15,17) triple = 17

**EXAMPLE 2** C2D #14 (Min of $x^2$-6$x$+14) → Vertex at $x = 3$ → 9 - 18 + 14 = 5

**TEST C2D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Competition Number Sense, Algebra & Geometry — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF(13, 14) × LCM(13, 14) = _____
2. GCF(15, 17) × LCM(15, 17) = _____
3. GCF(17, 20) × LCM(17, 20) = _____
4. GCF(19, 23) × LCM(19, 23) = _____
5. GCF(21, 26) × LCM(21, 26) = _____
6. GCF(23, 29) × LCM(23, 29) = _____
7. GCF(25, 32) × LCM(25, 32) = _____
8. $(31^2 - 5^2) \div (31 - 5)$ = _____
9. $(33^2 - 6^2) \div (33 - 6)$ = _____
10. $(35^2 - 7^2) \div (35 - 7)$ = _____
11. $(37^2 - 8^2) \div (37 - 8)$ = _____
12. $(39^2 - 9^2) \div (39 - 9)$ = _____
13. $(41^2 - 4^2) \div (41 - 4)$ = _____
14. $(43^2 - 5^2) \div (43 - 5)$ = _____
15. Number of positive factors of 24 = _____
16. Number of positive factors of 144 = _____
17. Number of positive factors of 864 = _____
18. Number of positive factors of 192 = _____
19. Number of positive factors of 1152 = _____
20. Number of positive factors of 6912 = _____

**TEST C2D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section C2 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. GCF(16, 16) × LCM(16, 16) = _____
2. GCF(18, 19) × LCM(18, 19) = _____
3. GCF(20, 22) × LCM(20, 22) = _____
4. GCF(22, 25) × LCM(22, 25) = _____
5. GCF(24, 28) × LCM(24, 28) = _____
6. GCF(26, 31) × LCM(26, 31) = _____
7. GCF(28, 34) × LCM(28, 34) = _____
8. $(34^2 - 5^2) \div (34 - 5)$ = _____
9. $(36^2 - 6^2) \div (36 - 6)$ = _____
10. $(38^2 - 7^2) \div (38 - 7)$ = _____
11. $(40^2 - 8^2) \div (40 - 8)$ = _____
12. $(42^2 - 9^2) \div (42 - 9)$ = _____
13. $(44^2 - 4^2) \div (44 - 4)$ = _____
14. $(46^2 - 5^2) \div (46 - 5)$ = _____
15. Number of positive factors of 144 = _____
16. Number of positive factors of 864 = _____
17. Number of positive factors of 192 = _____
18. Number of positive factors of 1152 = _____
19. Number of positive factors of 6912 = _____
20. Number of positive factors of 1536 = _____

# Section 9: Trigonometry, Vectors & Polar Coordinates

Special angles, right-triangle projections, inverse trig, and vector magnitudes. • Study the Champion Way & complete W1–W6 before starting Tests 9A1–9D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 9: TRIGONOMETRY, VECTORS & POLAR COORDINATES

**RULE 1** — **Rule 1 — The 3-Key Vector & Hypotenuse One-Liner ([→ϑ,r]): Never type $\sqrt{x^2 + y^2}$ or arctan(y/x) separately! Enter y [ENTER] x [yellow] [→ϑ,r] to get magnitude r in X and angle ϑ in Y!**

**RULE 2** — Rule 2 — Maximum of $a \cdot \sin(x) + b \cdot \cos(x)$ Is $\sqrt{a^2 + b^2}$: The amplitude (maximum value) of $a \cdot \sin(x) + b \cdot \cos(x)$ is simply the hypotenuse $\sqrt{a^2 + b^2}$ (e.g., max of 5 sin $x$ + 12 cos x is 13!).

**RULE 3** — Rule 3 — Pythagorean & Double-Angle Trig Identities: Spot $\sin^2\vartheta + \cos^2\vartheta = 1$, $\sec^2\vartheta - \tan^2\vartheta = 1$, and $2 \sin(\vartheta) \cos(\vartheta) = \sin(2\vartheta)$ before touching the calculator!

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Polar Magnitude & Angle One-Liner: Find magnitude and angle of ⟨3, 4⟩** Stack Keys: 4 ENTER 3 ↶ ▸θ,r

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute $\sqrt{3^2 + 4^2} = 5$, then clear and compute atan(4 ÷ 3) = 53.13°.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Key [4] [ENTER] [3] [yellow] [→ϑ,r] → Line X shows $r = 5$, Line Y shows $\vartheta = 53.13°$ simultaneously!**

**Archetype 2 — Linear Trig Combination: Evaluate 12 cos(45°) + 5 sin(45°)** Stack Keys: 1 7 ENTER 4 5 SIN ×

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 12 × cos(45°), write it down, compute 5 × sin(45°), add.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Since cos(45°) = sin(45°), factor it out: (12 + 5) × sin(45°) = 17 × sin(45°) = 12.0208 in 5 keys!**

**Archetype 3 — Wave Amplitude Shortcut: Find max value of 8 sin(x) - 15 cos(x)** Stack Keys: 1 5 ENTER 8 ↶ ▸θ,r

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Take derivatives or test angles.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Max of a $\sin(x) \pm b \cos(x)$ is $\sqrt{a^2 + b^2} = \sqrt{8^2 + 15^2} = 17$!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** Vector ⟨5, 12⟩ → Magnitude $r$ = _____

Shortcut Cue: Pythagorean triple or [12][ENTER][5][→ϑ,r]

**W2.** $5 \sin^2(20°) + 5 \cos^2(20°)$ = _____

Shortcut Cue: Factor $5(\sin^2 + \cos^2) = 5(1)$

**W3.** Area of Δ: $a = 8$, $b = 10$, $C = 30°$ → _____

Shortcut Cue: $1/2 \times a \times b \times \sin(30°) = 0.5 \times 80 \times 0.5$

**W4.** $2 \sin(15°) \cos(15°)$ = _____

Shortcut Cue: Double-angle identity: sin(30°)

**W5.** max value of $3 \sin(x) + 4 \cos(x)$ = _____

Shortcut Cue: Hypotenuse $\sqrt{3^2 + 4^2}$

**W6.** Law of Cosines: $a$=8, $b$=5, $C$=60° → $c$ = _____

Shortcut Cue: $c^2 = 64 + 25 - 2(8)(5)(0.5) = 49 \rightarrow c = 7$

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 9: Trigonometry, Vectors & Polar Coordinates

Review Worked Examples 1 & 2 below, then solve Test 9A1 (Level 1: Foundation (Sprint 1)) & Test 9A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** Vector $\langle 5, 12\rangle \rightarrow$ Magnitude $r = 13$ (Pythagorean triple or [12][ENTER][5][$\rightarrow \vartheta, r$])

**EXAMPLE 2** $5\sin^2(20°) + 5\cos^2(20°) = 5$ (Factor $5(\sin^2 + \cos^2) = 5(1)$)

**TEST 9A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 4 min

Exact Special-Angle Trig Ratios & Degree/Radian Conversions

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $\sin(0°) =$ _____
2. $\cos(0°) =$ _____
3. $\sin(30°) =$ _____
4. $\cos(60°) =$ _____
5. $\sin(90°) =$ _____
6. $\cos(90°) =$ _____
7. $\tan(0°) =$ _____
8. $\tan(45°) =$ _____
9. $2 \times \sin(30°) =$ _____
10. $4 \times \cos(60°) =$ _____
11. $\sin(30°) + \cos(60°) =$ _____
12. $5 \times \tan(45°) =$ _____
13. Acute angle $\vartheta$ (deg): $\sin(\vartheta) = 0.5 \rightarrow$ _____
14. Acute angle $\vartheta$ (deg): $\cos(\vartheta) = 0.5 \rightarrow$ _____
15. Acute angle $\vartheta$ (deg): $\tan(\vartheta) = 1 \rightarrow$ _____
16. Right triangle: $90° - 35° =$ _____
17. Right triangle: $90° - 62° =$ _____
18. $180°$ in radians (multiple of $\pi$) = _____
19. $90°$ in radians = _____
20. $60°$ in radians = _____

**TEST 9A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 4 min

Exact Special-Angle Trig Ratios & Degree/Radian Conversions — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $10 \times \sin(30°) =$ _____
2. $14 \times \cos(60°) =$ _____
3. Right triangle: $90° - 23° =$ _____
4. $22 \times \sin(30°) =$ _____
5. $26 \times \cos(60°) =$ _____
6. Right triangle: $90° - 32° =$ _____
7. $34 \times \sin(30°) =$ _____
8. $38 \times \cos(60°) =$ _____
9. Right triangle: $90° - 41° =$ _____
10. $46 \times \sin(30°) =$ _____
11. $50 \times \cos(60°) =$ _____
12. Right triangle: $90° - 50° =$ _____
13. $58 \times \sin(30°) =$ _____
14. $62 \times \cos(60°) =$ _____
15. Right triangle: $90° - 59° =$ _____
16. $70 \times \sin(30°) =$ _____
17. $74 \times \cos(60°) =$ _____
18. Right triangle: $90° - 68° =$ _____
19. $82 \times \sin(30°) =$ _____
20. $86 \times \cos(60°) =$ _____

# Section 9: Trigonometry, Vectors & Polar Coordinates

Review Worked Examples 1 & 2 below, then solve Test 9A3 (Level 1: Foundation (Sprint 3)) & Test 9B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** Area of $\Delta$: $a = 8$, $b = 10$, $C = 30° \rightarrow 20$ $(1/2 \times a \times b \times \sin(30°) = 0.5 \times 80 \times 0.5)$

**EXAMPLE 2** $2 \sin(15°) \cos(15°) = 0.5$ (Double-angle identity: $\sin(30°)$)

### TEST 9A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 4 min

Bridge from Level 1 to Level 2 (Right-Triangle Projections & Vec)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $16 \times \sin(30°) =$ _____
2. $20 \times \cos(60°) =$ _____
3. Right triangle: $90° - 25° =$ _____
4. $28 \times \sin(30°) =$ _____
5. $32 \times \cos(60°) =$ _____
6. Right triangle: $90° - 34° =$ _____
7. $40 \times \sin(30°) =$ _____
8. $44 \times \cos(60°) =$ _____
9. Right triangle: $90° - 43° =$ _____
10. $52 \times \sin(30°) =$ _____
11. $56 \times \cos(60°) =$ _____
12. Right triangle: $90° - 52° =$ _____
13. $64 \times \sin(30°) =$ _____
14. $68 \times \cos(60°) =$ _____
15. Right triangle: $90° - 61° =$ _____
16. $76 \times \sin(30°) =$ _____
17. $80 \times \cos(60°) =$ _____
18. Right triangle: $90° - 70° =$ _____
19. $88 \times \sin(30°) =$ _____
20. $92 \times \cos(60°) =$ _____

### TEST 9B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 5 min

Right-Triangle Projections & Vector Magnitudes

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Hyp $= 10$, $\vartheta = 30° \rightarrow$ Opp leg $y =$ _____
2. Hyp $= 12$, $\vartheta = 60° \rightarrow$ Adj leg $x =$ _____
3. Hyp $= 20$, $\vartheta = 30° \rightarrow$ Opp leg $y =$ _____
4. Hyp $= 16$, $\vartheta = 60° \rightarrow$ Adj leg $x =$ _____
5. Adj $= 9$, $\vartheta = 45° \rightarrow$ Opp leg $y =$ _____
6. Opp $= 7$, $\vartheta = 30° \rightarrow$ Hypotenuse $r =$ _____
7. Adj $= 5$, $\vartheta = 60° \rightarrow$ Hypotenuse $r =$ _____
8. Polar $(r = 10, \vartheta = 0°) \rightarrow x =$ _____
9. Polar $(r = 10, \vartheta = 0°) \rightarrow y =$ _____
10. Polar $(r = 8, \vartheta = 90°) \rightarrow y =$ _____
11. Polar $(r = 14, \vartheta = 60°) \rightarrow x =$ _____
12. Polar $(r = 18, \vartheta = 30°) \rightarrow y =$ _____
13. Vector $\langle 3, 4 \rangle \rightarrow$ Magnitude $r =$ _____
14. Vector $\langle 6, 8 \rangle \rightarrow$ Magnitude $r =$ _____
15. Vector $\langle 5, 12 \rangle \rightarrow$ Magnitude $r =$ _____
16. Vector $\langle 8, 6 \rangle \rightarrow$ Magnitude $r =$ _____
17. Vector $\langle 9, 12 \rangle \rightarrow$ Magnitude $r =$ _____
18. Vector $\langle 8, 15 \rangle \rightarrow$ Magnitude $r =$ _____
19. Vector $\langle 12, 16 \rangle \rightarrow$ Magnitude $r =$ _____
20. Vector $\langle 7, 24 \rangle \rightarrow$ Magnitude $r =$ _____

# Section 9: Trigonometry, Vectors & Polar Coordinates

Review Worked Examples 1 & 2 below, then solve Test 9B2 (Level 2: Core Fluency (Sprint 2)) & Test 9B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** — Archetype 1 — Polar Magnitude & Angle One-Liner: Find magnitude and angle of ⟨3, 4⟩ → Key [4] [ENTER] [3] [yellow] [→θ,r] → Line X shows $r = 5$, Line Y shows $\theta = 53.13°$ simultaneously!

**EXAMPLE 2** — max value of $3\sin(x) + 4\cos(x) = 5$ (Hypotenuse $\sqrt{3^2 + 4^2}$)

**TEST 9B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 5 min

Right-Triangle Projections & Vector Magnitudes — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Hyp = 16, $\theta = 30°$ → Opp leg $y$ = _____
2. Hyp = 20, $\theta = 60°$ → Adj leg $x$ = _____
3. Hyp = 24, $\theta = 30°$ → Opp leg $y$ = _____
4. Hyp = 28, $\theta = 60°$ → Adj leg $x$ = _____
5. Hyp = 32, $\theta = 30°$ → Opp leg $y$ = _____
6. Hyp = 36, $\theta = 60°$ → Adj leg $x$ = _____
7. Hyp = 40, $\theta = 30°$ → Opp leg $y$ = _____
8. Hyp = 44, $\theta = 60°$ → Adj leg $x$ = _____
9. Hyp = 48, $\theta = 30°$ → Opp leg $y$ = _____
10. Hyp = 52, $\theta = 60°$ → Adj leg $x$ = _____
11. Vector ⟨6, 8⟩ → Magnitude $r$ = _____
12. Vector ⟨15, 36⟩ → Magnitude $r$ = _____
13. Vector ⟨32, 60⟩ → Magnitude $r$ = _____
14. Vector ⟨7, 24⟩ → Magnitude $r$ = _____
15. Vector ⟨18, 80⟩ → Magnitude $r$ = _____
16. Vector ⟨9, 12⟩ → Magnitude $r$ = _____
17. Vector ⟨20, 48⟩ → Magnitude $r$ = _____
18. Vector ⟨8, 15⟩ → Magnitude $r$ = _____
19. Vector ⟨14, 48⟩ → Magnitude $r$ = _____
20. Vector ⟨27, 120⟩ → Magnitude $r$ = _____

**TEST 9B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 5 min

Bridge from Level 2 to Level 3 (Trigonometric Identities, Square)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Hyp = 20, $\theta = 30°$ → Opp leg $y$ = _____
2. Hyp = 24, $\theta = 60°$ → Adj leg $x$ = _____
3. Hyp = 28, $\theta = 30°$ → Opp leg $y$ = _____
4. Hyp = 32, $\theta = 60°$ → Adj leg $x$ = _____
5. Hyp = 36, $\theta = 30°$ → Opp leg $y$ = _____
6. Hyp = 40, $\theta = 60°$ → Adj leg $x$ = _____
7. Hyp = 44, $\theta = 30°$ → Opp leg $y$ = _____
8. Hyp = 48, $\theta = 60°$ → Adj leg $x$ = _____
9. Hyp = 52, $\theta = 30°$ → Opp leg $y$ = _____
10. Hyp = 56, $\theta = 60°$ → Adj leg $x$ = _____
11. Vector ⟨9, 12⟩ → Magnitude $r$ = _____
12. Vector ⟨20, 48⟩ → Magnitude $r$ = _____
13. Vector ⟨8, 15⟩ → Magnitude $r$ = _____
14. Vector ⟨14, 48⟩ → Magnitude $r$ = _____
15. Vector ⟨27, 120⟩ → Magnitude $r$ = _____
16. Vector ⟨12, 16⟩ → Magnitude $r$ = _____
17. Vector ⟨5, 12⟩ → Magnitude $r$ = _____
18. Vector ⟨16, 30⟩ → Magnitude $r$ = _____
19. Vector ⟨21, 72⟩ → Magnitude $r$ = _____
20. Vector ⟨36, 160⟩ → Magnitude $r$ = _____

# Section 9: Trigonometry, Vectors & Polar Coordinates

Review Worked Examples 1 & 2 below, then solve Test 9C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 9C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Linear Trig Combination: Evaluate $12\cos(45°) + 5\sin(45°)$ → Since $\cos(45°) = \sin(45°)$, factor it out: $(12 + 5) \times \sin(45°) = 17 \times \sin(45°) = 12.0208$ in 5 keys!

**EXAMPLE 2** 9D #1 ($12\cos 45° + 5\sin 45°$) → Factor $(12 + 5) \times \sin(45°)$: [17] [ENTER] [45] [SIN] [×] → 12.0208

**TEST 9C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Trigonometric Identities, Squares & Vector Operations

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $\sin^2(30°) =$ _____
2. $\cos^2(30°) =$ _____
3. $\sin^2(45°) =$ _____
4. $\cos^2(45°) =$ _____
5. $\tan^2(60°) =$ _____
6. $\sin^2(37°) + \cos^2(37°) =$ _____
7. $5\sin^2(20°) + 5\cos^2(20°) =$ _____
8. $4\cos^2(30°) =$ _____
9. $8\sin^2(45°) =$ _____
10. $2\sin(30°)\cos(60°) =$ _____
11. $\langle 2, 3\rangle + \langle 1, 1\rangle \rightarrow$ Magnitude $r =$ _____
12. $\langle 4, 1\rangle + \langle 2, 7\rangle \rightarrow$ Magnitude $r =$ _____
13. $\langle 8, 5\rangle - \langle 3, -7\rangle \rightarrow$ Magnitude $r =$ _____
14. Dot product $\langle 2, 3\rangle \cdot \langle 4, 5\rangle =$ _____
15. Dot product $\langle 3, -2\rangle \cdot \langle 4, 6\rangle =$ _____
16. Area of $\Delta$: $a = 8$, $b = 10$, $C = 30° \rightarrow$ _____
17. Area of $\Delta$: $a = 6$, $b = 12$, $C = 30° \rightarrow$ _____
18. $12\cos(60°) + 8\sin(30°) =$ _____
19. $20\sin(30°) - 6\tan(45°) =$ _____
20. $\sec(60°) =$ _____

**TEST 9C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Trigonometric Identities, Squares & Vector Operations — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $5\sin^2(15°) + 5\cos^2(15°) =$ _____
2. $7\sin^2(19°) + 7\cos^2(19°) =$ _____
3. $9\sin^2(23°) + 9\cos^2(23°) =$ _____
4. $11\sin^2(27°) + 11\cos^2(27°) =$ _____
5. $13\sin^2(31°) + 13\cos^2(31°) =$ _____
6. $15\sin^2(35°) + 15\cos^2(35°) =$ _____
7. $17\sin^2(39°) + 17\cos^2(39°) =$ _____
8. $19\sin^2(43°) + 19\cos^2(43°) =$ _____
9. Dot product $\langle 11, 6\rangle \cdot \langle 4, 3\rangle =$ _____
10. Dot product $\langle 12, 7\rangle \cdot \langle 5, 3\rangle =$ _____
11. Dot product $\langle 13, 3\rangle \cdot \langle 6, 3\rangle =$ _____
12. Dot product $\langle 14, 4\rangle \cdot \langle 7, 3\rangle =$ _____
13. Dot product $\langle 15, 5\rangle \cdot \langle 4, 3\rangle =$ _____
14. Dot product $\langle 16, 6\rangle \cdot \langle 5, 3\rangle =$ _____
15. Area of $\Delta$: $a = 6$, $b = 6$, $C = 30° \rightarrow$ _____
16. Area of $\Delta$: $a = 8$, $b = 8$, $C = 30° \rightarrow$ _____
17. Area of $\Delta$: $a = 10$, $b = 10$, $C = 30° \rightarrow$ _____
18. Area of $\Delta$: $a = 12$, $b = 12$, $C = 30° \rightarrow$ _____
19. Area of $\Delta$: $a = 14$, $b = 14$, $C = 30° \rightarrow$ _____
20. Area of $\Delta$: $a = 16$, $b = 16$, $C = 30° \rightarrow$ _____

# Section 9: Trigonometry, Vectors & Polar Coordinates

Review Worked Examples 1 & 2 below, then solve Test 9C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 9D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Wave Amplitude Shortcut: Find max value of $8\sin(x) - 15\cos(x)$ → Max of $a\sin(x) \pm b\cos(x)$ is $\sqrt{a^2 + b^2} = \sqrt{8^2 + 15^2} = 17$!

**EXAMPLE 2** 9D #5 (Angle of ⟨3, 4⟩) → [4] [ENTER] [3] [yellow] [→θ,r] [x⇄y] → 53.13°

## TEST 9C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3) Goal: 5 min

Bridge from Level 3 to Level 4 (4-Sig-Fig Calculator Projections)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $7\sin^2(18°) + 7\cos^2(18°) =$ _____
2. $9\sin^2(22°) + 9\cos^2(22°) =$ _____
3. $11\sin^2(26°) + 11\cos^2(26°) =$ _____
4. $13\sin^2(30°) + 13\cos^2(30°) =$ _____
5. $15\sin^2(34°) + 15\cos^2(34°) =$ _____
6. $17\sin^2(38°) + 17\cos^2(38°) =$ _____
7. $19\sin^2(42°) + 19\cos^2(42°) =$ _____
8. $21\sin^2(46°) + 21\cos^2(46°) =$ _____
9. Dot product $\langle 12, 6\rangle \cdot \langle 4, 4\rangle =$ _____
10. Dot product $\langle 13, 7\rangle \cdot \langle 5, 4\rangle =$ _____
11. Dot product $\langle 14, 3\rangle \cdot \langle 6, 4\rangle =$ _____
12. Dot product $\langle 15, 4\rangle \cdot \langle 7, 4\rangle =$ _____
13. Dot product $\langle 16, 5\rangle \cdot \langle 4, 4\rangle =$ _____
14. Dot product $\langle 17, 6\rangle \cdot \langle 5, 4\rangle =$ _____
15. Area of $\Delta$: $a = 8$, $b = 6$, $C = 30° \rightarrow$ _____
16. Area of $\Delta$: $a = 10$, $b = 8$, $C = 30° \rightarrow$ _____
17. Area of $\Delta$: $a = 12$, $b = 10$, $C = 30° \rightarrow$ _____
18. Area of $\Delta$: $a = 14$, $b = 12$, $C = 30° \rightarrow$ _____
19. Area of $\Delta$: $a = 16$, $b = 14$, $C = 30° \rightarrow$ _____
20. Area of $\Delta$: $a = 18$, $b = 16$, $C = 30° \rightarrow$ _____

## TEST 9D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1) Goal: 6 min

4-Sig-Fig Calculator Projections, Polar Angles & Law of Cosines

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $12\cos(45°) + 5\sin(45°)$ (4 dec) = _____
2. $10\cos(30°)$ (4 dec) = _____
3. $10\sin(45°)$ (4 dec) = _____
4. $10\tan(30°)$ (4 dec) = _____
5. Vector $\langle 3, 4\rangle$ angle $\vartheta$ (deg, 2 dec) = _____
6. Vector $\langle 4, 3\rangle$ angle $\vartheta$ (deg, 2 dec) = _____
7. Vector $\langle 5, 5\rangle$ angle $\vartheta$ (deg) = _____
8. Vector $\langle 0, -10\rangle$ angle $\vartheta$ (deg) = _____
9. Law of Cosines: $a$=3, $b$=5, $C$=60° → $c^2$ = _____
10. Law of Cosines: $a$=8, $b$=5, $C$=60° → $c$ = _____
11. Law of Sines: $a$=10, $A$=30°, $B$=90° → $b$ = _____
12. $2\sin(15°)\cos(15°) =$ _____
13. $\cos^2(15°) - \sin^2(15°)$ (4 dec) = _____
14. $1 + \tan^2(45°) =$ _____
15. $\sec^2(30°) - \tan^2(30°) =$ _____
16. max value of $3\sin(x) + 4\cos(x) =$ _____
17. max value of $5\sin(x) + 12\cos(x) =$ _____
18. max value of $8\sin(x) - 15\cos(x) =$ _____
19. Period of $\sin(2x)$ in degrees = _____
20. Period of $\cos(4x)$ in degrees = _____

# Section 9: Trigonometry, Vectors & Polar Coordinates

Review Worked Examples 1 & 2 below, then solve Test 9D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 9D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 9D #10 (Law of Cosines $a$=8,$b$=5,$C$=60°) → cos(60°)=0.5 cancels the 2 in $2ab\cos C$: $c^2 = a^2+b^2-ab = 64+25-40 = 49 \rightarrow 7$

**EXAMPLE 2** 9D #17 (max of 5 sin $x$ + 12 cos $x$) → Amplitude $\sqrt{5^2 + 12^2} = 13$

**TEST 9D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

4-Sig-Fig Calculator Projections, Polar Angles & Law of Cosines — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. max value of 6 sin($x$) + 8 cos($x$) = _____
2. max value of 15 sin($x$) + 36 cos($x$) = _____
3. max value of 8 sin($x$) + 15 cos($x$) = _____
4. max value of 14 sin($x$) + 48 cos($x$) = _____
5. max value of 27 sin($x$) + 120 cos($x$) = _____
6. max value of 3 sin($x$) + 4 cos($x$) = _____
7. max value of 10 sin($x$) + 24 cos($x$) = _____
8. max value of 24 sin($x$) + 45 cos($x$) = _____
9. max value of 7 sin($x$) + 24 cos($x$) = _____
10. max value of 18 sin($x$) + 80 cos($x$) = _____
11. Period of sin(3$x$) in degrees = _____
12. Period of sin(4$x$) in degrees = _____
13. Period of sin(5$x$) in degrees = _____
14. Period of sin(6$x$) in degrees = _____
15. Law of Cosines: $a$=8, $b$=9, $C$=60° → $c^2$ = _____
16. Period of sin(8$x$) in degrees = _____
17. Period of sin(9$x$) in degrees = _____
18. Period of sin(10$x$) in degrees = _____
19. Law of Cosines: $a$=12, $b$=13, $C$=60° → $c^2$ = _____
20. Period of sin(12$x$) in degrees = _____

**TEST 9D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 9 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. max value of 9 sin($x$) + 12 cos($x$) = _____
2. max value of 5 sin($x$) + 12 cos($x$) = _____
3. max value of 16 sin($x$) + 30 cos($x$) = _____
4. max value of 21 sin($x$) + 72 cos($x$) = _____
5. max value of 9 sin($x$) + 40 cos($x$) = _____
6. max value of 6 sin($x$) + 8 cos($x$) = _____
7. max value of 15 sin($x$) + 36 cos($x$) = _____
8. max value of 8 sin($x$) + 15 cos($x$) = _____
9. max value of 14 sin($x$) + 48 cos($x$) = _____
10. max value of 27 sin($x$) + 120 cos($x$) = _____
11. Period of sin(4$x$) in degrees = _____
12. Period of sin(5$x$) in degrees = _____
13. Period of sin(6$x$) in degrees = _____
14. Law of Cosines: $a$=8, $b$=8, $C$=60° → $c^2$ = _____
15. Period of sin(8$x$) in degrees = _____
16. Period of sin(9$x$) in degrees = _____
17. Period of sin(10$x$) in degrees = _____
18. Law of Cosines: $a$=12, $b$=12, $C$=60° → $c^2$ = _____
19. Period of sin(12$x$) in degrees = _____
20. Law of Cosines: $a$=14, $b$=14, $C$=60° → $c^2$ = _____

# Section 10: Exponents, Logarithms & Growth Models

Powers, fractional exponents, base-10 and natural logarithms, and exponential growth. • Study the Champion Way & complete W1–W6 before starting Tests 10A1–10D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 10: EXPONENTS, LOGARITHMS & GROWTH MODELS

**RULE 1** — **Rule 1 — Inside-Out Growth & Compounding Pipeline: For $P(1 + r/n)^{nt}$ or $P{\cdot}e^{rt}$, ALWAYS compute the inner rate or exponent first, apply [$y^x$] or [$e^x$], and multiply by P last!**

**RULE 2** — Rule 2 — Change-of-Base Logarithm Shortcut: To evaluate $\log_b(a)$, recognize exact powers $b^k = a$ (so $\log_4(64) = 3$), or key [$a$] [LN] [$b$] [LN] [÷] on the SC-32.

**RULE 3** — Rule 3 — Fractional Exponents Root-First Rule: To evaluate $a^{m/n}$ (like $27^{2/3}$ or $16^{3/4}$), always take the nth root FIRST to keep numbers small, then raise to power m ($27^{1/3} = 3 \rightarrow 3^2 = 9$)!

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Continuous Growth Pipeline: Evaluate $500 \times e^{0.04 \times 5}$** — Stack Keys: [•] [0] [4] [ENTER] [5] [×] [$e^x$] [5] [0] [0] [×]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Type 500 first, then struggle to compute $e^{0.04 \times 5}$ without losing 500.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Start at the exponent: [.04] [ENTER] [5] [×] [$e^x$] [500] [×] → 610.7014 in 7 keys!**

**Archetype 2 — Monthly Compound Interest: Evaluate $250 \times (1 + 0.06 \div 12)^{24}$** — Stack Keys: [•] [0] [6] [ENTER] [1] [2] [÷] [1] [+] [2] [4] [$y^x$] [2] [5] [0] [×]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 0.06 ÷ 12 on paper, add 1, raise to 24, multiply by 250.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Inside-out chain: [.06] [ENTER] [12] [÷] [1] [+] [24] [$y^x$] [250] [×] → 281.79!**

**Archetype 3 — Fractional Exponent Root-First: Evaluate $64^{2/3}$** — Stack Keys: [6] [4] [ENTER] [3] [$x^{-1}$] [2] [×] [$y^x$]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Square $64^2 = 4096$ first, then try to find the cube root of 4096.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Take the cube root of 64 first ($\sqrt[3]{64} = 4$), then square: $4^2 = 16$!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** $\log_{10}(4) + \log_{10}(25) =$ _____

Shortcut Cue: $\log_{10}(4 \times 25) = \log_{10}(100)$

**W2.** Half-life (3 periods): $100 \times (0.5)^3 =$ _____

Shortcut Cue: $100 \div 8 = 12.5$

**W3.** $27^{2/3} =$ _____

Shortcut Cue: Cube root first (3), then square ($3^2$)

**W4.** $16^{3/4} =$ _____

Shortcut Cue: 4th root first (2), then cube ($2^3$)

**W5.** Change of base: $\ln(125) \div \ln(5) =$ _____

Shortcut Cue: $\log_5(125) = 3$ since $5^3 = 125$

**W6.** Solve: $4^x = 32$ (decimal) = _____

Shortcut Cue: Base 2: $2^{2x} = 2^5 \rightarrow 2x = 5$

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 10: Exponents, Logarithms & Growth Models

Review Worked Examples 1 & 2 below, then solve Test 10A1 (Level 1: Foundation (Sprint 1)) & Test 10A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $\log_{10}(4) + \log_{10}(25) = 2$ $(\log_{10}(4 \times 25) = \log_{10}(100))$

**EXAMPLE 2** Half-life (3 periods): $100 \times (0.5)^3 = 12.5$ $(100 \div 8 = 12.5)$

## TEST 10A1 • LEVEL 1: FOUNDATION (SPRINT 1) Goal: 4 min

Integer Powers, Zero/Negative Exponents & Roots

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $2^0 =$ | _____ |
| 2. | $e^0 =$ | _____ |
| 3. | $2^4 =$ | _____ |
| 4. | $2^5 =$ | _____ |
| 5. | $2^6 =$ | _____ |
| 6. | $2^8 =$ | _____ |
| 7. | $2^{10} =$ | _____ |
| 8. | $3^3 =$ | _____ |
| 9. | $3^4 =$ | _____ |
| 10. | $5^3 =$ | _____ |
| 11. | $5^4 =$ | _____ |
| 12. | $10^3 =$ | _____ |
| 13. | $10^4 =$ | _____ |
| 14. | $2^{-1}$ (decimal) $=$ | _____ |
| 15. | $2^{-2}$ (decimal) $=$ | _____ |
| 16. | $5^{-1}$ (decimal) $=$ | _____ |
| 17. | $10^{-2}$ (decimal) $=$ | _____ |
| 18. | $16^{1/2} =$ | _____ |
| 19. | $27^{1/3} =$ | _____ |
| 20. | $8^{2/3} =$ | _____ |

## TEST 10A2 • LEVEL 1: FOUNDATION (SPRINT 2) Goal: 4 min

Integer Powers, Zero/Negative Exponents & Roots — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

| | | |
|---|---|---|
| 1. | $2^3 + 1 =$ | _____ |
| 2. | $2^4 + 1 =$ | _____ |
| 3. | $2^5 + 1 =$ | _____ |
| 4. | $2^6 + 1 =$ | _____ |
| 5. | $2^7 + 1 =$ | _____ |
| 6. | $2^8 + 1 =$ | _____ |
| 7. | $2^9 + 1 =$ | _____ |
| 8. | $2^{10} + 1 =$ | _____ |
| 9. | $2^3 + 1 =$ | _____ |
| 10. | $2^4 + 1 =$ | _____ |
| 11. | $27^{1/3} =$ | _____ |
| 12. | $64^{1/3} =$ | _____ |
| 13. | $125^{1/3} =$ | _____ |
| 14. | $216^{1/3} =$ | _____ |
| 15. | $27^{1/3} =$ | _____ |
| 16. | $64^{1/3} =$ | _____ |
| 17. | $125^{1/3} =$ | _____ |
| 18. | $216^{1/3} =$ | _____ |
| 19. | $27^{1/3} =$ | _____ |
| 20. | $64^{1/3} =$ | _____ |

# Section 10: Exponents, Logarithms & Growth Models

Review Worked Examples 1 & 2 below, then solve Test 10A3 (Level 1: Foundation (Sprint 3)) & Test 10B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $27^{2/3} = 9$ **(Cube root first (3), then square ($3^2$))**

**EXAMPLE 2** $16^{3/4} = 8$ **(4th root first (2), then cube ($2^3$))**

## TEST 10A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 4 min

Bridge from Level 1 to Level 2 (Base-10 Logs, Binary Logs & Natu)

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. $2^4 + 2 =$ _____
2. $2^5 + 2 =$ _____
3. $2^6 + 2 =$ _____
4. $2^7 + 2 =$ _____
5. $2^8 + 2 =$ _____
6. $2^9 + 2 =$ _____
7. $2^{10} + 2 =$ _____
8. $2^{11} + 2 =$ _____
9. $2^4 + 2 =$ _____
10. $2^5 + 2 =$ _____
11. $64^{1/3} =$ _____
12. $125^{1/3} =$ _____
13. $216^{1/3} =$ _____
14. $343^{1/3} =$ _____
15. $64^{1/3} =$ _____
16. $125^{1/3} =$ _____
17. $216^{1/3} =$ _____
18. $343^{1/3} =$ _____
19. $64^{1/3} =$ _____
20. $125^{1/3} =$ _____

## TEST 10B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 5 min

Base-10 Logs, Binary Logs & Natural Logarithm Identities

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. $\log_{10}(1) =$ _____
2. $\log_{10}(10) =$ _____
3. $\log_{10}(100) =$ _____
4. $\log_{10}(1000) =$ _____
5. $\log_{10}(10000) =$ _____
6. $\log_{10}(0.1) =$ _____
7. $\log_{10}(0.01) =$ _____
8. $\ln(1) =$ _____
9. $\ln(e^1) =$ _____
10. $\ln(e^3) =$ _____
11. $\ln(e^{-2}) =$ _____
12. $e^{\ln 5} =$ _____
13. $e^{2 \ln 3} =$ _____
14. $10^{\log 7} =$ _____
15. $\log_2(8) =$ _____
16. $\log_2(32) =$ _____
17. $\log_2(64) =$ _____
18. $\log_3(27) =$ _____
19. $\log_5(125) =$ _____
20. $\log_{10}(2) + \log_{10}(5) =$ _____

# Section 10: Exponents, Logarithms & Growth Models

Review Worked Examples 1 & 2 below, then solve Test 10B2 (Level 2: Core Fluency (Sprint 2)) & Test 10B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** **Archetype 1 — Continuous Growth Pipeline: Evaluate $500 \times e^{0.04 \times 5}$ → Start at the exponent: [.04] [ENTER] [5] [×] [$e^x$] [500] [×] → 610.7014 in 7 keys!**

**EXAMPLE 2** **Change of base: $\ln(125) \div \ln(5) = 3$ ($\log_5(125) = 3$ since $5^3 = 125$)**

**TEST 10B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** **Goal: 5 min**

Base-10 Logs, Binary Logs & Natural Logarithm Identities — Speed Builder

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. $\ln(e^2) =$ _____
2. $\ln(e^3) =$ _____
3. $\ln(e^4) =$ _____
4. $\ln(e^5) =$ _____
5. $\ln(e^6) =$ _____
6. $\ln(e^7) =$ _____
7. $\ln(e^8) =$ _____
8. $\ln(e^9) =$ _____
9. $\log_2(8) =$ _____
10. $\log_2(16) =$ _____
11. $\log_2(32) =$ _____
12. $\log_2(64) =$ _____
13. $\log_2(128) =$ _____
14. $\log_2(256) =$ _____
15. $e^{\ln 6} =$ _____
16. $e^{\ln 8} =$ _____
17. $e^{\ln 10} =$ _____
18. $e^{\ln 12} =$ _____
19. $e^{\ln 14} =$ _____
20. $e^{\ln 16} =$ _____

**TEST 10B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** **Goal: 5 min**

Bridge from Level 2 to Level 3 (Logarithm Laws, Half-Life Decay )

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. $\ln(e^3) =$ _____
2. $\ln(e^4) =$ _____
3. $\ln(e^5) =$ _____
4. $\ln(e^6) =$ _____
5. $\ln(e^7) =$ _____
6. $\ln(e^8) =$ _____
7. $\ln(e^9) =$ _____
8. $\ln(e^{10}) =$ _____
9. $\log_2(16) =$ _____
10. $\log_2(32) =$ _____
11. $\log_2(64) =$ _____
12. $\log_2(128) =$ _____
13. $\log_2(256) =$ _____
14. $\log_2(512) =$ _____
15. $e^{\ln 9} =$ _____
16. $e^{\ln 11} =$ _____
17. $e^{\ln 13} =$ _____
18. $e^{\ln 15} =$ _____
19. $e^{\ln 17} =$ _____
20. $e^{\ln 19} =$ _____

# Section 10: Exponents, Logarithms & Growth Models

Review Worked Examples 1 & 2 below, then solve Test 10C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 10C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Monthly Compound Interest: Evaluate $250 \times (1 + 0.06 \div 12)^{24}$ → Inside-out chain: [.06] [ENTER] [12] [÷] [1] [+] [24] [$y^x$] [250] [×] → 281.79!

**EXAMPLE 2** 10D #1 (ln(2) ÷ 0.07) → [2] [LN] [.07] [÷] → 9.9021

**TEST 10C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Logarithm Laws, Half-Life Decay & Compound Growth

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $\log_{10}(4) + \log_{10}(25) =$ _____
2. $\log_{10}(20) + \log_{10}(50) =$ _____
3. $\log_{10}(200) - \log_{10}(2) =$ _____
4. $\log_{10}(5000) - \log_{10}(5) =$ _____
5. $\log_{2}(24) - \log_{2}(3) =$ _____
6. $\log_{3}(18) - \log_{3}(2) =$ _____
7. Solve: $2^x = 64$ _____
8. Solve: $3^x = 81$ _____
9. Solve: $5^{x-1} = 25$ _____
10. Solve: $\log_{10}(x) = 3$ _____
11. Solve: $\log_{2}(x) = 5$ _____
12. Half-life (1 period): $100 \times (0.5)^1 =$ _____
13. Half-life (2 periods): $100 \times (0.5)^2 =$ _____
14. Half-life (3 periods): $100 \times (0.5)^3 =$ _____
15. Half-life (4 periods): $160 \times (0.5)^4 =$ _____
16. Doubling (3 periods): $50 \times 2^3 =$ _____
17. Tripling (2 periods): $20 \times 3^2 =$ _____
18. 1 year at 10%: $1000 \times 1.10 =$ _____
19. 2 years at 10%: $1000 \times (1.10)^2 =$ _____
20. 2 years at 5%: $1000 \times (1.05)^2 =$ _____

**TEST 10C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Logarithm Laws, Half-Life Decay & Compound Growth — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Half-life (1 periods): $120 \times (0.5)^1 =$ _____
2. Half-life (2 periods): $160 \times (0.5)^2 =$ _____
3. Half-life (3 periods): $200 \times (0.5)^3 =$ _____
4. Half-life (1 periods): $240 \times (0.5)^1 =$ _____
5. Half-life (2 periods): $280 \times (0.5)^2 =$ _____
6. Half-life (3 periods): $320 \times (0.5)^3 =$ _____
7. Half-life (1 periods): $360 \times (0.5)^1 =$ _____
8. Half-life (2 periods): $400 \times (0.5)^2 =$ _____
9. Half-life (3 periods): $440 \times (0.5)^3 =$ _____
10. Half-life (1 periods): $480 \times (0.5)^1 =$ _____
11. Doubling (2 periods): $100 \times 2^2 =$ _____
12. Doubling (3 periods): $125 \times 2^3 =$ _____
13. Doubling (4 periods): $150 \times 2^4 =$ _____
14. Doubling (2 periods): $175 \times 2^2 =$ _____
15. Doubling (3 periods): $200 \times 2^3 =$ _____
16. Doubling (4 periods): $225 \times 2^4 =$ _____
17. Doubling (2 periods): $250 \times 2^2 =$ _____
18. Doubling (3 periods): $275 \times 2^3 =$ _____
19. Doubling (4 periods): $300 \times 2^4 =$ _____
20. Doubling (2 periods): $325 \times 2^2 =$ _____

# Section 10: Exponents, Logarithms & Growth Models

Review Worked Examples 1 & 2 below, then solve Test 10C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 10D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Fractional Exponent Root-First: Evaluate $64^{2/3}$ → Take the cube root of 64 first ($\sqrt[3]{64} = 4$), then square: $4^2 = 16$!

**EXAMPLE 2** 10D #3 ($500 \times e^{0.04\times5}$) → [.04] [ENTER] [5] [×] [$e^x$] [500] [×] → 610.7014

**TEST 10C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Continuous Compounding, Doubling)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Half-life (1 periods): $160 \times (0.5)^1 =$ _____
2. Half-life (2 periods): $200 \times (0.5)^2 =$ _____
3. Half-life (3 periods): $240 \times (0.5)^3 =$ _____
4. Half-life (1 periods): $280 \times (0.5)^1 =$ _____
5. Half-life (2 periods): $320 \times (0.5)^2 =$ _____
6. Half-life (3 periods): $360 \times (0.5)^3 =$ _____
7. Half-life (1 periods): $400 \times (0.5)^1 =$ _____
8. Half-life (2 periods): $440 \times (0.5)^2 =$ _____
9. Half-life (3 periods): $480 \times (0.5)^3 =$ _____
10. Half-life (1 periods): $520 \times (0.5)^1 =$ _____
11. Doubling (2 periods): $125 \times 2^2 =$ _____
12. Doubling (3 periods): $150 \times 2^3 =$ _____
13. Doubling (4 periods): $175 \times 2^4 =$ _____
14. Doubling (2 periods): $200 \times 2^2 =$ _____
15. Doubling (3 periods): $225 \times 2^3 =$ _____
16. Doubling (4 periods): $250 \times 2^4 =$ _____
17. Doubling (2 periods): $275 \times 2^2 =$ _____
18. Doubling (3 periods): $300 \times 2^3 =$ _____
19. Doubling (4 periods): $325 \times 2^4 =$ _____
20. Doubling (2 periods): $350 \times 2^2 =$ _____

**TEST 10D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Continuous Compounding, Doubling Time & Change of Base

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Doubling time: $\ln(2) \div 0.07$ (4 dec) = _____
2. Doubling time: $\ln(2) \div 0.05$ (4 dec) = _____
3. Continuous: $500 \times e^{0.04 \times 5}$ (4 dec) = _____
4. Continuous: $1500 \times e^{0.05 \times 10}$ (2 dec) = _____
5. Monthly: $250 \times (1 + 0.06 \div 12)^{24}$ (2 dec) = _____
6. 3 years at 10%: $1000 \times (1.10)^3 =$ _____
7. Change of base: $\ln(8) \div \ln(2) =$ _____
8. Change of base: $\ln(81) \div \ln(3) =$ _____
9. Change of base: $\ln(125) \div \ln(5) =$ _____
10. Change of base: $\log_{10}(32) \div \log_{10}(2) =$ _____
11. $\log_4(64) =$ _____
12. $\log_8(64) =$ _____
13. $\log_4(8)$ (decimal) = _____
14. $\log_9(27)$ (decimal) = _____
15. $27^{2/3} =$ _____
16. $16^{3/4} =$ _____
17. $32^{3/5} =$ _____
18. $64^{2/3} =$ _____
19. Solve: $4^x = 32$ (decimal) = _____
20. Solve: $9^x = 27$ (decimal) = _____

# Section 10: Exponents, Logarithms & Growth Models

Review Worked Examples 1 & 2 below, then solve Test 10D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 10D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 10D #5 $(250 \times (1+0.06/12)^{24}) \rightarrow$ [.06] [ENTER] [12] [÷] [1] [+] [24] [$y^x$] [250] [×] → 281.79

**EXAMPLE 2** 10D #18 $(64^{2/3}) \rightarrow (\sqrt[3]{64})^2 = 4^2 = 16$

**TEST 10D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Continuous Compounding, Doubling Time & Change of Base — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Change of base: $\ln(8) \div \ln(2) =$ _____
2. Change of base: $\ln(81) \div \ln(3) =$ _____
3. Change of base: $\ln(16) \div \ln(4) =$ _____
4. Change of base: $\ln(125) \div \ln(5) =$ _____
5. Change of base: $\ln(16) \div \ln(2) =$ _____
6. Change of base: $\ln(9) \div \ln(3) =$ _____
7. Change of base: $\ln(64) \div \ln(4) =$ _____
8. Change of base: $\ln(625) \div \ln(5) =$ _____
9. Change of base: $\ln(4) \div \ln(2) =$ _____
10. Change of base: $\ln(27) \div \ln(3) =$ _____
11. $27^{2/3} =$ _____
12. $64^{3/3} =$ _____
13. $125^{2/3} =$ _____
14. $27^{3/3} =$ _____
15. $64^{2/3} =$ _____
16. $125^{3/3} =$ _____
17. $27^{2/3} =$ _____
18. $64^{3/3} =$ _____
19. $125^{2/3} =$ _____
20. $27^{3/3} =$ _____

**TEST 10D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 10 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Change of base: $\ln(16) \div \ln(2) =$ _____
2. Change of base: $\ln(9) \div \ln(3) =$ _____
3. Change of base: $\ln(64) \div \ln(4) =$ _____
4. Change of base: $\ln(625) \div \ln(5) =$ _____
5. Change of base: $\ln(4) \div \ln(2) =$ _____
6. Change of base: $\ln(27) \div \ln(3) =$ _____
7. Change of base: $\ln(256) \div \ln(4) =$ _____
8. Change of base: $\ln(25) \div \ln(5) =$ _____
9. Change of base: $\ln(8) \div \ln(2) =$ _____
10. Change of base: $\ln(81) \div \ln(3) =$ _____
11. $64^{2/3} =$ _____
12. $125^{3/3} =$ _____
13. $216^{2/3} =$ _____
14. $64^{3/3} =$ _____
15. $125^{2/3} =$ _____
16. $216^{3/3} =$ _____
17. $64^{2/3} =$ _____
18. $125^{3/3} =$ _____
19. $216^{2/3} =$ _____
20. $64^{3/3} =$ _____

# Section 11: Combinatorics, Probability & Statistics

Factorials, permutations, combinations, probability ratios, and dataset statistics. • Study the Champion Way & complete W1–W6 before starting Tests 11A1–11D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 11: COMBINATORICS, PROBABILITY & STATISTICS

| | |
|---|---|
| RULE 1 | **Rule 1 — Cancel Factorials Before Multiplying: Never compute 8! and 5! separately for 8C3! Cancel 5! immediately: 8C3 = (8 × 7 × 6) ÷ (3 × 2 × 1) = 8 × 7 = 56 (since 3×2 cancels 6)!** |
| RULE 2 | Rule 2 — Pascal's Row Sum Identity: The sum of all combinations in Row n (nC0 + nC1 + ... + nCn) is always $2^n$ (so Row 5 sum = $2^5$ = 32; Row 6 sum = $2^6$ = 64). |
| RULE 3 | Rule 3 — Symmetric Dataset SD & Mean Shortcuts: For three evenly spaced numbers {a - d, a, a + d}, the Mean is a and the Sample Standard Deviation is simply d! |

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

### Archetype 1 — Instant Combination Cancellation: Evaluate 8C3

Stack Keys: `8` `ENTER` `7` `×`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 8! = 40,320, 3! = 6, 5! = 120, multiply 6 × 120 = 720, divide.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Write 3 descending factors (8 × 7 × 6) ÷ 6 → cancel 6 to get 8 × 7 = 56 in 1 second!**

### Archetype 2 — Word Arrangements with Repeated Letters: Arrangements of LEVEL

Stack Keys: `5` `↶` `x!` `4` `÷`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Write out permutations by hand.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: 5 letters with two Ls (2!) and two Es (2!): 5! ÷ (2! × 2!) = 120 ÷ 4 = 30!**

### Archetype 3 — Evenly Spaced Sample Standard Deviation: Sample SD of {10, 14, 18}

Stack Keys: `1` `0` `Σ+` `1` `4` `Σ+` `1` `8` `Σ+`

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute deviations $(-4)^2$, $0^2$, $4^2$, sum to 32, divide by $n$-1 = 2 to get 16, take $\sqrt{16}$.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: For any 3 evenly spaced numbers {$a$ - $d$, a, $a$ + $d$}, Sample SD = step $d$ = 4 immediately!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** **7C2 =** _____

Shortcut Cue: (7 × 6) ÷ 2 = 7 × 3

**W2.** **8C3 =** _____

Shortcut Cue: (8 × 7 × 6) ÷ 6 = 8 × 7

**W3.** **Arrangements of BOOK (4! ÷ 2!) =** _____

Shortcut Cue: 24 ÷ 2

**W4.** **Row 5 sum: 5C0 + ... + 5C5 =** _____

Shortcut Cue: Pascal's Row Sum = $2^5$

**W5.** **5 coins: P(exactly 2 Heads) =** _____

Shortcut Cue: 5C2 ÷ $2^5$ = 10 ÷ 32

**W6.** **Sample SD of {2, 5, 8} =** _____

Shortcut Cue: 3 evenly spaced numbers → step $d$ = 3

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 11: Combinatorics, Probability & Statistics

Review Worked Examples 1 & 2 below, then solve Test 11A1 (Level 1: Foundation (Sprint 1)) & Test 11A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $7C2 = 21$ $((7 \times 6) \div 2 = 7 \times 3)$

**EXAMPLE 2** $8C3 = 56$ $((8 \times 7 \times 6) \div 6 = 8 \times 7)$

**TEST 11A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 4 min

Factorials, Sample Means, Medians & Ranges

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $0!$ = _____
2. $1!$ = _____
3. $2!$ = _____
4. $3!$ = _____
5. $4!$ = _____
6. $5!$ = _____
7. $6!$ = _____
8. Sum: $1 + 2 + 3 + 4 + 5$ = _____
9. Mean of $\{2, 4, 6, 8, 10\}$ = _____
10. Mean of $\{10, 20, 30\}$ = _____
11. Mean of $\{4, 7, 10, 11\}$ = _____
12. Median of $\{3, 5, 9, 12, 15\}$ = _____
13. Median of $\{2, 6, 10, 14\}$ = _____
14. Range of $\{4, 9, 15, 22\}$ = _____
15. Mode of $\{2, 5, 5, 8, 9\}$ = _____
16. Fair coin $P$(Heads) (decimal) = _____
17. Fair 6-sided die P(roll = 6) = _____
18. Fair 6-sided die $P$(even) (decimal) = _____
19. Fair 6-sided die P(roll $>$ 4) = _____
20. $3 \times 4$ Choices (Fundamental Rule) = _____

**TEST 11A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 4 min

Factorials, Sample Means, Medians & Ranges — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Mean of $\{6, 8, 10\}$ = _____
2. Mean of $\{7, 10, 13\}$ = _____
3. Mean of $\{8, 12, 16\}$ = _____
4. Mean of $\{9, 14, 19\}$ = _____
5. Mean of $\{14, 16, 18\}$ = _____
6. Mean of $\{15, 18, 21\}$ = _____
7. Mean of $\{16, 20, 24\}$ = _____
8. Mean of $\{17, 22, 27\}$ = _____
9. Mean of $\{22, 24, 26\}$ = _____
10. Mean of $\{23, 26, 29\}$ = _____
11. Range of $\{3, 7, 12, 17\}$ = _____
12. Range of $\{5, 9, 14, 22\}$ = _____
13. Range of $\{7, 11, 16, 27\}$ = _____
14. Range of $\{9, 13, 18, 32\}$ = _____
15. Range of $\{11, 15, 20, 37\}$ = _____
16. Range of $\{13, 17, 22, 42\}$ = _____
17. Range of $\{15, 19, 24, 47\}$ = _____
18. Range of $\{17, 21, 26, 52\}$ = _____
19. Range of $\{19, 23, 28, 57\}$ = _____
20. Range of $\{21, 25, 30, 62\}$ = _____

# Section 11: Combinatorics, Probability & Statistics

Review Worked Examples 1 & 2 below, then solve Test 11A3 (Level 1: Foundation (Sprint 3)) & Test 11B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** Arrangements of BOOK $(4! \div 2!) = 12$ $(24 \div 2)$

**EXAMPLE 2** Row 5 sum: $5C0 + \ldots + 5C5 = 32$ (Pascal's Row Sum $= 2^5$)

**TEST 11A3 • LEVEL 1: FOUNDATION (SPRINT 3)** Goal: 4 min

Bridge from Level 1 to Level 2 (Permutations nPr, Combinations n)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Mean of {9, 11, 13} = _____
2. Mean of {10, 13, 16} = _____
3. Mean of {11, 15, 19} = _____
4. Mean of {12, 17, 22} = _____
5. Mean of {17, 19, 21} = _____
6. Mean of {18, 21, 24} = _____
7. Mean of {19, 23, 27} = _____
8. Mean of {20, 25, 30} = _____
9. Mean of {25, 27, 29} = _____
10. Mean of {26, 29, 32} = _____
11. Range of {4, 8, 13, 20} = _____
12. Range of {6, 10, 15, 25} = _____
13. Range of {8, 12, 17, 30} = _____
14. Range of {10, 14, 19, 35} = _____
15. Range of {12, 16, 21, 40} = _____
16. Range of {14, 18, 23, 45} = _____
17. Range of {16, 20, 25, 50} = _____
18. Range of {18, 22, 27, 55} = _____
19. Range of {20, 24, 29, 60} = _____
20. Range of {22, 26, 31, 65} = _____

**TEST 11B1 • LEVEL 2: CORE FLUENCY (SPRINT 1)** Goal: 5 min

Permutations nPr, Combinations nCr & Factorial Quotients

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $5! \div 4!$ = _____
2. $6! \div 5!$ = _____
3. $5P2$ = _____
4. $6P2$ = _____
5. $6P3$ = _____
6. $7P2$ = _____
7. $4C2$ = _____
8. $5C2$ = _____
9. $5C3$ = _____
10. $6C2$ = _____
11. $6C3$ = _____
12. $7C2$ = _____
13. $7C3$ = _____
14. $8C2$ = _____
15. $8C3$ = _____
16. $9C2$ = _____
17. $10C2$ = _____
18. $10C3$ = _____
19. $12C2$ = _____
20. $100C1$ = _____

# Section 11: Combinatorics, Probability & Statistics

Review Worked Examples 1 & 2 below, then solve Test 11B2 (Level 2: Core Fluency (Sprint 2)) & Test 11B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** **Archetype 1 — Instant Combination Cancellation: Evaluate 8C3 → Write 3 descending factors (8 × 7 × 6) ÷ 6 → cancel 6 to get 8 × 7 = 56 in 1 second!**

**EXAMPLE 2** **5 coins: P(exactly 2 Heads) = 0.3125 (5C2 ÷ $2^5$ = 10 ÷ 32)**

**TEST 11B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** **Goal: 5 min**

Permutations nPr, Combinations nCr & Factorial Quotients — Speed Builder

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | **6P2** = | |
| 2. | **7C3** = | |
| 3. | **8C2** = | |
| 4. | **9P2** = | |
| 5. | **10C2** = | |
| 6. | **11C3** = | |
| 7. | **12P2** = | |
| 8. | **13C3** = | |
| 9. | **6C2** = | |
| 10. | **7P2** = | |
| 11. | **8C2** = | |
| 12. | **9C3** = | |
| 13. | **10P2** = | |
| 14. | **11C3** = | |
| 15. | **12C2** = | |
| 16. | **13P2** = | |
| 17. | **6C2** = | |
| 18. | **7C3** = | |
| 19. | **8P2** = | |
| 20. | **9C3** = | |

**TEST 11B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** **Goal: 5 min**

Bridge from Level 2 to Level 3 (Independent Events, Arrangements)

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

| | | |
|---|---|---|
| 1. | **7P2** = | |
| 2. | **8C3** = | |
| 3. | **9C2** = | |
| 4. | **10P2** = | |
| 5. | **11C2** = | |
| 6. | **12C3** = | |
| 7. | **13P2** = | |
| 8. | **14C3** = | |
| 9. | **7C2** = | |
| 10. | **8P2** = | |
| 11. | **9C2** = | |
| 12. | **10C3** = | |
| 13. | **11P2** = | |
| 14. | **12C3** = | |
| 15. | **13C2** = | |
| 16. | **14P2** = | |
| 17. | **7C2** = | |
| 18. | **8C3** = | |
| 19. | **9P2** = | |
| 20. | **10C3** = | |

# Section 11: Combinatorics, Probability & Statistics

Review Worked Examples 1 & 2 below, then solve Test 11C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 11C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Word Arrangements with Repeated Letters: Arrangements of LEVEL → 5 letters with two Ls (2!) and two Es (2!): $5! \div (2! \times 2!) = 120 \div 4 = 30!$

**EXAMPLE 2** 11B #15 ($8C3$) → $(8 \times 7 \times 6) \div 6 = 56$

**TEST 11C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Independent Events, Arrangements & Weighted Averages

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 2 coins: P(both Heads) (decimal) = _____
2. 3 coins: P(all 3 Heads) (decimal) = _____
3. 3 coins: P(at least 1 Tail) (decimal) = _____
4. 2 fair 6-sided dice: Total outcomes = _____
5. 2 dice: Ways to roll sum = 7 = _____
6. 2 dice: P(sum = 7) = _____
7. 2 dice: P(sum = 2) = _____
8. 2 dice: P(sum = 12) = _____
9. Arrangements of letters in CAT = _____
10. Arrangements of letters in MATH = _____
11. Arrangements of letters in STACK = _____
12. Distinct arrangements of MOM = _____
13. Distinct arrangements of BOOK = _____
14. Distinct arrangements of LEVEL = _____
15. Handshakes among 6 people = _____
16. Handshakes among 8 people = _____
17. Diagonals in a hexagon = _____
18. Mean of 5 numbers is 12 → Sum = _____
19. Mean of {4, 8, 12, x} is 10 → $x$ = _____
20. 1 ÷ 6C2 (4 dec) = _____

**TEST 11C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Independent Events, Arrangements & Weighted Averages — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Handshakes among 7 people = _____
2. Handshakes among 8 people = _____
3. Handshakes among 9 people = _____
4. Handshakes among 10 people = _____
5. Handshakes among 11 people = _____
6. Handshakes among 12 people = _____
7. Handshakes among 13 people = _____
8. Handshakes among 14 people = _____
9. Handshakes among 15 people = _____
10. Handshakes among 16 people = _____
11. Mean of 4 numbers is 13 → Sum = _____
12. Mean of 5 numbers is 15 → Sum = _____
13. Mean of 6 numbers is 17 → Sum = _____
14. Mean of 7 numbers is 19 → Sum = _____
15. Mean of 8 numbers is 21 → Sum = _____
16. Mean of 4 numbers is 23 → Sum = _____
17. Mean of 5 numbers is 25 → Sum = _____
18. Mean of 6 numbers is 27 → Sum = _____
19. Mean of 7 numbers is 29 → Sum = _____
20. Mean of 8 numbers is 31 → Sum = _____

# Section 11: Combinatorics, Probability & Statistics

Review Worked Examples 1 & 2 below, then solve Test 11C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 11D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** — Archetype 3 — Evenly Spaced Sample Standard Deviation: Sample SD of {10, 14, 18} → For any 3 evenly spaced numbers {$a$ - $d$, a, $a$ + $d$}, Sample SD = step $d$ = 4 immediately!

**EXAMPLE 2** — 11C #14 (Arrangements of LEVEL) → 5! ÷ (2! × 2!) = 120 ÷ 4 = 30

**TEST 11C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Binomial Probability, Pascal's T)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Handshakes among 9 people = _____
2. Handshakes among 10 people = _____
3. Handshakes among 11 people = _____
4. Handshakes among 12 people = _____
5. Handshakes among 13 people = _____
6. Handshakes among 14 people = _____
7. Handshakes among 15 people = _____
8. Handshakes among 16 people = _____
9. Handshakes among 17 people = _____
10. Handshakes among 18 people = _____
11. Mean of 4 numbers is 16 → Sum = _____
12. Mean of 5 numbers is 18 → Sum = _____
13. Mean of 6 numbers is 20 → Sum = _____
14. Mean of 7 numbers is 22 → Sum = _____
15. Mean of 8 numbers is 24 → Sum = _____
16. Mean of 4 numbers is 26 → Sum = _____
17. Mean of 5 numbers is 28 → Sum = _____
18. Mean of 6 numbers is 30 → Sum = _____
19. Mean of 7 numbers is 32 → Sum = _____
20. Mean of 8 numbers is 34 → Sum = _____

**TEST 11D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Binomial Probability, Pascal's Triangle & Variance

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Row 4 sum: 4C0+4C1+4C2+4C3+4C4 = _____
2. Row 5 sum: 5C0 + 5C1 + ... + 5C5 = _____
3. Row 6 sum: 6C0 + 6C1 + ... + 6C6 = _____
4. 4 coins: P(exactly 2 Heads) (decimal) = _____
5. 4 coins: P(exactly 3 Heads) (decimal) = _____
6. 5 coins: Ways to get 2 Heads = _____
7. 5 coins: P(exactly 2 Heads) (decimal) = _____
8. Bag (3 Red, 2 Blue), draw 2: P(both Red) = _____
9. Bag (4 Red, 2 Blue), draw 2: P(both Red) = _____
10. Committee of 2 Boys (from 4) & 2 Girls (from 3) = _____
11. Circular arrangements of 5 people = _____
12. Circular arrangements of 6 people = _____
13. Population variance $\sigma^2$ of {2, 4, 6} = _____
14. Sample SD of {2, 5, 8} = _____
15. Sample SD of {10, 14, 18} = _____
16. $z$-score: $x = 85$, $\mu = 75$, $\sigma = 5$ → _____
17. $z$-score: $x = 64$, $\mu = 70$, $\sigma = 4$ → _____
18. Expected value: 0.5(10) + 0.5(20) = _____
19. Expected value of 1 fair die (1..6) = _____
20. Expected value of 2 fair dice sum = _____

# Section 11: Combinatorics, Probability & Statistics

Review Worked Examples 1 & 2 below, then solve Test 11D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 11D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 11D #3 (Row 6 sum 6C0+...+6C6) → $2^6 = 64$

**EXAMPLE 2** 11D #15 (Sample SD of {10,14,18}) → 3 evenly spaced values with step $d = 4$ → SD = 4

**TEST 11D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Binomial Probability, Pascal's Triangle & Variance — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Row 4 sum: 4C0 + ... + 4C4 = _____
2. Row 5 sum: 5C0 + ... + 5C5 = _____
3. Row 6 sum: 6C0 + ... + 6C6 = _____
4. Row 7 sum: 7C0 + ... + 7C7 = _____
5. Row 8 sum: 8C0 + ... + 8C8 = _____
6. Row 9 sum: 9C0 + ... + 9C9 = _____
7. Row 10 sum: 10C0 + ... + 10C10 = _____
8. Sample SD of {21, 25, 29} = _____
9. Sample SD of {22, 27, 32} = _____
10. Sample SD of {23, 29, 35} = _____
11. Sample SD of {24, 31, 38} = _____
12. Sample SD of {25, 33, 41} = _____
13. Sample SD of {26, 35, 44} = _____
14. Sample SD of {27, 37, 47} = _____
15. $z$-score: $x = 72$, $\mu = 60$, $\sigma = 4$ → _____
16. $z$-score: $x = 80$, $\mu = 65$, $\sigma = 5$ → _____
17. $z$-score: $x = 88$, $\mu = 70$, $\sigma = 6$ → _____
18. $z$-score: $x = 96$, $\mu = 75$, $\sigma = 7$ → _____
19. $z$-score: $x = 92$, $\mu = 80$, $\sigma = 4$ → _____
20. $z$-score: $x = 100$, $\mu = 85$, $\sigma = 5$ → _____

**TEST 11D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 11 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Row 5 sum: 5C0 + ... + 5C5 = _____
2. Row 6 sum: 6C0 + ... + 6C6 = _____
3. Row 7 sum: 7C0 + ... + 7C7 = _____
4. Row 8 sum: 8C0 + ... + 8C8 = _____
5. Row 9 sum: 9C0 + ... + 9C9 = _____
6. Row 10 sum: 10C0 + ... + 10C10 = _____
7. Row 11 sum: 11C0 + ... + 11C11 = _____
8. Sample SD of {23, 28, 33} = _____
9. Sample SD of {24, 30, 36} = _____
10. Sample SD of {25, 32, 39} = _____
11. Sample SD of {26, 34, 42} = _____
12. Sample SD of {27, 36, 45} = _____
13. Sample SD of {28, 38, 48} = _____
14. Sample SD of {29, 40, 51} = _____
15. $z$-score: $x = 76$, $\mu = 60$, $\sigma = 4$ → _____
16. $z$-score: $x = 85$, $\mu = 65$, $\sigma = 5$ → _____
17. $z$-score: $x = 94$, $\mu = 70$, $\sigma = 6$ → _____
18. $z$-score: $x = 103$, $\mu = 75$, $\sigma = 7$ → _____
19. $z$-score: $x = 96$, $\mu = 80$, $\sigma = 4$ → _____
20. $z$-score: $x = 105$, $\mu = 85$, $\sigma = 5$ → _____

# Section 12: Base Conversions, Binary & Bitwise Logic

Binary, octal, and hexadecimal place value, bit shifting, and boolean gates. • Study the Champion Way & complete W1–W6 before starting Tests 12A1–12D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — SECTION 12: BASE CONVERSIONS, BINARY & BITWISE LOGIC

**RULE 1** **Rule 1 — Group Binary into 4-Bit Nibbles (8-4-2-1) for Instant Hex Conversion: Every 4 binary bits map directly to 1 Hex digit (0000=0 .. 1010=*A* .. 1111=*F*). Example: 1010 $1100_2$ = $AC_{16}$!**

**RULE 2** Rule 2 — Bitwise Shifts Are Powers of Two: Left shift $x << k$ multiplies x by $2^k$; right shift $x >> k$ divides x by $2^k$ (integer part).

**RULE 3** Rule 3 — Two's Complement & Bitwise NOT Shortcuts: In $n$-bit unsigned binary, NOT($x$) = ($2^n$ - 1) - $x$, and 8-bit Two's Complement of -$k$ is (256 - $k$) in Hex (so -1 = $FF_{16}$, -2 = $FE_{16}$).

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — 8-Bit Binary to Hexadecimal: Convert $10101100_2$ to Hex** Stack Keys: BIN 1 0 1 0 1 1 0 0 ↰ BASE HEX

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Convert to decimal (128 + 32 + 8 + 4 = 172), then divide by 16 to get 10 R 12.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Split into two 4-bit nibbles: $1010_2$ = 10 = *A* and $1100_2$ = 12 = *C* → $AC_{16}$ instantly!**

**Archetype 2 — Bitwise AND / XOR Evaluation: Evaluate 12 AND 10 and 12 XOR 10** Stack Keys: 1 2 ENTER 1 0 AND

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Guess decimal operations.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Compare 4-bit $1100_2$ (12) and $1010_2$ (10): shared 1-bit is $1000_2$ = 8 (AND); differing bits are $0110_2$ = 6 (XOR)!**

**Archetype 3 — Alternating Bit Mask XOR: Evaluate 170 XOR 85** Stack Keys: 1 7 0 ENTER 8 5 XOR

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Convert 170 and 85 bit-by-bit on scratch paper.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: 170 ($10101010_2$) and 85 ($01010101_2$) are exact bit complements adding to 255, so 170 XOR 85 = 255!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** **Hex $2F_{16}$ in decimal =** _____

Shortcut Cue: 2 × 16 + 15 = 32 + 15

**W2.** **12 ($1100_2$) XOR 10 ($1010_2$) =** _____

Shortcut Cue: Differing bits: $0110_2$

**W3.** **Left shift: 7 << 3 =** _____

Shortcut Cue: $7 \times 2^3 = 7 \times 8$

**W4.** **Binary $10101100_2$ in hexadecimal =** _____

Shortcut Cue: Nibbles 1010 (A) and 1100 (C)

**W5.** **Decimal 4095 ($2^{12} - 1$) in Hex =** _____

Shortcut Cue: Three 4-bit 1111 nibbles → FFF

**W6.** **8-bit Two's Complement of -1 (Hex) =** _____

Shortcut Cue: 256 - 1 = 255 = $FF_{16}$

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Section 12: Base Conversions, Binary & Bitwise Logic

Review Worked Examples 1 & 2 below, then solve Test 12A1 (Level 1: Foundation (Sprint 1)) & Test 12A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** Hex $2F_{16}$ in decimal = 47 ($2 \times 16 + 15 = 32 + 15$)

**EXAMPLE 2** 12 ($1100_2$) XOR 10 ($1010_2$) = 6 (Differing bits: $0110_2$)

## TEST 12A1 • LEVEL 1: FOUNDATION (SPRINT 1) Goal: 4 min

4-Bit Binary & Decimal Conversions

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. Binary $0001_2$ in decimal = _____
2. Binary $0010_2$ in decimal = _____
3. Binary $0011_2$ in decimal = _____
4. Binary $0100_2$ in decimal = _____
5. Binary $0101_2$ in decimal = _____
6. Binary $0110_2$ in decimal = _____
7. Binary $0111_2$ in decimal = _____
8. Binary $1000_2$ in decimal = _____
9. Binary $1001_2$ in decimal = _____
10. Binary $1010_2$ in decimal = _____
11. Binary $1100_2$ in decimal = _____
12. Binary $1111_2$ in decimal = _____
13. Decimal 5 in 4-bit binary = _____
14. Decimal 6 in 4-bit binary = _____
15. Decimal 9 in 4-bit binary = _____
16. Decimal 10 in 4-bit binary = _____
17. Decimal 12 in 4-bit binary = _____
18. Decimal 13 in 4-bit binary = _____
19. Decimal 14 in 4-bit binary = _____
20. Decimal 15 in 4-bit binary = _____

## TEST 12A2 • LEVEL 1: FOUNDATION (SPRINT 2) Goal: 4 min

4-Bit Binary & Decimal Conversions — Fluency Builder

**Date:** _____ **Time:** _____ **Score:** ___/20 **Contest Pts:** ____

1. Binary $00110_2$ in decimal = _____
2. Decimal 5 in 4-bit binary = _____
3. Binary $01100_2$ in decimal = _____
4. Decimal 7 in 4-bit binary = _____
5. Binary $10010_2$ in decimal = _____
6. Decimal 9 in 4-bit binary = _____
7. Binary $11000_2$ in decimal = _____
8. Decimal 11 in 4-bit binary = _____
9. Binary $11110_2$ in decimal = _____
10. Decimal 13 in 4-bit binary = _____
11. Binary $00110_2$ in decimal = _____
12. Decimal 15 in 4-bit binary = _____
13. Binary $01100_2$ in decimal = _____
14. Decimal 2 in 4-bit binary = _____
15. Binary $10010_2$ in decimal = _____
16. Decimal 4 in 4-bit binary = _____
17. Binary $11000_2$ in decimal = _____
18. Decimal 6 in 4-bit binary = _____
19. Binary $11110_2$ in decimal = _____
20. Decimal 8 in 4-bit binary = _____

# Section 12: Base Conversions, Binary & Bitwise Logic

Review Worked Examples 1 & 2 below, then solve Test 12A3 (Level 1: Foundation (Sprint 3)) & Test 12B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** Left shift: $7 << 3 = 56$ $(7 \times 2^3 = 7 \times 8)$

**EXAMPLE 2** Binary $10101100_2$ in hexadecimal = AC (Nibbles 1010 ($A$) and 1100 ($C$))

### TEST 12A3 • LEVEL 1: FOUNDATION (SPRINT 3) Goal: 4 min

Bridge from Level 1 to Level 2 (8-Bit Binary, Octal (Base 8) & H)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $01011_2$ in decimal = _____
2. Decimal 8 in 4-bit binary = _____
3. Binary $10001_2$ in decimal = _____
4. Decimal 10 in 4-bit binary = _____
5. Binary $10111_2$ in decimal = _____
6. Decimal 12 in 4-bit binary = _____
7. Binary $11101_2$ in decimal = _____
8. Decimal 14 in 4-bit binary = _____
9. Binary $00101_2$ in decimal = _____
10. Decimal 1 in 4-bit binary = _____
11. Binary $01011_2$ in decimal = _____
12. Decimal 3 in 4-bit binary = _____
13. Binary $10001_2$ in decimal = _____
14. Decimal 5 in 4-bit binary = _____
15. Binary $10111_2$ in decimal = _____
16. Decimal 7 in 4-bit binary = _____
17. Binary $11101_2$ in decimal = _____
18. Decimal 9 in 4-bit binary = _____
19. Binary $00101_2$ in decimal = _____
20. Decimal 11 in 4-bit binary = _____

### TEST 12B1 • LEVEL 2: CORE FLUENCY (SPRINT 1) Goal: 5 min

8-Bit Binary, Octal (Base 8) & Hexadecimal (Base 16)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $10000_2$ in decimal = _____
2. Binary $10100_2$ in decimal = _____
3. Binary $11111_2$ in decimal = _____
4. Binary $100000_2$ in decimal = _____
5. Binary $1000000_2$ in decimal = _____
6. Binary $11111111_2$ in decimal = _____
7. Hex $A_{16}$ in decimal = _____
8. Hex $C_{16}$ in decimal = _____
9. Hex $F_{16}$ in decimal = _____
10. Hex $10_{16}$ in decimal = _____
11. Hex $1A_{16}$ in decimal = _____
12. Hex $20_{16}$ in decimal = _____
13. Hex $2F_{16}$ in decimal = _____
14. Hex $FF_{16}$ in decimal = _____
15. Decimal 16 in hexadecimal = _____
16. Decimal 31 in hexadecimal = _____
17. Decimal 64 in hexadecimal = _____
18. Decimal 255 in hexadecimal = _____
19. Decimal 8 in octal (Base 8) = _____
20. Decimal 64 in octal (Base 8) = _____

# Section 12: Base Conversions, Binary & Bitwise Logic

Review Worked Examples 1 & 2 below, then solve Test 12B2 (Level 2: Core Fluency (Sprint 2)) & Test 12B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — 8-Bit Binary to Hexadecimal: Convert $10101100_2$ to Hex → Split into two 4-bit nibbles: $1010_2 = 10 = A$ and $1100_2 = 12 = C$ → $AC_{16}$ instantly!

**EXAMPLE 2** Decimal 4095 ($2^{12}$ - 1) in Hex = FFF (Three 4-bit 1111 nibbles → FFF)

**TEST 12B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 5 min

8-Bit Binary, Octal (Base 8) & Hexadecimal (Base 16) — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Hex $18_{16}$ in decimal = _____
2. Decimal 35 in hexadecimal = _____
3. Hex $2E_{16}$ in decimal = _____
4. Decimal 57 in hexadecimal = _____
5. Hex $44_{16}$ in decimal = _____
6. Decimal 79 in hexadecimal = _____
7. Hex $5A_{16}$ in decimal = _____
8. Decimal 101 in hexadecimal = _____
9. Hex $70_{16}$ in decimal = _____
10. Decimal 123 in hexadecimal = _____
11. Hex $86_{16}$ in decimal = _____
12. Decimal 145 in hexadecimal = _____
13. Hex $9C_{16}$ in decimal = _____
14. Decimal 167 in hexadecimal = _____
15. Hex $B2_{16}$ in decimal = _____
16. Decimal 189 in hexadecimal = _____
17. Hex $C8_{16}$ in decimal = _____
18. Decimal 211 in hexadecimal = _____
19. Hex $DE_{16}$ in decimal = _____
20. Decimal 233 in hexadecimal = _____

**TEST 12B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 5 min

Bridge from Level 2 to Level 3 (Bitwise Boolean Logic (AND, OR, )

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Hex $1F_{16}$ in decimal = _____
2. Decimal 42 in hexadecimal = _____
3. Hex $35_{16}$ in decimal = _____
4. Decimal 64 in hexadecimal = _____
5. Hex $4B_{16}$ in decimal = _____
6. Decimal 86 in hexadecimal = _____
7. Hex $61_{16}$ in decimal = _____
8. Decimal 108 in hexadecimal = _____
9. Hex $77_{16}$ in decimal = _____
10. Decimal 130 in hexadecimal = _____
11. Hex $8D_{16}$ in decimal = _____
12. Decimal 152 in hexadecimal = _____
13. Hex $A3_{16}$ in decimal = _____
14. Decimal 174 in hexadecimal = _____
15. Hex $B9_{16}$ in decimal = _____
16. Decimal 196 in hexadecimal = _____
17. Hex $CF_{16}$ in decimal = _____
18. Decimal 218 in hexadecimal = _____
19. Hex $E5_{16}$ in decimal = _____
20. Decimal 240 in hexadecimal = _____

# Section 12: Base Conversions, Binary & Bitwise Logic

Review Worked Examples 1 & 2 below, then solve Test 12C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test 12C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Bitwise AND / XOR Evaluation: Evaluate 12 AND 10 and 12 XOR 10 → Compare 4-bit $1100_2$ (12) and $1010_2$ (10): shared 1-bit is $1000_2 = 8$ (AND); differing bits are $0110_2 = 6$ (XOR)!

**EXAMPLE 2** 12C #15 (192 AND 240) → 192 = 128 + 64, both inside top mask 240 → 192

**TEST 12C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 5 min

Bitwise Boolean Logic (AND, OR, XOR) & Bit Shifts

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 1 AND 1 = _____
2. 1 AND 0 = _____
3. 1 OR 0 = _____
4. 1 XOR 1 = _____
5. 1 XOR 0 = _____
6. 12 AND 10 = _____
7. 12 OR 10 = _____
8. 12 XOR 10 = _____
9. 6 AND 3 = _____
10. 6 OR 3 = _____
11. 6 XOR 3 = _____
12. 15 AND 9 = _____
13. 15 XOR 9 = _____
14. 255 AND 15 = _____
15. 192 AND 240 = _____
16. Left shift: 5 << 1 = _____
17. Left shift: 3 << 2 = _____
18. Left shift: 7 << 3 = _____
19. Right shift: 24 >> 2 = _____
20. Right shift: 64 >> 3 = _____

**TEST 12C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 5 min

Bitwise Boolean Logic (AND, OR, XOR) & Bit Shifts — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 8 AND 4 = _____
2. 9 OR 6 = _____
3. 10 XOR 8 = _____
4. Left shift: 11 << 1 = _____
5. 12 AND 12 = _____
6. 13 OR 14 = _____
7. 14 XOR 4 = _____
8. Left shift: 15 << 2 = _____
9. 16 AND 8 = _____
10. 17 OR 10 = _____
11. 18 XOR 12 = _____
12. Left shift: 19 << 3 = _____
13. 20 AND 4 = _____
14. 21 OR 6 = _____
15. 22 XOR 8 = _____
16. Left shift: 23 << 1 = _____
17. 24 AND 12 = _____
18. 25 OR 14 = _____
19. 26 XOR 4 = _____
20. Left shift: 27 << 2 = _____

# Section 12: Base Conversions, Binary & Bitwise Logic

Review Worked Examples 1 & 2 below, then solve Test 12C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test 12D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Alternating Bit Mask XOR: Evaluate 170 XOR 85 → 170 ($10101010_2$) and 85 ($01010101_2$) are exact bit complements adding to 255, so 170 XOR 85 = 255!

**EXAMPLE 2** 12D #9 ($10101100_2$ in Hex) → $1010_2 = A$, $1100_2 = C$ → AC

**TEST 12C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 5 min

Bridge from Level 3 to Level 4 (Mixed-Base Arithmetic, Two's Com)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 11 AND 5 = _____
2. 12 OR 7 = _____
3. 13 XOR 9 = _____
4. Left shift: 14 << 1 = _____
5. 15 AND 13 = _____
6. 16 OR 3 = _____
7. 17 XOR 5 = _____
8. Left shift: 18 << 2 = _____
9. 19 AND 9 = _____
10. 20 OR 11 = _____
11. 21 XOR 13 = _____
12. Left shift: 22 << 3 = _____
13. 23 AND 5 = _____
14. 24 OR 7 = _____
15. 25 XOR 9 = _____
16. Left shift: 26 << 1 = _____
17. 27 AND 13 = _____
18. 28 OR 3 = _____
19. 29 XOR 5 = _____
20. Left shift: 30 << 2 = _____

**TEST 12D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Mixed-Base Arithmetic, Two's Complement & Hex Masks

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary sum: $1010_2 + 0101_2$ (in dec) = _____
2. Binary sum: $1100_2 + 0110_2$ (in dec) = _____
3. Hex sum: $A_{16} + 6_{16}$ (in Hex) = _____
4. Hex sum: $F_{16} + 1_{16}$ (in Hex) = _____
5. Hex sum: $1A_{16} + 16_{16}$ (in Hex) = _____
6. Octal $17_8$ in decimal = _____
7. Octal $77_8$ in decimal = _____
8. Octal $100_8$ in hexadecimal = _____
9. Binary $10101100_2$ in hexadecimal = _____
10. Binary $11110000_2$ in hexadecimal = _____
11. Hex $B4_{16}$ in 8-bit binary = _____
12. Decimal 4095 in hexadecimal = _____
13. Max unsigned 8-bit integer = _____
14. Max unsigned 10-bit integer = _____
15. 4-bit unsigned bitwise NOT of 5 = _____
16. 8-bit unsigned bitwise NOT of 200 = _____
17. 8-bit Two's Complement of -1 (in Hex) = _____
18. 8-bit Two's Complement of -2 (in Hex) = _____
19. (255 AND 240) >> 4 = _____
20. 170 XOR 85 = _____

# Section 12: Base Conversions, Binary & Bitwise Logic

Review Worked Examples 1 & 2 below, then solve Test 12D2 (Level 4: Advanced Challenge (Sprint 2)) & Test 12D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** 12D #12 (4095 in Hex) → $2^{12} - 1 = 12$ ones = three F nibbles → FFF

**EXAMPLE 2** 12D #20 (170 XOR 85) → Complementary bit masks $10101010_2$ ^ $01010101_2 = 11111111_2 = 255$

**TEST 12D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Mixed-Base Arithmetic, Two's Complement & Hex Masks — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $00110111_2$ in hexadecimal = _____
2. Binary $01001010_2$ in hexadecimal = _____
3. Binary $01011101_2$ in hexadecimal = _____
4. Binary $01110000_2$ in hexadecimal = _____
5. Binary $10000011_2$ in hexadecimal = _____
6. Binary $10010110_2$ in hexadecimal = _____
7. Binary $10101001_2$ in hexadecimal = _____
8. Binary $10111100_2$ in hexadecimal = _____
9. Binary $11001111_2$ in hexadecimal = _____
10. Binary $11100010_2$ in hexadecimal = _____
11. 8-bit Two's Complement of -4 (in Hex) = _____
12. 8-bit Two's Complement of -6 (in Hex) = _____
13. 8-bit Two's Complement of -8 (in Hex) = _____
14. 8-bit Two's Complement of -10 (in Hex) = _____
15. 8-bit Two's Complement of -12 (in Hex) = _____
16. 8-bit Two's Complement of -14 (in Hex) = _____
17. 8-bit Two's Complement of -16 (in Hex) = _____
18. 8-bit Two's Complement of -18 (in Hex) = _____
19. 8-bit Two's Complement of -20 (in Hex) = _____
20. 8-bit Two's Complement of -22 (in Hex) = _____

**TEST 12D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section 12 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $01000100_2$ in hexadecimal = _____
2. Binary $01010111_2$ in hexadecimal = _____
3. Binary $01101010_2$ in hexadecimal = _____
4. Binary $01111101_2$ in hexadecimal = _____
5. Binary $10010000_2$ in hexadecimal = _____
6. Binary $10100011_2$ in hexadecimal = _____
7. Binary $10110110_2$ in hexadecimal = _____
8. Binary $11001001_2$ in hexadecimal = _____
9. Binary $11011100_2$ in hexadecimal = _____
10. Binary $11101111_2$ in hexadecimal = _____
11. 8-bit Two's Complement of -5 (in Hex) = _____
12. 8-bit Two's Complement of -7 (in Hex) = _____
13. 8-bit Two's Complement of -9 (in Hex) = _____
14. 8-bit Two's Complement of -11 (in Hex) = _____
15. 8-bit Two's Complement of -13 (in Hex) = _____
16. 8-bit Two's Complement of -15 (in Hex) = _____
17. 8-bit Two's Complement of -17 (in Hex) = _____
18. 8-bit Two's Complement of -19 (in Hex) = _____
19. 8-bit Two's Complement of -21 (in Hex) = _____
20. 8-bit Two's Complement of -23 (in Hex) = _____

# Cumulative Review C3: All-Level Championship (Sections 1–12)

Grand championship speed tests spanning arithmetic, algebra, trig, logs, stats, and CS. • Study the Champion Way & complete W1–W6 before starting Tests C3A1–C3D3.

## 1. CHAMPIONSHIP SPEED RULES & SHORTCUTS — CUMULATIVE REVIEW C3: ALL-LEVEL CHAMPIONSHIP (SECTIONS 1–12)

**RULE 1** **Rule 1 — Championship Triage: Solve 1-second identity problems (Vieta's roots, $\sin^2+\cos^2=1$, $a^2-b^2$, GCF×LCM, Pascal $2^n$, nibble Hex) mentally, saving stack keystrokes for multi-step transcendentals.**

**RULE 2** Rule 2 — Zero Scratch-Paper Transcription on Transcendentals: Keep 64-bit precision on the stack from the deepest inner expression outward, rounding only the final display to 2 or 4 decimal places.

**RULE 3** Rule 3 — Use [→∂,r] for Every Hypotenuse, Distance, and Wave Amplitude Problem: 2D distance, vector magnitude, and max(a sin $x$ + $b$ cos $x$) are all solved by a single [→∂,r] operation.

## 2. SLOW WAY VS. CHAMPION WAY — WORKED SPEED ARCHETYPES & HARDWARE PIPELINES

**Archetype 1 — Mixed Fraction Quotient + Sum: Evaluate (3 1/3 ÷ 1 1/4) + 2/5** Stack Keys: [3] [.] [1] [.] [3] [ENTER] [1] [.] [1] [.] [4] [÷] [.] [2] [.] [5] [+]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Convert 10/3 ÷ 5/4 = 40/15 on paper, add 2/5, convert to decimal.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Key [3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+] → 3.0667 in 5 entries!**

**Archetype 2 — Continuous Compounding Sprint: Evaluate 1500 × $e^{0.05 \times 10}$** Stack Keys: [.] [0] [5] [ENTER] [1] [0] [×] [$e^x$] [1] [5] [0] [0] [×]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Compute 0.05 × 10, write 0.5, compute $e^{0.5}$, multiply by 1500.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: Start at the exponent: [.05] [ENTER] [10] [×] [$e^x$] [1500] [×] → 2473.08!**

**Archetype 3 — Grand Championship Multi-Identity Check: Evaluate GCF(24, 36) × LCM(24, 36)** Stack Keys: [2] [4] [ENTER] [3] [6] [×]

**SLOW / AMATEUR WAY (Avoid):**

Slow Way: Factor 24 and 36, find GCF = 12 and LCM = 72, multiply.

**CHAMPION SPEED WAY (Use This!):**

**Champion Way: GCF(a, b) × LCM(a, b) = $a \times b$ = 24 × 36 = (30 - 6)(30 + 6) = $30^2$ - $6^2$ = 900 - 36 = 864!**

## 3. UNTIMED GUIDED WARM-UP DRILL — PRACTICE THE CHAMPION SHORTCUTS BEFORE TIMING

**W1.** **12 cos(45°) + 5 sin(45°) (4 dec) =** _____

Shortcut Cue: (12 + 5) × sin(45°) = 17 × 0.7071

**W2.** **Monthly: 250 × (1 + 0.06 ÷ 12)$^{24}$ =** _____

Shortcut Cue: [.06][ENTER][12][÷][1][+][24][$y^x$][250][×]

**W3.** **1 + 2 + 3 + ... + 20 =** _____

Shortcut Cue: Gauss rule: 20 × 21 ÷ 2

**W4.** **GCF(24, 36) × LCM(24, 36) =** _____

Shortcut Cue: 24 × 36 = $30^2$ - $6^2$ = 900 - 36

**W5.** **max value of 5 sin($x$) + 12 cos($x$) =** _____

Shortcut Cue: Amplitude $\sqrt{5^2 + 12^2}$

**W6.** **5 coins: P(exactly 2 Heads) =** _____

Shortcut Cue: 5C2 ÷ 32 = 10 ÷ 32

**Warm-Up Check & Champion Pipelines: Solve on paper/stack first! Complete answer keys & keystrokes at stackcalc.io/curriculum/answers#speed-fluency**

# Cumulative Review C3: All-Level Championship (Sections 1–12)

Review Worked Examples 1 & 2 below, then solve Test C3A1 (Level 1: Foundation (Sprint 1)) & Test C3A2 (Level 1: Foundation (Sprint 2)).

**EXAMPLE 1** $12\cos(45°) + 5\sin(45°)$ (4 dec) $= 12.0208$ ((12 + 5) × sin(45°) = 17 × 0.7071)

**EXAMPLE 2** Monthly: $250 \times (1 + 0.06 \div 12)^{24} = 281.79$ ([.06][ENTER][12][÷][1][+][24][$y^x$][250][×])

**TEST C3A1 • LEVEL 1: FOUNDATION (SPRINT 1)** Goal: 5 min

Core Cross-Topic Speed: Arithmetic, Fractions, Powers & Binary

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $(18 - 6) \times 3 =$ _____
2. $3/4 + 1/2$ (decimal) = _____
3. GCF of 18 and 24 = _____
4. LCM of 6 and 8 = _____
5. Solve: $3x - 5 = 16$ _____
6. Hypotenuse for legs 5 and 12 = _____
7. $\sin(30°) + \cos(60°) =$ _____
8. $\tan(45°) \times 8 =$ _____
9. $2^5 =$ _____
10. $\log_{10}(1000) =$ _____
11. $\ln(e^4) =$ _____
12. 5! = _____
13. 5C2 = _____
14. Mean of {4, 8, 12, 16} = _____
15. Binary $1010_2$ in decimal = _____
16. Decimal 15 in 4-bit binary = _____
17. Hex $10_{16}$ in decimal = _____
18. 12 AND 10 = _____
19. $15^2 =$ _____
20. 1000 − 375 = _____

**TEST C3A2 • LEVEL 1: FOUNDATION (SPRINT 2)** Goal: 5 min

Core Cross-Topic Speed: Arithmetic, Fractions, Powers & Binary — Fluency Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $00110_2$ in decimal = _____
2. Decimal 5 in 4-bit binary = _____
3. Binary $01100_2$ in decimal = _____
4. Decimal 7 in 4-bit binary = _____
5. Binary $10010_2$ in decimal = _____
6. Decimal 9 in 4-bit binary = _____
7. Binary $11000_2$ in decimal = _____
8. Decimal 11 in 4-bit binary = _____
9. Binary $11110_2$ in decimal = _____
10. Decimal 13 in 4-bit binary = _____
11. Binary $00110_2$ in decimal = _____
12. Decimal 15 in 4-bit binary = _____
13. Binary $01100_2$ in decimal = _____
14. Decimal 2 in 4-bit binary = _____
15. Binary $10010_2$ in decimal = _____
16. Decimal 4 in 4-bit binary = _____
17. Binary $11000_2$ in decimal = _____
18. Decimal 6 in 4-bit binary = _____
19. Binary $11110_2$ in decimal = _____
20. Decimal 8 in 4-bit binary = _____

# Cumulative Review C3: All-Level Championship (Sections 1–12)

Review Worked Examples 1 & 2 below, then solve Test C3A3 (Level 1: Foundation (Sprint 3)) & Test C3B1 (Level 2: Core Fluency (Sprint 1)).

**EXAMPLE 1** $1 + 2 + 3 + \ldots + 20 = 210$ (Gauss rule: $20 \times 21 \div 2$)

**EXAMPLE 2** $\text{GCF}(24, 36) \times \text{LCM}(24, 36) = 864$ ($24 \times 36 = 30^2 - 6^2 = 900 - 36$)

**TEST C3A3 • LEVEL 1: FOUNDATION (SPRINT 3)** Goal: 5 min

Bridge from Level 1 to Level 2 (Intermediate Championship: Quadr)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $01011_2$ in decimal = _____
2. Decimal 8 in 4-bit binary = _____
3. Binary $10001_2$ in decimal = _____
4. Decimal 10 in 4-bit binary = _____
5. Binary $10111_2$ in decimal = _____
6. Decimal 12 in 4-bit binary = _____
7. Binary $11101_2$ in decimal = _____
8. Decimal 14 in 4-bit binary = _____
9. Binary $00101_2$ in decimal = _____
10. Decimal 1 in 4-bit binary = _____
11. Binary $01011_2$ in decimal = _____
12. Decimal 3 in 4-bit binary = _____
13. Binary $10001_2$ in decimal = _____
14. Decimal 5 in 4-bit binary = _____
15. Binary $10111_2$ in decimal = _____
16. Decimal 7 in 4-bit binary = _____
17. Binary $11101_2$ in decimal = _____
18. Decimal 9 in 4-bit binary = _____
19. Binary $00101_2$ in decimal = _____
20. Decimal 11 in 4-bit binary = _____

**TEST C3B1 • LEVEL 2: CORE FLUENCY (SPRINT 1)** Goal: 5 min

Intermediate Championship: Quadratics, Vectors, Logs & Combinatorics

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Sum of double 6 and near-double $(8 + 9)$ = _____
2. $((35 - 7) \times 4) \div (14 + 2)$ = _____
3. $(1/3 + 1/6) \times 18$ = _____
4. Greater root of $x^2 + 5x + 6 = 0$ : _____
5. Positive root of $x^2 - 25 = 0$ : _____
6. Slope from (-2, 1) to (2, 13) = _____
7. Vector $\langle 8, 15 \rangle \rightarrow$ Magnitude $r$ = _____
8. Hyp = 20, $\vartheta = 30°$ → Opp leg $y$ = _____
9. $2^{10}$ = _____
10. $\log_2(64)$ = _____
11. $1000 \times (1.05)^2$ = _____
12. Half-life (3 periods): $100 \times (0.5)^3$ = _____
13. 6P3 = _____
14. 7C3 = _____
15. 8C3 = _____
16. Decimal 255 in hexadecimal = _____
17. Decimal 64 in octal (Base 8) = _____
18. 12 OR 10 = _____
19. 12 XOR 10 = _____
20. 192 AND 240 = _____

# Cumulative Review C3: All-Level Championship (Sections 1–12)

Review Worked Examples 1 & 2 below, then solve Test C3B2 (Level 2: Core Fluency (Sprint 2)) & Test C3B3 (Level 2: Core Fluency (Sprint 3)).

**EXAMPLE 1** Archetype 1 — Mixed Fraction Quotient + Sum: Evaluate (3 1/3 ÷ 1 1/4) + 2/5 → Key [3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+] → 3.0667 in 5 entries!

**EXAMPLE 2** max value of $5\sin(x) + 12\cos(x) = 13$ (Amplitude $\sqrt{5^2 + 12^2}$)

**TEST C3B2 • LEVEL 2: CORE FLUENCY (SPRINT 2)** Goal: 5 min

Intermediate Championship: Quadratics, Vectors, Logs & Combinatorics — Speed Builder

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Hex $18_{16}$ in decimal = _____
2. Decimal 35 in hexadecimal = _____
3. Hex $2E_{16}$ in decimal = _____
4. Decimal 57 in hexadecimal = _____
5. Hex $44_{16}$ in decimal = _____
6. Decimal 79 in hexadecimal = _____
7. Hex $5A_{16}$ in decimal = _____
8. Decimal 101 in hexadecimal = _____
9. Hex $70_{16}$ in decimal = _____
10. Decimal 123 in hexadecimal = _____
11. Hex $86_{16}$ in decimal = _____
12. Decimal 145 in hexadecimal = _____
13. Hex $9C_{16}$ in decimal = _____
14. Decimal 167 in hexadecimal = _____
15. Hex $B2_{16}$ in decimal = _____
16. Decimal 189 in hexadecimal = _____
17. Hex $C8_{16}$ in decimal = _____
18. Decimal 211 in hexadecimal = _____
19. Hex $DE_{16}$ in decimal = _____
20. Decimal 233 in hexadecimal = _____

**TEST C3B3 • LEVEL 2: CORE FLUENCY (SPRINT 3)** Goal: 5 min

Bridge from Level 2 to Level 3 (Advanced Championship: Multi-Ste)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Hex $1F_{16}$ in decimal = _____
2. Decimal 42 in hexadecimal = _____
3. Hex $35_{16}$ in decimal = _____
4. Decimal 64 in hexadecimal = _____
5. Hex $4B_{16}$ in decimal = _____
6. Decimal 86 in hexadecimal = _____
7. Hex $61_{16}$ in decimal = _____
8. Decimal 108 in hexadecimal = _____
9. Hex $77_{16}$ in decimal = _____
10. Decimal 130 in hexadecimal = _____
11. Hex $8D_{16}$ in decimal = _____
12. Decimal 152 in hexadecimal = _____
13. Hex $A3_{16}$ in decimal = _____
14. Decimal 174 in hexadecimal = _____
15. Hex $B9_{16}$ in decimal = _____
16. Decimal 196 in hexadecimal = _____
17. Hex $CF_{16}$ in decimal = _____
18. Decimal 218 in hexadecimal = _____
19. Hex $E5_{16}$ in decimal = _____
20. Decimal 240 in hexadecimal = _____

# Cumulative Review C3: All-Level Championship (Sections 1–12)

Review Worked Examples 1 & 2 below, then solve Test C3C1 (Level 3: Multi-Step Speed (Sprint 1)) & Test C3C2 (Level 3: Multi-Step Speed (Sprint 2)).

**EXAMPLE 1** Archetype 2 — Continuous Compounding Sprint: Evaluate $1500 \times e^{0.05 \times 10}$ → Start at the exponent: [.05] [ENTER] [10] [×] [$e^x$] [1500] [×] → 2473.08!

**EXAMPLE 2** C3C #1 ((3 1/3 ÷ 1 1/4) + 2/5) → [3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+] → 3.0667

**TEST C3C1 • LEVEL 3: MULTI-STEP SPEED (SPRINT 1)** Goal: 6 min

Advanced Championship: Multi-Step Rationals, Trig & Growth

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. (3 1/3 ÷ 1 1/4) + 2/5 (4 dec) = _____
2. 1 ÷ (1/4 + 1/6) (decimal) = _____
3. Distance from (-3, 5) to (5, -10) = _____
4. Evaluate $2x^2 + 4x - 5$ at $x = 3$ : _____
5. Sum of roots of $x^2 - 11x + 28 = 0$ : _____
6. 12 cos(45°) + 5 sin(45°) (4 dec) = _____
7. $\sin^2(25°) + \cos^2(25°)$ = _____
8. Dot product ⟨3, 4⟩ · ⟨5, -2⟩ = _____
9. Doubling time: ln(2) ÷ 0.07 (4 dec) = _____
10. Continuous: $500 \times e^{0.04 \times 5}$ (4 dec) = _____
11. Continuous: $1500 \times e^{0.05 \times 10}$ (2 dec) = _____
12. Monthly: $250 \times (1 + 0.06 \div 12)^{24}$ (2 dec) = _____
13. 10C3 = _____
14. Distinct arrangements of LEVEL = _____
15. 2 dice: P(sum = 7) = _____
16. Decimal 4095 in hexadecimal = _____
17. Hex $2F_{16}$ in decimal = _____
18. Left shift: 9 << 3 = _____
19. 24 × 26 = _____
20. $35^2$ = _____

**TEST C3C2 • LEVEL 3: MULTI-STEP SPEED (SPRINT 2)** Goal: 6 min

Advanced Championship: Multi-Step Rationals, Trig & Growth — Inside-Out Mastery

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 8 AND 4 = _____
2. 9 OR 6 = _____
3. 10 XOR 8 = _____
4. Left shift: 11 << 1 = _____
5. 12 AND 12 = _____
6. 13 OR 14 = _____
7. 14 XOR 4 = _____
8. Left shift: 15 << 2 = _____
9. 16 AND 8 = _____
10. 17 OR 10 = _____
11. 18 XOR 12 = _____
12. Left shift: 19 << 3 = _____
13. 20 AND 4 = _____
14. 21 OR 6 = _____
15. 22 XOR 8 = _____
16. Left shift: 23 << 1 = _____
17. 24 AND 12 = _____
18. 25 OR 14 = _____
19. 26 XOR 4 = _____
20. Left shift: 27 << 2 = _____

# Cumulative Review C3: All-Level Championship (Sections 1–12)

Review Worked Examples 1 & 2 below, then solve Test C3C3 (Level 3: Multi-Step Speed (Sprint 3)) & Test C3D1 (Level 4: Advanced Challenge (Sprint 1)).

**EXAMPLE 1** Archetype 3 — Grand Championship Multi-Identity Check: Evaluate GCF(24, 36) × LCM(24, 36) → GCF(a, b) × LCM(a, b) = $a \times b = 24 \times 36 = (30 - 6)(30 + 6) = 30^2 - 6^2 = 900 - 36 = 864$!

**EXAMPLE 2** C3C #6 (12 cos 45° + 5 sin 45°) → 17 × sin(45°) → 12.0208

**TEST C3C3 • LEVEL 3: MULTI-STEP SPEED (SPRINT 3)** Goal: 6 min

Bridge from Level 3 to Level 4 (Grand Invitational Speed Challen)

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. 11 AND 5 = _____
2. 12 OR 7 = _____
3. 13 XOR 9 = _____
4. Left shift: 14 << 1 = _____
5. 15 AND 13 = _____
6. 16 OR 3 = _____
7. 17 XOR 5 = _____
8. Left shift: 18 << 2 = _____
9. 19 AND 9 = _____
10. 20 OR 11 = _____
11. 21 XOR 13 = _____
12. Left shift: 22 << 3 = _____
13. 23 AND 5 = _____
14. 24 OR 7 = _____
15. 25 XOR 9 = _____
16. Left shift: 26 << 1 = _____
17. 27 AND 13 = _____
18. 28 OR 3 = _____
19. 29 XOR 5 = _____
20. Left shift: 30 << 2 = _____

**TEST C3D1 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 1)** Goal: 6 min

Grand Invitational Speed Challenge

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. $1 + 2 + 3 + \ldots + 20 =$ _____
2. $99 + 98 + 97 + 96 + 95 =$ _____
3. $25^2 - 15^2 =$ _____
4. $24 \times (5/6 - 3/8) =$ _____
5. GCF(24, 36) × LCM(24, 36) = _____
6. Number of positive factors of 36 = _____
7. 3D diagonal $\sqrt{3^2 + 4^2 + 12^2} =$ _____
8. Minimum value of $y = x^2 - 8x + 19$ : _____
9. Solve for x: $2x + y = 15$, $x - y = 3$ : _____
10. Law of Cosines: $a$=8, $b$=5, $C$=60° → $c$ = _____
11. max value of 5 sin($x$) + 12 cos($x$) = _____
12. Change of base: ln(125) ÷ ln(5) = _____
13. $64^{2/3} =$ _____
14. Solve: $4^x = 32$ (decimal) = _____
15. Row 6 sum: 6C0 + 6C1 + ... + 6C6 = _____
16. 5 coins: P(exactly 2 Heads) (decimal) = _____
17. Sample SD of {10, 14, 18} = _____
18. Binary $10101100_2$ in hexadecimal = _____
19. 8-bit Two's Complement of -1 (in Hex) = _____
20. (170 XOR 85) = _____

# Cumulative Review C3: All-Level Championship (Sections 1–12)

Review Worked Examples 1 & 2 below, then solve Test C3D2 (Level 4: Advanced Challenge (Sprint 2)) & Test C3D3 (Level 4: Championship Trial (Sprint 3)).

**EXAMPLE 1** C3D #5 (GCF(24,36)×LCM(24,36)) → 24 × 36 = (30 - 6)(30 + 6) = 900 - 36 = 864

**EXAMPLE 2** C3D #10 (Law of Cosines $a$=8,$b$=5,$C$=60°) → $c^2$ = 64 + 25 - 40 = 49 → $c$ = 7

**TEST C3D2 • LEVEL 4: ADVANCED CHALLENGE (SPRINT 2)** Goal: 6 min

Grand Invitational Speed Challenge — Contest Pacing

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $00110111_2$ in hexadecimal = _____
2. Binary $01001010_2$ in hexadecimal = _____
3. Binary $01011101_2$ in hexadecimal = _____
4. Binary $01110000_2$ in hexadecimal = _____
5. Binary $10000011_2$ in hexadecimal = _____
6. Binary $10010110_2$ in hexadecimal = _____
7. Binary $10101001_2$ in hexadecimal = _____
8. Binary $10111100_2$ in hexadecimal = _____
9. Binary $11001111_2$ in hexadecimal = _____
10. Binary $11100010_2$ in hexadecimal = _____
11. 8-bit Two's Complement of -4 (in Hex) = _____
12. 8-bit Two's Complement of -6 (in Hex) = _____
13. 8-bit Two's Complement of -8 (in Hex) = _____
14. 8-bit Two's Complement of -10 (in Hex) = _____
15. 8-bit Two's Complement of -12 (in Hex) = _____
16. 8-bit Two's Complement of -14 (in Hex) = _____
17. 8-bit Two's Complement of -16 (in Hex) = _____
18. 8-bit Two's Complement of -18 (in Hex) = _____
19. 8-bit Two's Complement of -20 (in Hex) = _____
20. 8-bit Two's Complement of -22 (in Hex) = _____

**TEST C3D3 • LEVEL 4: CHAMPIONSHIP TRIAL (SPRINT 3)** Goal: 5 min

Section C3 Cumulative Tournament Sprint

Date: _____ Time: _____ Score: ___/20 Contest Pts: ____

1. Binary $01000100_2$ in hexadecimal = _____
2. Binary $01010111_2$ in hexadecimal = _____
3. Binary $01101010_2$ in hexadecimal = _____
4. Binary $01111101_2$ in hexadecimal = _____
5. Binary $10010000_2$ in hexadecimal = _____
6. Binary $10100011_2$ in hexadecimal = _____
7. Binary $10110110_2$ in hexadecimal = _____
8. Binary $11001001_2$ in hexadecimal = _____
9. Binary $11011100_2$ in hexadecimal = _____
10. Binary $11101111_2$ in hexadecimal = _____
11. 8-bit Two's Complement of -5 (in Hex) = _____
12. 8-bit Two's Complement of -7 (in Hex) = _____
13. 8-bit Two's Complement of -9 (in Hex) = _____
14. 8-bit Two's Complement of -11 (in Hex) = _____
15. 8-bit Two's Complement of -13 (in Hex) = _____
16. 8-bit Two's Complement of -15 (in Hex) = _____
17. 8-bit Two's Complement of -17 (in Hex) = _____
18. 8-bit Two's Complement of -19 (in Hex) = _____
19. 8-bit Two's Complement of -21 (in Hex) = _____
20. 8-bit Two's Complement of -23 (in Hex) = _____

# Inside-Out Stack Reference Sheet (Part 1)

Five zero-memory RPN stack patterns that eliminate parentheses and avoid memory register round-trips.

---

### 1. DEEPEST INNER EXPRESSION FIRST (ZERO PARENTHESES)

Evaluate the deepest nested group first so intermediate results wait in Register Y. Example: 100 ÷ (3 + 4×5):

**Hardware Pipeline:** 

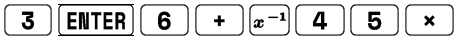

### 2. INSTANT DOUBLING & SQUARING ([ENTER] [+] AND [ENTER] [×])

Pressing [ENTER] duplicates Register X into Register Y without retyping digits. To double 18 and add 5:

**Hardware Pipeline:** 

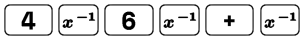

### 3. NON-COMMUTATIVE SWAP (X SWAP Y) & RECIPROCAL ([1/x])

When denominator D is computed first, type numerator N and press [x swap y] [÷]—or invert D with [1/x] and multiply:

**Hardware Pipeline:** 


### 4. PARALLEL RESISTANCE & HARMONIC SUMS (AUTOMATIC STACK LIFT)

For 1 / (1/4 + 1/6), automatic stack lift pushes the first reciprocal into Y while the second is computed:

**Hardware Pipeline:** 


### 5. 2D VECTOR MAGNITUDE, HYPOTENUSE & WAVE AMPLITUDE ([→ϑ,r])

Find the hypotenuse $\sqrt{5^2 + 12^2}$ or wave amplitude max(a sin $x$ + $b$ cos $x$) in one operation on the SC-32:

**Hardware Pipeline:** 


### THE 4-REGISTER AUTOMATIC LIFT & DROP CYCLE (SC-6 VS. SC-32)

- The Stack Lift (Entry): Typing a number and pressing [ENTER] pushes Register Z into T, Y into Z, and X into Y. The typed number sits in Register X ready for immediate operation.
- The Stack Drop (Binary Ops): Pressing any 2-input operator (+, -, ×, ÷, $y^x$) computes f(Y, X) → $X$, drops Z into Y, and copies T into Z while preserving T at the top of the stack.
- Zero Scratchpad Memory: Complex formulas like (a + b) × (c + d) ÷ (e - f) never require parentheses or memory store/recall buttons—every intermediate quantity is automatically held in hardware registers!

**REGISTER STATES DURING COMPOUND EVALUATION: (2 + 3) × (4 + 5)**

```
1. Key [2] [ENTER] [3] [+] --> Stack: [Y = 0, X = 5]
2. Key [4] [ENTER] [5] [+] --> Stack: [Y = 5, X = 9] (Note: first result 5 automatically waited in Y!)
3. Key [×] --> Stack: [X = 45] (Parentheses eliminated, 0 memory variables used)
```

# Championship Speed & Mental Math Identities (Part 2)

Ten high-speed algebraic, number sense, and geometric identities for competition math.

| TEN ESSENTIAL CHAMPIONSHIP SPEED IDENTITIES & SHORTCUT FORMULAS | |
|---|---|
| **1. Gauss Series & Odd Sum** | $1 + 2 + ... + n = n(n + 1) \div 2$ \| Sum of first k odd numbers $(1 + 3 + 5 + ...) = k^2$ |
| **2. Sum of Squares & Cubes** | $1^2 + 2^2 + ... + n^2 = n(n + 1)(2n + 1) \div 6$ \| $1^3 + 2^3 + ... + n^3 = [n(n + 1) \div 2]^2$ |
| **3. Difference of Squares** | $a^2 - b^2 = (a - b)(a + b)$ \| Symmetric Product: $(m - d)(m + d) = m^2 - d^2$ (e.g., $28 \times 32 = 30^2 - 2^2 = 896$) |
| **4. Squares Ending in 5** | $(10t + 5)^2 = 100 \times t(t + 1) + 25$ (e.g., $35^2$: $3 \times 4 = 12 \rightarrow 1225$; $75^2$: $7 \times 8 = 56 \rightarrow 5625$) |
| **5. Numbers Near 100** | $(100 + a)(100 + b) = 100(100 + a + b) + ab$ (e.g., $104 \times 107 = 100 \times 111 + 28 = 11{,}128$) |
| **6. GCF × LCM & Divisors** | GCF(a, b) × LCM(a, b) = $a \times b$ \| Number of divisors of $p^a \times q^b \times r^c = (a + 1)(b + 1)(c + 1)$ |
| **7. Vieta's Quadratic Rules** | For $ax^2 + bx + c = 0$: Sum of Roots $(r_1 + r_2) = -b / a$ \| Product of Roots $(r_1 \times r_2) = c / a$ |
| **8. 2×2 Sum/Difference System** | If $x + y = S$ and $x - y = D$, then $x = (S + D) \div 2$ (Average) and $y = (S - D) \div 2$ (Half-Difference) |
| **9. 3D Diagonal & Equilateral** | 3D Box Diagonal $d = \sqrt{l^2 + w^2 + h^2}$ \| Equilateral Triangle Area $A = (\sqrt{3} \div 4)s^2 \approx 0.43301 \times s^2$ |
| **10. Pythagorean Triples** | Primitive Families: (3, 4, 5), (5, 12, 13), (7, 24, 25), (8, 15, 17), (9, 40, 41), (20, 21, 29) |

### MAGNITUDE ESTIMATION & MENTAL SANITY-CHECK PROTOCOL

Before recording any calculated answer on a timed test, perform a 1-second order-of-magnitude sanity check:
- Leading-Digit Truncation: $48.7 \times 312 \approx 50 \times 300 = 15{,}000$. If your calculator shows 1,519.4 or 151,940, catch the decimal point error immediately!
- Sign and Range Checks: Sum of positive terms must exceed each addend. Square roots of numbers $> 1$ must be smaller than the radicand. Sines and cosines must remain within $[-1, +1]$.
- Contest Pen-Down Rule: In UIL and Number Sense tournaments, erasing is prohibited. Write only when certain of your first digit!

# Sprint Drill 1: Complement Arithmetic & Base Bridges

Base 10, 100, and 1,000 Complements • Goal: Under 2 Minutes • 20 Problems

**Date:** _____ **Time Goal:** < 2 min **Actual Time:** _____ **Score:** _____ / 20

1. **100 − 37** _____
   Shortcut: Complement to 100: tens add to 9, ones add to 10
2. **100 − 64** _____
   Shortcut: Complement to 100: tens add to 9, ones add to 10
3. **100 − 19** _____
   Shortcut: Complement to 100: tens add to 9, ones add to 10
4. **100 − 82** _____
   Shortcut: Complement to 100: tens add to 9, ones add to 10
5. **1,000 − 456** _____
   Shortcut: All from 9, last from 10 (left-to-right complement)
6. **1,000 − 728** _____
   Shortcut: All from 9, last from 10 (left-to-right complement)
7. **1,000 − 183** _____
   Shortcut: All from 9, last from 10 (left-to-right complement)
8. **1,000 − 891** _____
   Shortcut: All from 9, last from 10 (left-to-right complement)
9. **47 + 58** _____
   Shortcut: Round addend to next multiple of 10, then adjust
10. **86 + 39** _____
    Shortcut: Round addend to next multiple of 10, then adjust
11. **65 + 47** _____
    Shortcut: Round addend to next multiple of 10, then adjust
12. **94 + 78** _____
    Shortcut: Round addend to next multiple of 10, then adjust
13. **132 − 48** _____
    Shortcut: Bridge through 100/1,000 using distance on number line
14. **154 − 67** _____
    Shortcut: Bridge through 100/1,000 using distance on number line
15. **200 − 73** _____
    Shortcut: Bridge through 100/1,000 using distance on number line
16. **500 − 164** _____
    Shortcut: Bridge through 100/1,000 using distance on number line
17. **2,000 − 645** _____
    Shortcut: Bridge through 100/1,000 using distance on number line
18. **10,000 − 3,420** _____
    Shortcut: Bridge through 100/1,000 using distance on number line
19. **389 + 198** _____
    Shortcut: Round addend to next multiple of 10, then adjust
20. **742 − 299** _____
    Shortcut: Bridge through 100/1,000 using distance on number line

**Self-Check Online: Worked solutions and Champion Pipelines at https://stackcalc.io/curriculum/answers#speed-fluency**

# Sprint Drill 2: Instant Doubling, Halving & Distribution

Multiplication by 5, 11, 15, 25, 50 & Factoring • Goal: Under 2 Minutes • 20 Problems

**Date:** _____ **Time Goal:** < **2 min** **Actual Time:** _____ **Score:** _____ / **20**

**1.** **24 × 5** _____

Shortcut: Halve the number, then multiply by 10 (append 0)

**2.** **48 × 5** _____

Shortcut: Halve the number, then multiply by 10 (append 0)

**3.** **86 × 5** _____

Shortcut: Halve the number, then multiply by 10 (append 0)

**4.** **32 × 25** _____

Shortcut: Divide by 4 (halve twice), then multiply by 100

**5.** **44 × 25** _____

Shortcut: Divide by 4 (halve twice), then multiply by 100

**6.** **68 × 25** _____

Shortcut: Divide by 4 (halve twice), then multiply by 100

**7.** **84 × 50** _____

Shortcut: Halve the number, then multiply by 100 (append 00)

**8.** **36 × 50** _____

Shortcut: Halve the number, then multiply by 100 (append 00)

**9.** **16 × 15** _____

Shortcut: Multiply by 10, then add half of that product

**10.** **24 × 15** _____

Shortcut: Multiply by 10, then add half of that product

**11.** **32 × 15** _____

Shortcut: Multiply by 10, then add half of that product

**12.** **18 × 12** _____

Shortcut: Symmetric difference of squares: $(m-d)(m+d) = m^2 - d^2$

**13.** **22 × 18** _____

Shortcut: Symmetric difference of squares: $(m-d)(m+d) = m^2 - d^2$

**14.** **45 × 11** _____

Shortcut: Split outer digits and insert their sum in the middle

**15.** **73 × 11** _____

Shortcut: Split outer digits and insert their sum in the middle

**16.** **89 × 11** _____

Shortcut: Split outer digits and insert their sum in the middle

**17.** **64 ÷ 4** _____

Shortcut: Halve repeatedly or scale both terms to divide cleanly

**18.** **144 ÷ 8** _____

Shortcut: Halve repeatedly or scale both terms to divide cleanly

**19.** **240 ÷ 15** _____

Shortcut: Halve repeatedly or scale both terms to divide cleanly

**20.** **350 ÷ 25** _____

Shortcut: Halve repeatedly or scale both terms to divide cleanly

**Self-Check Online: Worked solutions and Champion Pipelines at https://stackcalc.io/curriculum/answers#speed-fluency**

# Sprint Drill 3: Rapid Squares, Square Roots & Cubes

Two-Digit Squares, Root Extraction & Cube Sums • Goal: Under 2 Minutes • 20 Problems

**Date:** _____ **Time Goal:** < 2 min **Actual Time:** _____ **Score:** _____ / 20

1. $15^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

2. $25^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

3. $35^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

4. $45^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

5. $65^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

6. $75^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

7. $85^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

8. $95^2$ _____

   Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

9. $11^2$ _____

   Shortcut: Base-10 binomial square: $(10 + d)^2 = 100 + 20d + d^2$

10. $12^2$ _____

    Shortcut: Base-10 binomial square: $(10 + d)^2 = 100 + 20d + d^2$

11. $13^2$ _____

    Shortcut: Base-10 binomial square: $(10 + d)^2 = 100 + 20d + d^2$

12. $14^2$ _____

    Shortcut: Base-10 binomial square: $(10 + d)^2 = 100 + 20d + d^2$

13. $16^2$ _____

    Shortcut: Base-10 binomial square: $(10 + d)^2 = 100 + 20d + d^2$

14. $20^2$ _____

    Shortcut: Base-10 binomial square: $(10 + d)^2 = 100 + 20d + d^2$

15. $\sqrt{144}$ _____

    Shortcut: Extract exact principal square root from perfect square

16. $\sqrt{196}$ _____

    Shortcut: Extract exact principal square root from perfect square

17. $\sqrt{225}$ _____

    Shortcut: Extract exact principal square root from perfect square

18. $\sqrt{400}$ _____

    Shortcut: Extract exact principal square root from perfect square

19. $2^3 + 3^3$ _____

    Shortcut: Evaluate perfect cubes ($2^3$=8, $3^3$=27, $4^3$=64) first, then combine

20. $4^3 - 5^2$ _____

    Shortcut: Ends-in-5 square: multiply tens digit t × (t + 1), append 25

**Self-Check Online: Worked solutions and Champion Pipelines at https://stackcalc.io/curriculum/answers#speed-fluency**

# Sprint Drill 4: Fraction, Decimal & Percent Benchmark Equivalents

Unit Fractions, Eighths, Thirds, Sixths & Percents • Goal: Under 2 Minutes • 20 Problems

**Date:** _____ **Time Goal:** < **2 min** **Actual Time:** _____ **Score:** _____ / **20**

1. **1/2 to Decimal** _____
   Shortcut: Convert denominator to 100 for instant decimal/percent
2. **1/4 to Percent** _____
   Shortcut: Convert denominator to 100 for instant decimal/percent
3. **3/4 to Decimal** _____
   Shortcut: Convert denominator to 100 for instant decimal/percent
4. **1/5 to Percent** _____
   Shortcut: Fifths benchmark: double numerator for tenths (20% per fifth)
5. **2/5 to Decimal** _____
   Shortcut: Fifths benchmark: double numerator for tenths (20% per fifth)
6. **3/5 to Percent** _____
   Shortcut: Fifths benchmark: double numerator for tenths (20% per fifth)
7. **4/5 to Decimal** _____
   Shortcut: Fifths benchmark: double numerator for tenths (20% per fifth)
8. **1/8 to Decimal** _____
   Shortcut: Eighths ladder: 1/8 = 0.125 → multiply by numerator
9. **3/8 to Decimal** _____
   Shortcut: Eighths ladder: 1/8 = 0.125 → multiply by numerator
10. **5/8 to Decimal** _____
    Shortcut: Eighths ladder: 1/8 = 0.125 → multiply by numerator
11. **7/8 to Decimal** _____
    Shortcut: Eighths ladder: 1/8 = 0.125 → multiply by numerator
12. **1/3 to Percent** _____
    Shortcut: Thirds & sixths benchmark: 1/3 = 33.3%, 1/6 = 0.167
13. **2/3 to Percent** _____
    Shortcut: Thirds & sixths benchmark: 1/3 = 33.3%, 1/6 = 0.167
14. **1/6 to Decimal** _____
    Shortcut: Thirds & sixths benchmark: 1/3 = 33.3%, 1/6 = 0.167
15. **5/6 to Decimal** _____
    Shortcut: Thirds & sixths benchmark: 1/3 = 33.3%, 1/6 = 0.167
16. **1/10 to Percent** _____
    Shortcut: Convert denominator to 100 for instant decimal/percent
17. **7/10 to Percent** _____
    Shortcut: Convert denominator to 100 for instant decimal/percent
18. **1/20 to Percent** _____
    Shortcut: Convert denominator to 100 for instant decimal/percent
19. **0.375 to Fraction** _____
    Shortcut: Eighths ladder: 1/8 = 0.125 → multiply by numerator
20. **0.625 to Fraction** _____
    Shortcut: Eighths ladder: 1/8 = 0.125 → multiply by numerator

**Self-Check Online: Worked solutions and Champion Pipelines at https://stackcalc.io/curriculum/answers#speed-fluency**

# Sprint Drill 5: Proportions, Rates & Unit Conversions

Equivalent Ratios, Speed, Time & Scale Conversions • Goal: Under 2 Minutes • 20 Problems

**Date:** _____ **Time Goal:** < 2 min **Actual Time:** _____ **Score:** _____ / 20

1. **$3/4 = x/20$** _____
   Shortcut: Scale numerator and denominator by the constant ratio factor

2. **$5/8 = 15/x$** _____
   Shortcut: Scale numerator and denominator by the constant ratio factor

3. **$2/7 = x/35$** _____
   Shortcut: Scale numerator and denominator by the constant ratio factor

4. **$9/12 = 12/x$** _____
   Shortcut: Scale numerator and denominator by the constant ratio factor

5. **4 ft in inches** _____
   Shortcut: Multiply or divide by exact unit conversion factor

6. **3 yd in feet** _____
   Shortcut: Multiply or divide by exact unit conversion factor

7. **2.5 m in cm** _____
   Shortcut: Multiply or divide by exact unit conversion factor

8. **4.2 km in meters** _____
   Shortcut: Multiply or divide by exact unit conversion factor

9. **3 hr in minutes** _____
   Shortcut: Multiply or divide by exact unit conversion factor

10. **150 sec in minutes** _____
    Shortcut: Multiply or divide by exact unit conversion factor

11. **50% of 84** _____
    Shortcut: Decompose percent into 50%, 25%, 10%, or 5% mental blocks

12. **25% of 120** _____
    Shortcut: Decompose percent into 50%, 25%, 10%, or 5% mental blocks

13. **10% of 350** _____
    Shortcut: Decompose percent into 50%, 25%, 10%, or 5% mental blocks

14. **15% of 60** _____
    Shortcut: Decompose percent into 50%, 25%, 10%, or 5% mental blocks

15. **20% of 45** _____
    Shortcut: Decompose percent into 50%, 25%, 10%, or 5% mental blocks

16. **5% of 80** _____
    Shortcut: Decompose percent into 50%, 25%, 10%, or 5% mental blocks

17. **60 mi/hr in mi/min** _____
    Shortcut: Multiply or divide by exact unit conversion factor

18. **12 gal in quarts** _____
    Shortcut: Multiply or divide by exact unit conversion factor

19. **32 oz in pounds** _____
    Shortcut: Multiply or divide by exact unit conversion factor

20. **1.5 kg in grams** _____
    Shortcut: Multiply or divide by exact unit conversion factor

**Self-Check Online: Worked solutions and Champion Pipelines at https://stackcalc.io/curriculum/answers#speed-fluency**

# Sprint Drill 6: Polynomials, Factoring & Quadratic Shortcuts

Binomial Products, Difference of Squares & Roots • Goal: Under 2 Minutes • 20 Problems

**Date:** _____ **Time Goal:** < 2 min **Actual Time:** _____ **Score:** _____ / 20

1. $(x+2)(x+3)$ _____

   Shortcut: Expand binomial product using area model or FOIL identity

2. $(x+4)(x+5)$ _____

   Shortcut: Expand binomial product using area model or FOIL identity

3. $(x-3)(x+3)$ _____

   Shortcut: Expand binomial product using area model or FOIL identity

4. $(x-7)(x+7)$ _____

   Shortcut: Expand binomial product using area model or FOIL identity

5. $(x+5)^2$ _____

   Shortcut: Expand binomial product using area model or FOIL identity

6. $(x-4)^2$ _____

   Shortcut: Expand binomial product using area model or FOIL identity

7. Factor: $x^2+7x+12$ _____

   Shortcut: Find two integers whose sum is b and product is c

8. Factor: $x^2+9x+20$ _____

   Shortcut: Find two integers whose sum is b and product is c

9. Factor: $x^2-5x+6$ _____

   Shortcut: Find two integers whose sum is b and product is c

10. Factor: $x^2-8x+15$ _____

    Shortcut: Find two integers whose sum is b and product is c

11. Factor: $x^2-25$ _____

    Shortcut: Find two integers whose sum is b and product is c

12. Factor: $x^2-64$ _____

    Shortcut: Find two integers whose sum is b and product is c

13. Roots: $x^2+6x+8=0$ _____

    Shortcut: Factor trinomial or apply [POLY] solver to extract roots

14. Roots: $x^2-7x+10=0$ _____

    Shortcut: Factor trinomial or apply [POLY] solver to extract roots

15. Roots: $x^2-9=0$ _____

    Shortcut: Factor trinomial or apply [POLY] solver to extract roots

16. Roots: $2x^2+5x+2=0$ _____

    Shortcut: Factor trinomial or apply [POLY] solver to extract roots

17. Vieta Sum: $x^2-11x+24$ _____

    Shortcut: Vieta's formulas: Sum of roots $=-b/a$, Product $=c/a$

18. Vieta Prod: $x^2-11x+24$ _____

    Shortcut: Vieta's formulas: Sum of roots $=-b/a$, Product $=c/a$

19. Discriminant: $x^2+4x+4$ _____

    Shortcut: Compute $\Delta=b^2-4ac$ to test root multiplicity

20. Discriminant: $x^2+5x+6$ _____

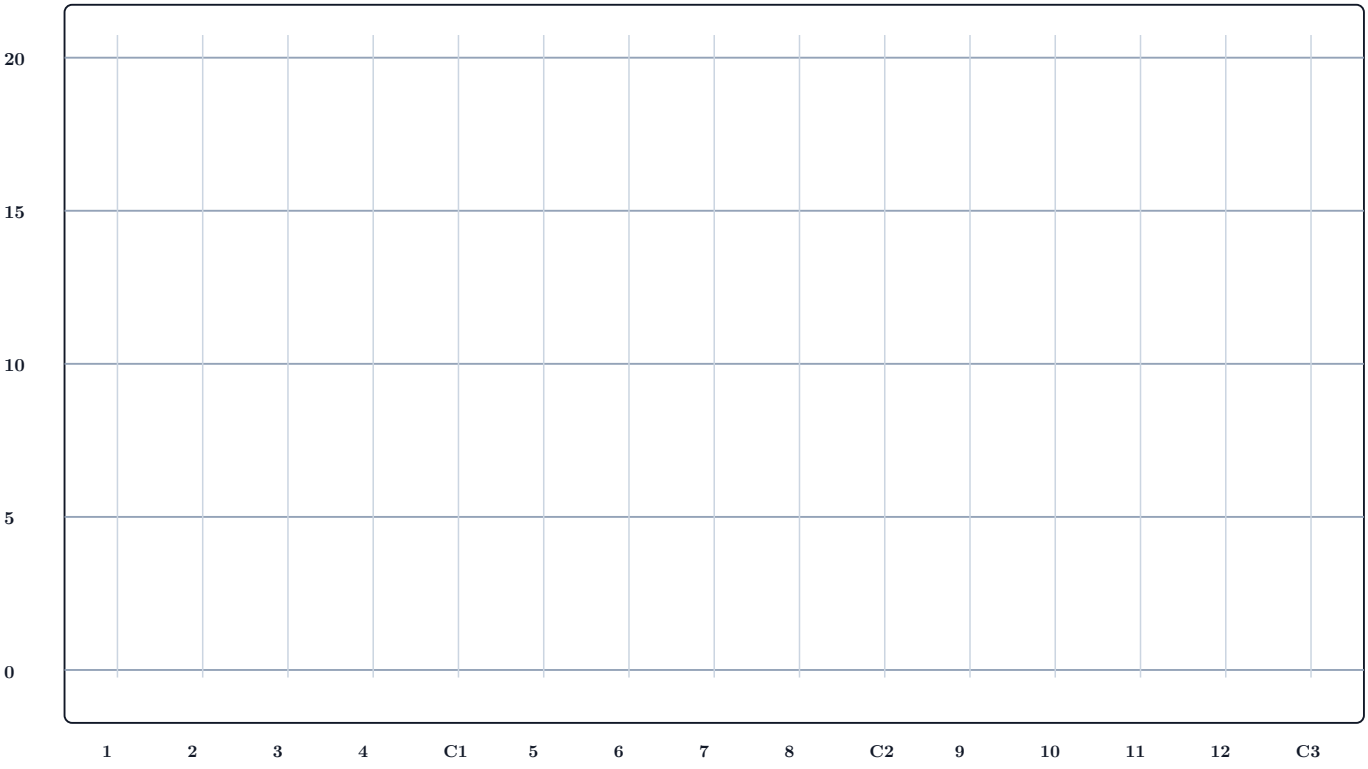
    Shortcut: Compute $\Delta=b^2-4ac$ to test root multiplicity

**Self-Check Online: Worked solutions and Champion Pipelines at https://stackcalc.io/curriculum/answers#speed-fluency**

# Speed & Fluency Score Progression Graph

Plot your highest score (out of 20) across all 15 sections and trace your growth curve.

### 1. 15-SECTION SCORE PROGRESSION GRAPH (PLOT YOUR PERSONAL BEST FROM 0 TO 20)




### 2. SPEED & ACCURACY VELOCITY BENCHMARKS (PROBLEMS PER MINUTE)

| | |
|---|---|
| **Novice Tier** | **1–3 problems / min (Score: 10–14 / 20 in 6 min)** |

Focus on accuracy and CRA block models before racing.

| | |
|---|---|
| **Apprentice Tier** | **4–5 problems / min (Score: 15–17 / 20 in 4 min)** |

Consistent stack execution without retyping digits.

| | |
|---|---|
| **Skilled Tier** | **6–8 problems / min (Score: 18–19 / 20 in 3 min)** |

Zero memory keystrokes; automatic reciprocal and square shortcuts.

| | |
|---|---|
| **Master Tier** | **9–10 problems / min (Score: 20 / 20 in 2–2.5 min)** |

Instant mental estimation and algebraic identity pattern matching.

| | |
|---|---|
| **Champion Tier** | **12+ problems / min (Score: 20 / 20 in < 2 min)** |

Tournament-level Number Sense & UIL Calculator fluency.

# Personal Best Log & 180-Day Consistency Heatmap

Record your all-time lowest times and color each day's square in the 36-week school year streak calendar.

## 1. ALL-TIME PERSONAL BESTS BY DIFFICULTY LEVEL

| | | | |
|---|---|---|---|
| **Level 1: Foundation (Tests A1–A3)** | **Fastest Time:** _____m _____s | Date: _____ | Section: _____ |
| **Level 2: Core Fluency (Tests B1–B3)** | **Fastest Time:** _____m _____s | Date: _____ | Section: _____ |
| **Level 3: Multi-Step Speed (Tests C1–C3)** | **Fastest Time:** _____m _____s | Date: _____ | Section: _____ |
| **Level 4: Tournament Challenge (Tests D1–D3)** | **Fastest Time:** _____m _____s | Date: _____ | Section: _____ |

## 2. 180-DAY CONSISTENCY HEATMAP • COLOR 1 BOX FOR EACH COMPLETED TIMED TEST (T001–T180)

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Week 01: | T001 | T002 | T003 | T004 | T005 | Week 19: | T091 | T092 | T093 | T094 | T095 |
| Week 02: | T006 | T007 | T008 | T009 | T010 | Week 20: | T096 | T097 | T098 | T099 | T100 |
| Week 03: | T011 | T012 | T013 | T014 | T015 | Week 21: | T101 | T102 | T103 | T104 | T105 |
| Week 04: | T016 | T017 | T018 | T019 | T020 | Week 22: | T106 | T107 | T108 | T109 | T110 |
| Week 05: | T021 | T022 | T023 | T024 | T025 | Week 23: | T111 | T112 | T113 | T114 | T115 |
| Week 06: | T026 | T027 | T028 | T029 | T030 | Week 24: | T116 | T117 | T118 | T119 | T120 |
| Week 07: | T031 | T032 | T033 | T034 | T035 | Week 25: | T121 | T122 | T123 | T124 | T125 |
| Week 08: | T036 | T037 | T038 | T039 | T040 | Week 26: | T126 | T127 | T128 | T129 | T130 |
| Week 09: | T041 | T042 | T043 | T044 | T045 | Week 27: | T131 | T132 | T133 | T134 | T135 |
| Week 10: | T046 | T047 | T048 | T049 | T050 | Week 28: | T136 | T137 | T138 | T139 | T140 |
| Week 11: | T051 | T052 | T053 | T054 | T055 | Week 29: | T141 | T142 | T143 | T144 | T145 |
| Week 12: | T056 | T057 | T058 | T059 | T060 | Week 30: | T146 | T147 | T148 | T149 | T150 |
| Week 13: | T061 | T062 | T063 | T064 | T065 | Week 31: | T151 | T152 | T153 | T154 | T155 |
| Week 14: | T066 | T067 | T068 | T069 | T070 | Week 32: | T156 | T157 | T158 | T159 | T160 |
| Week 15: | T071 | T072 | T073 | T074 | T075 | Week 33: | T161 | T162 | T163 | T164 | T165 |
| Week 16: | T076 | T077 | T078 | T079 | T080 | Week 34: | T166 | T167 | T168 | T169 | T170 |
| Week 17: | T081 | T082 | T083 | T084 | T085 | Week 35: | T171 | T172 | T173 | T174 | T175 |
| Week 18: | T086 | T087 | T088 | T089 | T090 | Week 36: | T176 | T177 | T178 | T179 | T180 |

# Section Milestone & Sprint Drill Mastery Log

Record your Sprint Drill 1–6 times and verify completion of all 15 Speed Sections before signing your certificate.

### 1. APPENDIX B MENTAL MATH SPRINT DRILL LOG (PP. 110–115 • GOAL: < 2 MIN PER SPRINT)

| | | | |
|---|---|---|---|
| Sprint Drill 1: Complement Arithmetic & Base Bridges | Score: _____ / 20 | Time: _____m _____s | [ ] Pass |
| Sprint Drill 2: Instant Doubling, Halving & Distribution | Score: _____ / 20 | Time: _____m _____s | [ ] Pass |
| Sprint Drill 3: Rapid Squares, Square Roots & Cubes | Score: _____ / 20 | Time: _____m _____s | [ ] Pass |
| Sprint Drill 4: Fraction, Decimal & Percent Benchmark Equivalents | Score: _____ / 20 | Time: _____m _____s | [ ] Pass |
| Sprint Drill 5: Proportions, Rates & Unit Conversions | Score: _____ / 20 | Time: _____m _____s | [ ] Pass |
| Sprint Drill 6: Polynomials, Factoring & Quadratic Shortcuts | Score: _____ / 20 | Time: _____m _____s | [ ] Pass |

### 2. 15-SECTION CHAMPION MASTERY CHECKLIST (VERIFY ALL 12 TESTS A1–D3 PER SECTION)

| | | | |
|---|---|---|---|
| Section 1: Step Counting, Benchmarks & Sequences | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 2: Number Bonds & Missing Addends | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 3: Complements & Make-10 Bridging | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 4: Doubles, Equal Groups & Squares | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Cumulative Review C1: Number Sense (Sections 1–4) | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 5: Multi-Step Operations & Grouping | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 6: Fractions, Decimals & Reciprocals | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 7: Primes, GCF/LCM & Nested Expressions | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 8: Algebra, Quadratics & Coordinate Geometry | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Cumulative Review C2: Intermediate Math (Sections 1–8) | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 9: Trigonometry, Vectors & Polar Coordinates | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 10: Exponents, Logarithms & Growth Models | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 11: Combinatorics, Probability & Statistics | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Section 12: Base Conversions, Binary & Bitwise Logic | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |
| Cumulative Review C3: All-Level Championship (Sections 1–12) | Warm-Up: [ ] | Best (/20): _____ | Fastest: _____m |

# Official Mastery Award Certificate

Official recognition for completing 15 speed lessons, 90 warm-up drills, and 180 progressive timed tests.

THE STACKCALC LIBRARY • OFFICIAL MASTERY CITATION

## CERTIFICATE OF MATHEMATICAL SPEED & FLUENCY

**180-TEST FULL-YEAR EDITION • 3,690 ACTIVE PROBLEMS MASTERED**

This is to officially certify that

(Student Name / Math Team Member)

has rigorously completed all 15 Speed Lessons, 90 Guided Shortcut Warm-Up Drills, and 180 Progressive Timed Tests (3,690 problems) in Stack Speed & Fluency. The student has demonstrated proven speed, automatic zero-parenthesis RPN stack pipelines, magnitude estimation accuracy, and mathematical confidence across all four difficulty levels:

**[ ] Level 1: Foundation (Tests A1–A3)** **[ ] Level 2: Core Fluency (Tests B1–B3)**

**[ ] Level 3: Multi-Step Speed (Tests C1–C3)** **[ ] Level 4: Tournament Challenge (Tests D1–D3)**

**Official School / Team Seal**

**Teacher / Parent / Coach Signature**

**Date Completed**

**Next Steps in The StackCalc Library: Explore Vol. I & II Trade Books, Books 2A–4 & 3D Block Studios at https://stackcalc.io!**

(This page intentionally left blank to allow clean removal and framing of the certificate on page 119.)

# SC-6 & SC-32 Speed Pipelines Directory

Permanent quick-reference directory of zero-memory RPN speed pipelines for daily training and competitions.

## STACKCALC SC-6 & SC-32 SPEED PIPELINES QUICK-REFERENCE DIRECTORY

| 1. ARITHMETIC & DUPLICATION PIPELINES | | STACK REGISTER MECHANISM |
|---|---|---|
| Double a Number: | X ENTER + | Clones X into Y; adds X + X in 2 keys |
| Square a Number: | X ENTER × | Clones X into Y; multiplies X × X |
| Compute Fourth Power ($x^4$): | X ENTER × ENTER × | Squares twice ($x \to x^2 \to x^4$) |
| Reciprocal (1/x): | X f 1/x | Inverts X in place (1 ÷ X) |

| 2. ORDER OF OPERATIONS (NO PARENTHESES) | | STACK REGISTER MECHANISM |
|---|---|---|
| (a + b) × (c + d): | A ENTER b + C ENTER D + × | Y holds (a+b) while X computes (c+d) |
| a ÷ (b + c): | b ENTER C + A ↑↓ ÷ | Inner sum first, then swap & divide |
| $a^2 + b^2$ (Sum of Squares): | A ENTER × b ENTER × + | $a^2$ waits in Y while $b^2$ forms in X |
| a / (b × c): | b ENTER C × A ↑↓ ÷ | Denominator product first, then $[x \rightleftarrows y]$ |

| 3. SC-32 POLAR, VECTOR & HARMONIC PIPELINES | | STACK REGISTER MECHANISM |
|---|---|---|
| Hypotenuse $\sqrt{x^2 + y^2}$: | Y ENTER X ↰ ▸θ,r | Resolves magnitude r into X instantly |
| Parallel Resistance $1/(1/R_1 + 1/R_2)$: | R1 $x^{-1}$ R2 $x^{-1}$ + $x^{-1}$ | Sum of reciprocals inverted in place |
| 2D Vector Angle $\vartheta$: | Y ENTER X ↰ ▸θ,r $x \rightleftarrows y$ | Swaps heading angle $\vartheta$ from Y into X |
| Compound Growth $P(1 + r)^t$: | 1 ENTER r + T $y^x$ P × | Evaluates base & power before × P |

| 4. COMPETITION TIME-SAVERS & TRICKS | | STACK REGISTER MECHANISM |
|---|---|---|
| Percent of Number (P% of N): | N ENTER P × 1 0 0 ÷ | Scales N × P and shifts decimal ÷ 100 |
| Average of Two Numbers: | A ENTER b + 2 ÷ | Midpoint (a + b) ÷ 2 with zero parens |
| Difference of Squares ($a^2$ - $b^2$): | A ENTER b - A ENTER b + × | Factored $(a - b)(a + b)$ on stack |
| Quadratic Mean / RMS: | A ENTER × b ENTER × + 2 ÷ f $\sqrt{x}$ | Root-mean-square $\sqrt{(a^2+b^2)/2}$ |

# Mental Math & Contest Shortcut Directory

High-speed number sense identities, fraction-decimal-percent benchmarks, and geometry constants.

## MENTAL MATH & CONTEST SHORTCUT FORMULA DIRECTORY

### 1. RAPID MULTIPLICATION TRICKS

| | |
|---|---|
| **Multiply by 5:** | Half the number, then append 0 (or multiply by 10 then divide by 2) |
| **Multiply by 25:** | Divide by 4, then append 00 (e.g. $48 \times 25 = 12 \rightarrow 1{,}200$) |
| **Multiply by 15:** | Add half the number to itself, then multiply by 10 (e.g. $24 \times 15 = 36 \rightarrow 360$) |
| **Multiply by 11:** | Sum adjacent digit pairs (e.g. $45 \times 11 = 4\ [4{+}5]\ 5 = 495$) |

### 2. SPECIAL SQUARES & PRODUCTS

| | |
|---|---|
| **Square Ending in 5:** | $(10t + 5)^2 = 100 \times t(t + 1) + 25$ (e.g. $75^2$: $7 \times 8 = 56 \rightarrow 5{,}625$) |
| **Numbers Near 100:** | $(100 + a)(100 + b) = 100(100 + a + b) + ab$ (e.g. $103 \times 106 = 10{,}918$) |
| **Symmetric Products:** | $(m - d)(m + d) = m^2 - d^2$ (e.g. $27 \times 33 = 30^2 - 3^2 = 891$) |
| **Difference of Squares:** | $a^2 - b^2 = (a - b)(a + b)$ (e.g. $52^2 - 48^2 = 4 \times 100 = 400$) |

### 3. FRACTION-DECIMAL-PERCENT BENCHMARKS

| | |
|---|---|
| **Eighths (1/8 to 7/8):** | $1/8 = 0.125$ (12.5%) $3/8 = 0.375$ (37.5%) $5/8 = 0.625$ $7/8 = 0.875$ |
| **Thirds & Sixths:** | $1/3 = 33.3\%$ $2/3 = 66.7\%$ $1/6 = 0.1667$ $5/6 = 0.8333$ |
| **Twelfths (1/12):** | $1/12 = 0.0833$ (8.33%) $5/12 = 0.4167$ $7/12 = 0.5833$ $11/12 = 0.9167$ |
| **Sixteenths (1/16):** | $1/16 = 0.0625$ (6.25%) $3/16 = 0.1875$ $5/16 = 0.3125$ $7/16 = 0.4375$ |

### 4. HIGH-YIELD GEOMETRY CONSTANTS

| | |
|---|---|
| **Equilateral Triangle Area:** | $A = (\sqrt{3} \div 4)s^2 \approx 0.43301 \times s^2$ |
| **Circle Area / Circumference:** | $A = \pi{\cdot}r^2 \approx 3.14159{\cdot}r^2$ \| $C = 2{\cdot}\pi{\cdot}r \approx 6.28318{\cdot}r$ |
| **30-60-90 Special Triangle:** | Sides ratio: $1 : \sqrt{3} : 2$ ($\sqrt{3} \approx 1.73205$) |
| **45-45-90 Special Triangle:** | Sides ratio: $1 : 1 : \sqrt{2}$ ($\sqrt{2} \approx 1.41421$) |

# Why 3D Block Studios Build Unbeatable Speed

How the physical Place Value, Fraction, and Prime Factor Studios power mental fluency.

### 1. TACTILE INTUITION FIRST: WHY PHYSICAL 3D BLOCKS MAKE MENTAL MATH AUTOMATIC

True mathematical speed never comes from blind memorization—it comes from seeing the physical structure of numbers immediately in your mind's eye. When you train with the three official StackCalc 3D Manipulative Block Studios alongside your timed tests, complements to 100, fraction-decimal benchmarks, and prime factorizations become effortless:

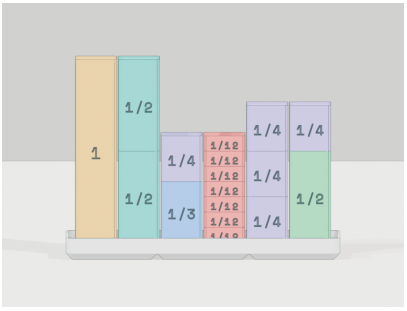
#### 1. Place Value Studio




**Sections 01–03 & 05: Complements & Powers of 10**

2-Row Trays (Y & X), Digit Blocks (0–9), and Gold [+1] Carry Tokens make Make-10/Make-100 complements and decimal shifts instant.

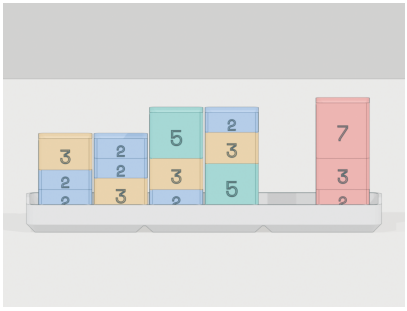
#### 2. Fraction Studio




**Sections 04 & 06: Fraction-Decimal Benchmarks**

84.0 mm proportional fraction towers (1 Whole down to 1/12) let you see equivalent fractions, unit rates, and harmonic sums at a glance.

#### 3. Prime Factor Studio




**Sections 07–09: Factoring, Squares & Logs**

Logarithmic prime blocks ($h(p) = 7.0 \cdot \log_2(p)$ mm for 2, 3, 5, 7) turn prime factorization, GCF/LCM, and logarithm laws into physical height addition.

### 2. TOURNAMENT & DAILY TIMED-SPRINT READINESS CHECKLIST

**Step 1 — Tactile Warm-Up (2 Minutes):**

Before starting your timer on a new section, build the core identity once with your Place Value, Fraction, or Prime Factor Studio blocks and work the 2 Page-Top Worked Examples by hand.

**Step 2 — Clear Registers & Check Mode:**

Press [CLEAR] before the clock starts so Registers T, Z, Y, and X are 0. For trigonometry tests (Sections 10–12 & C1–C3), verify your SC-32 angle mode (DEG vs. RAD).

**Step 3 — Inside-Out Execution (Zero Memory Keys):**

Evaluate the deepest inner expression first, let intermediate results ride on Waiting Shelf Y, and use [$x \rightleftarrows y$] or [$1/x$] instead of slow scratchpad re-entry.

**Step 4 — Post-Sprint Keystroke Audit:**

After grading your test at stackcalc.io/curriculum/answers#speed-fluency, re-key any missed or slow problem 3 times using the section's Optimal Champion Pipeline.

APPENDIX E (PART 4 OF 4) • OFFICIAL STACKCALC HARDWARE ECOSYSTEM & ORDERING DIRECTORY

# Official StackCalc Block Studios & Hardware Guide

Complete your hands-on math lab with our 3D Manipulative Block Studios and RPN calculators at https://stackcalc.io.

## 1. OFFICIAL STACKCALC 3D MANIPULATIVE BLOCK STUDIOS & CALCULATORS (STACKCALC.IO)

| | | |
|---|---|---|
| STUDIO 1 | **Place Value Studio** | **Speed Kit • K–6** |
| 6×2 Whole & Decimal Trays, 40 Digit Blocks (0–9), Gold [+1] Carry Tokens & Base-10 Set | | |
| STUDIO 2 | **Fraction Studio** | **Speed Kit • 2–8** |
| 78 Proportional Fraction Towers (84.0 mm Whole down to 1/12) & 12-Socket Comparator Lid | | |
| STUDIO 3 | **Prime Factor Studio** | **Speed Kit • 3–10** |
| Logarithmic Prime Blocks ($h(p) = 7.0 \cdot \log 2(p)$ mm for 2, 3, 5, 7) & GCF/LCM Comparator Lid | | |
| SC-6 | **Elementary RPN Calculator** | **Secs 1–8 • K–8** |
| 4-Register Visible Stack (T, Z, Y, X), Two-Dot Fractions (.a.b), [PV], [FACT], [GCF] + Web Sim | | |
| SC-32 | **Scientific STEM Calculator** | **Secs 9–C3 • 8–12** |
| 32-Key Scientific RPN: [POLY], [$\rightarrow \theta, r$], [SIN/COS/TAN], [LN/LOG], [BASE] + iOS/watchOS App | | |

## 2. COMPANION TRADE BOOKS & WORKBOOKS IN THE STACKCALC LIBRARY (GRADES K–12 & AP)

| | |
|---|---|
| **Volume I & II • Illustrated Trade Books:** | Vol. I: From Blocks to the Stack (Grades K–6) • Vol. II: Thinking in the Stack (Grades 7–12 & AP) |
| **Book 1 • Teaching with the Stack:** | Master K–12 Teacher & Homeschool Guide, Block Games & Coaching Scripts (46 pp.) |
| **Book 2A & 2B • Elementary Math Labs:** | Book 2A: Early Elementary (K–2) • Book 2B: Upper Elementary (3–5) (10 Units + 4 Exams • 840 Probs) |
| **Book 3 • Middle School Math Lab:** | Grades 6–8 Two-Method Student Workbook (SC-6 • 10 Units & 45 Sections • 1,000 Problems) |
| **Book 4 • High School & STEM Lab:** | Grades 9–12 & AP Scientific Workbook (SC-32 • 10 Units + 4 Cumulative Exams • 840 Problems) |

## 3. ONLINE ANSWER KEY, FREE SIMULATOR & HARDWARE ORDERING PORTAL

- Online Worked Solutions & Champion Pipelines (All 180 Tests / 3,690 Problems): https://stackcalc.io/curriculum/answers#speed-fluency
- Order Official 3D Manipulative Block Studios & Distraction-Free SC-6 / SC-32 Calculators: https://stackcalc.io
- Free Web Simulator & iOS/watchOS App: Practice on the interactive SC-6 / SC-32 simulator or download StackCalc 32 on the App Store!